Exemple d'utilisation du package DegradSim

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Préparation

Installation du package depuis GitHub :

```
remotes::install_github("rytracetito/levy_packages", subdir = "DegradSim")
```

Les parties I et II utilisent les sept **processus de Lévy** du package (`A = NULL` par défaut). La partie III montre les **processus de Lévy à échelle de temps** (*time-scaled Lévy processes*) $X_t = Y_{A(t)}$, obtenus avec

une fonction d'échelle de temps A (*time-scaling function*) : gamma, Wiener et inverse gaussien. La partie IV reprend ces simulations avec la **méthode Bridge** (`method = "Bridge"`), pour les cinq processus qui ont un pont dans le package `levyBridge`.

Paramètres utilisés :

Processus	Fonction	Paramètres
Gamma	`sim_gamma`	`a = 2`, `b = 3` (taux)
Wiener	`sim_wiener`	`mu = 1.5`, `sigma = 2`
Inverse gaussien	`sim_ig`	`mu = 0.5`, `eta = 2`
Variance-gamma	`sim_vg`	`theta = 0.5`, `sigma = 1.5`, `nu = 2`
Meixner	`sim_meixner`	`a = 1`, `b = 0.3`, `d = 1.5`, `m = 0.2`
Normal inverse gaussien	`sim_nig`	`alpha = 2`, `beta = 0.5`, `delta = 1`, `mu = 0.2`
Poisson composé	`sim_cpoisson`	`lambda = 0.5`, sauts exponentiels de moyenne 1 (`rjump` par défaut)

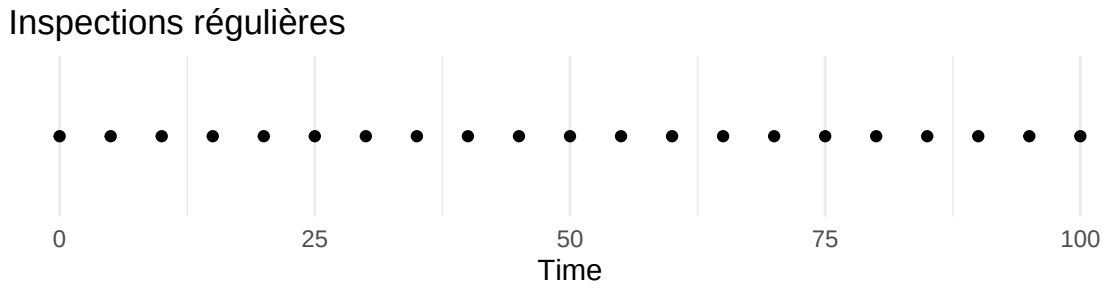
Dans chaque cas, les trajectoires simulées sont d'abord **stockées** dans un objet `x`, puis utilisées : premières valeurs, accroissements de la première trajectoire (arrondis à 3 décimales), puis graphique. Les accroissements montrent qu'un palier presque plat sur le graphique correspond à de petits accroissements, pas à des accroissements nuls (sauf pour le Poisson composé, constant entre deux chocs). Quand toutes les trajectoires ont les mêmes instants d'inspection, `x` est une matrice : une colonne `t`, puis une colonne par trajectoire (`X1`, `X2`, ...).

I - Une trajectoire

Une trajectoire inspectée 20 fois entre 0 et 100. `x` est une matrice à deux colonnes : `t` et `X1`.

Cas 1 : inspections régulières

```
t1 <- inspection_times("regular", n = 20, tmax = 100)$time
plot_plan(t1, "Inspections régulières")
```



Modèle 1 : gamma

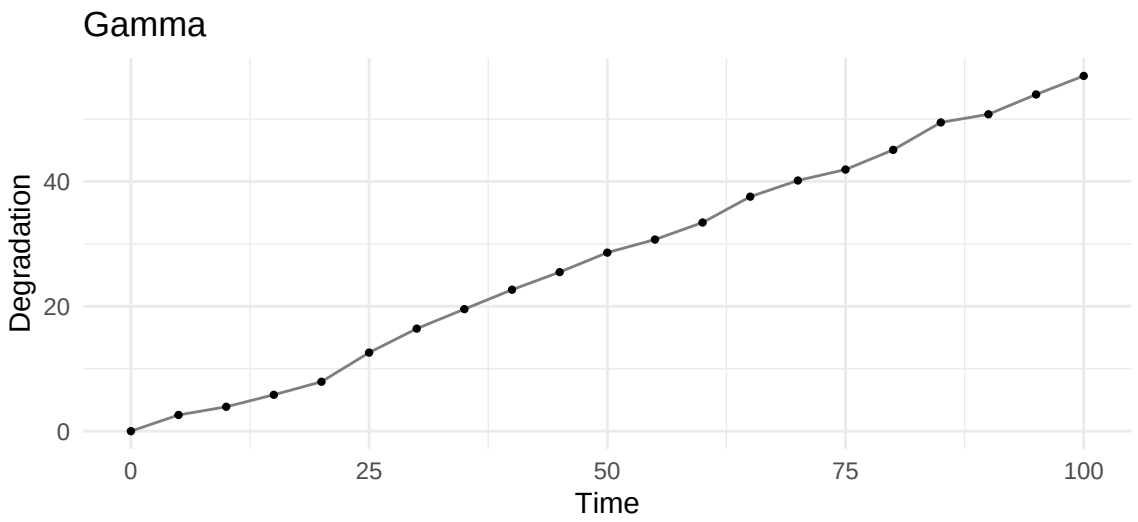
```
set.seed(11)
x <- sim_gamma(t1, a = 2, b = 3)
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  2.588550
## [3,] 10  3.900945
## [4,] 15  5.822353
## [5,] 20  7.910152
## [6,] 25 12.582689
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.589 1.312 1.921 2.088 4.673 3.841 3.120 3.116 2.819 3.123 2.095 2.725
## [13] 4.144 2.588 1.757 3.150 4.395 1.307 3.174 2.977
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

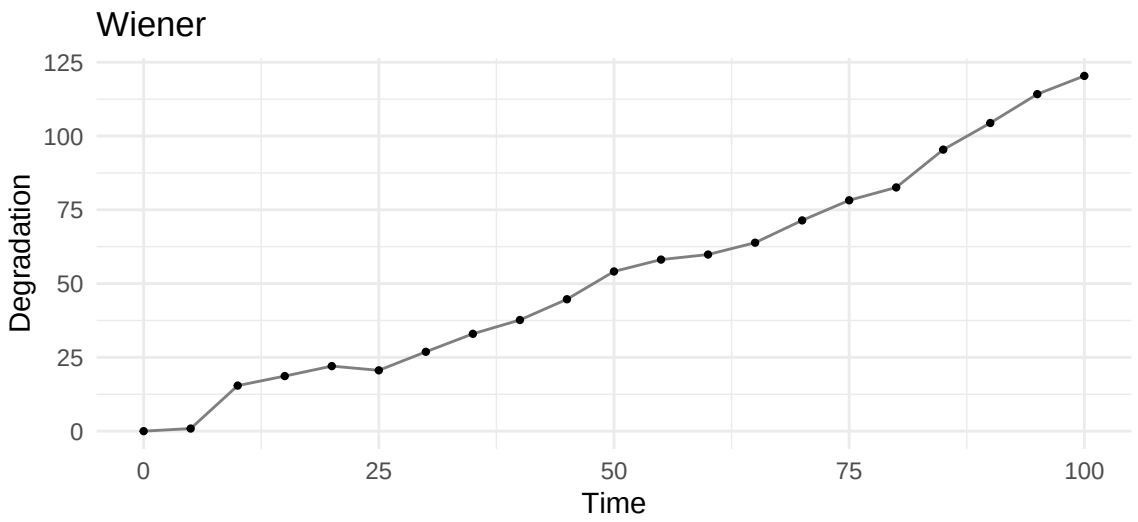
```
set.seed(12)
x <- sim_wiener(t1, mu = 1.5, sigma = 2)
head(x)
```

```
##       t         X1
## [1,]  0  0.0000000
## [2,]  5  0.8787004
## [3,] 10 15.4320167
## [4,] 15 18.6533253
## [5,] 20 22.0389368
## [6,] 25 20.6052097
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.879 14.553  3.221  3.386 -1.434  6.282  6.090  4.690  7.024  9.414
## [11]  4.022  1.714  4.014  7.553  6.818  4.354 12.817  9.023  9.767  6.188
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

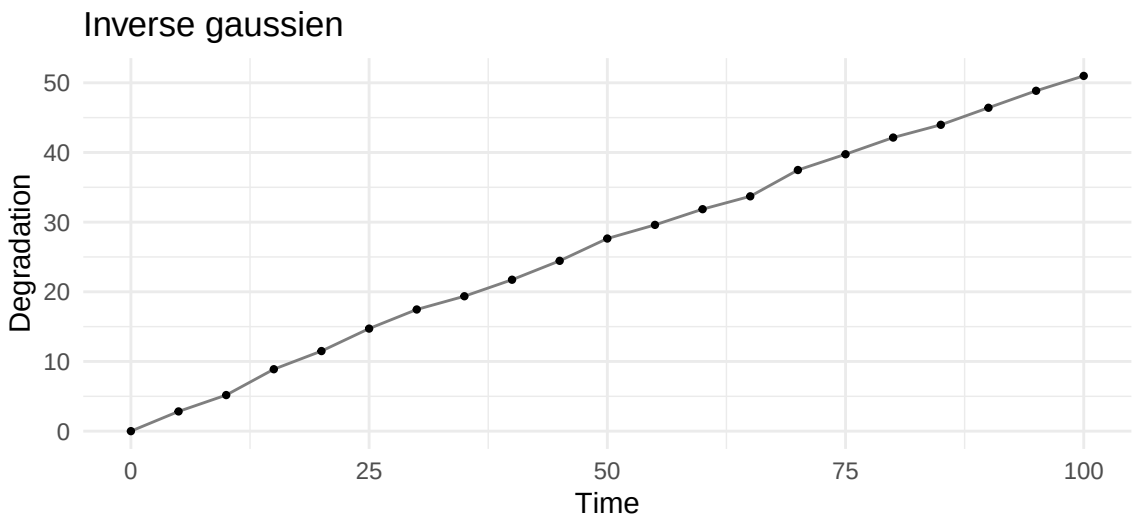
```
set.seed(13)
x <- sim_ig(t1, mu = 0.5, eta = 2)
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  2.829678
## [3,] 10  5.177833
## [4,] 15  8.886486
## [5,] 20 11.493417
## [6,] 25 14.718884
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.830 2.348 3.709 2.607 3.225 2.743 1.901 2.371 2.713 3.199 1.959 2.255
## [13] 1.846 3.775 2.266 2.394 1.832 2.444 2.437 2.137
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

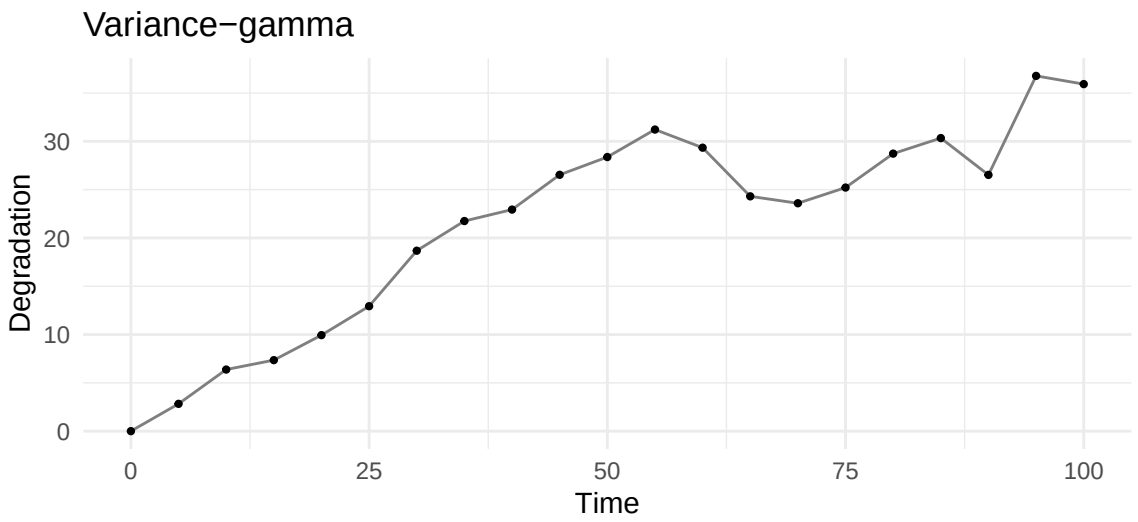
```
set.seed(14)
x <- sim_vg(t1, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  2.826550
## [3,] 10  6.375562
## [4,] 15  7.349937
## [5,] 20  9.939181
## [6,] 25 12.934933
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.827  3.549  0.974  2.589  2.996  5.745  3.070  1.187  3.601  1.838
## [11]  2.847 -1.879 -5.037 -0.715  1.625  3.514  1.609 -3.813 10.250 -0.850
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

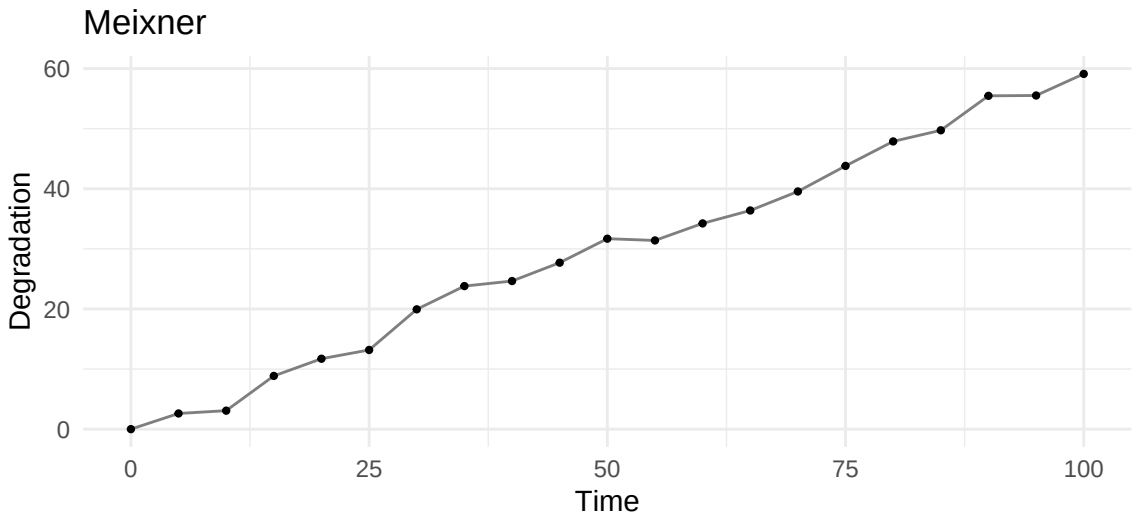
```
set.seed(15)
x <- sim_meixner(t1, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  2.609114
## [3,] 10  3.073932
## [4,] 15  8.858509
## [5,] 20 11.718557
## [6,] 25 13.175516
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.609  0.465  5.785  2.860  1.457  6.765  3.865  0.841  3.056  3.990
## [11] -0.291  2.835  2.153  3.165  4.254  4.075  1.855  5.735  0.053  3.594
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

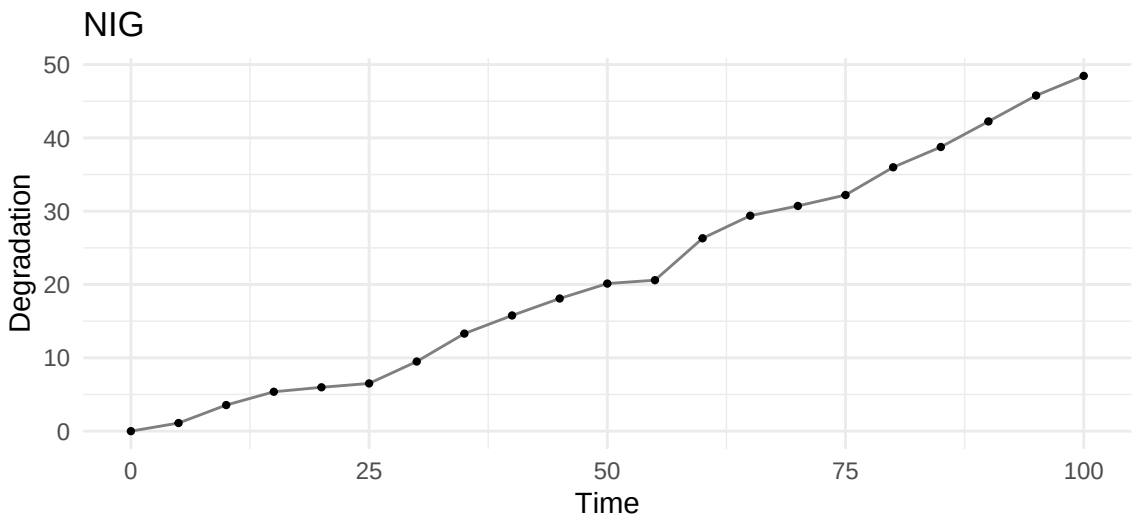
```
set.seed(16)
x <- sim_nig(t1, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##       t       X1
## [1,]  0 0.000000
## [2,]  5 1.106802
## [3,] 10 3.555632
## [4,] 15 5.369281
## [5,] 20 5.980709
## [6,] 25 6.508225
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.107 2.449 1.814 0.611 0.528 2.987 3.809 2.473 2.314 2.036 0.464 5.717
## [13] 3.078 1.331 1.489 3.784 2.767 3.484 3.534 2.677
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

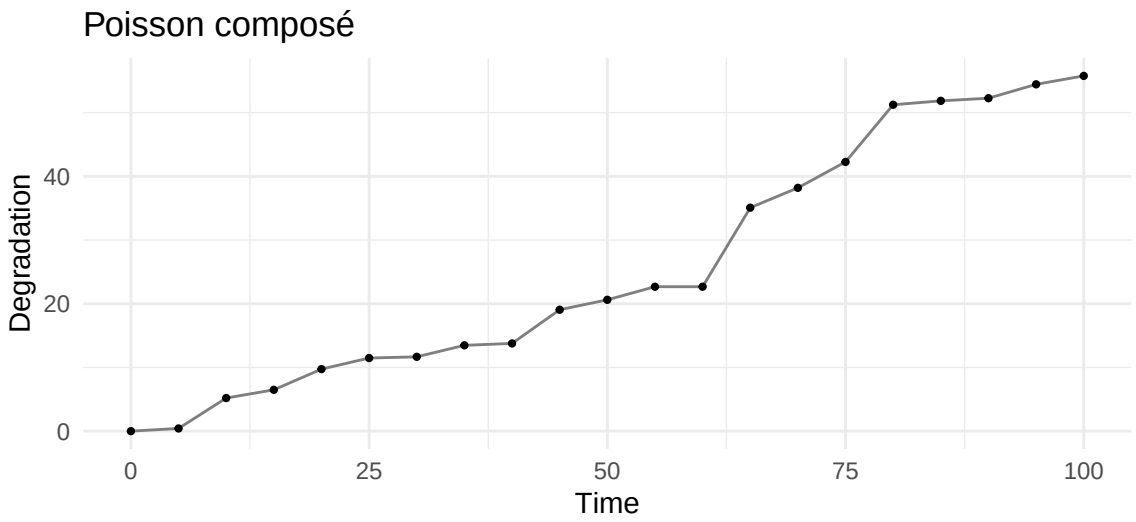
```
set.seed(17)
x <- sim_cpoisson(t1, lambda = 0.5)
head(x)
```

```
##       t         X1
## [1,]  0  0.0000000
## [2,]  5  0.4210759
## [3,] 10  5.1945031
## [4,] 15  6.4873965
## [5,] 20  9.7368579
## [6,] 25 11.4769556
```

```
round(diff(x[, "X1"]), 3)
```

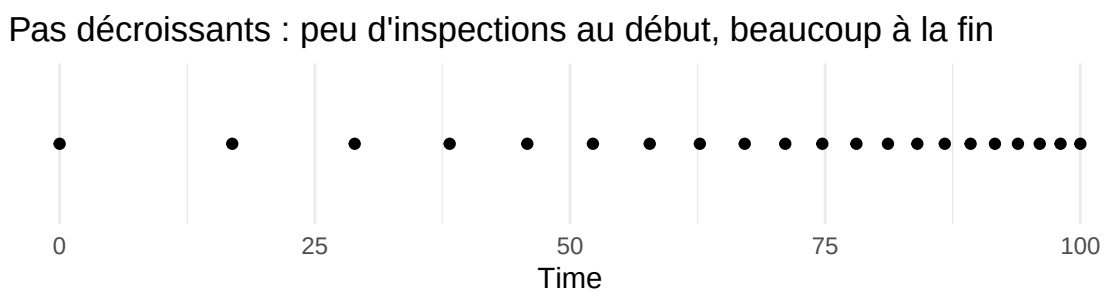
```
##  [1]  0.421  4.773  1.293  3.249  1.740  0.191  1.804  0.298  5.292  1.554
## [11]  2.055  0.000 12.406  3.120  4.068  8.975  0.614  0.413  2.177  1.325
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
0
20
40
0
25
50
75
100
Time

Cas 2 : pas décroissants

```
t2 <- inspection_times("decreasing", n = 20, tmax = 100)$time
plot_plan(t2, "Pas décroissants : peu d'inspections au début, beaucoup à la fin")
```



Modèle 1 : gamma

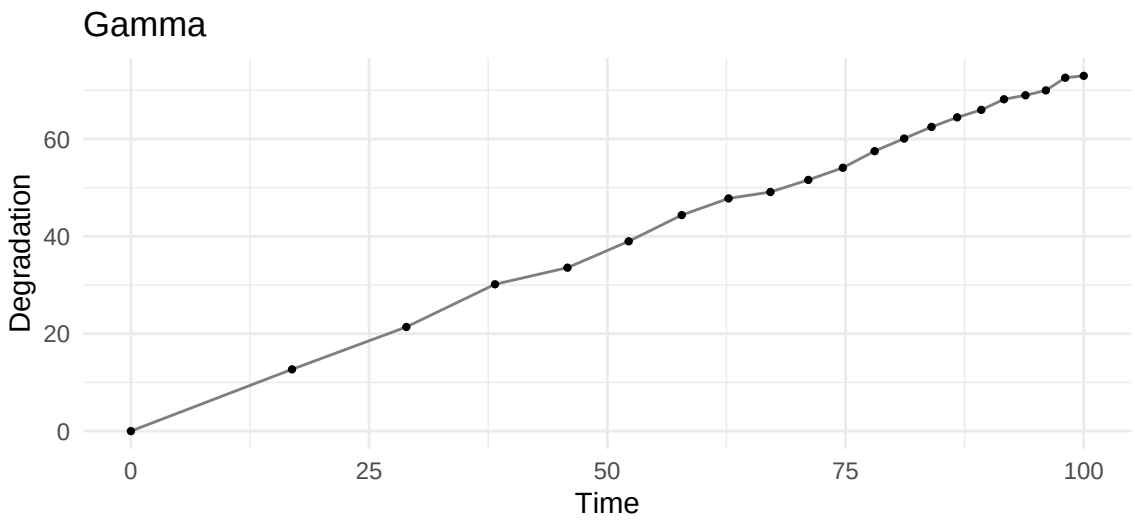
```
set.seed(21)
x <- sim_gamma(t2, a = 2, b = 3)
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 12.68436
## [3,] 28.90648 21.38241
## [4,] 38.21229 30.15090
## [5,] 45.81569 33.56269
## [6,] 52.24427 38.98223
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 12.684  8.698  8.768  3.412  5.420  5.385  3.423  1.320  2.481  2.496
## [11]  3.416  2.586  2.386  1.966  1.530  2.184  0.821  0.999  2.611  0.380
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

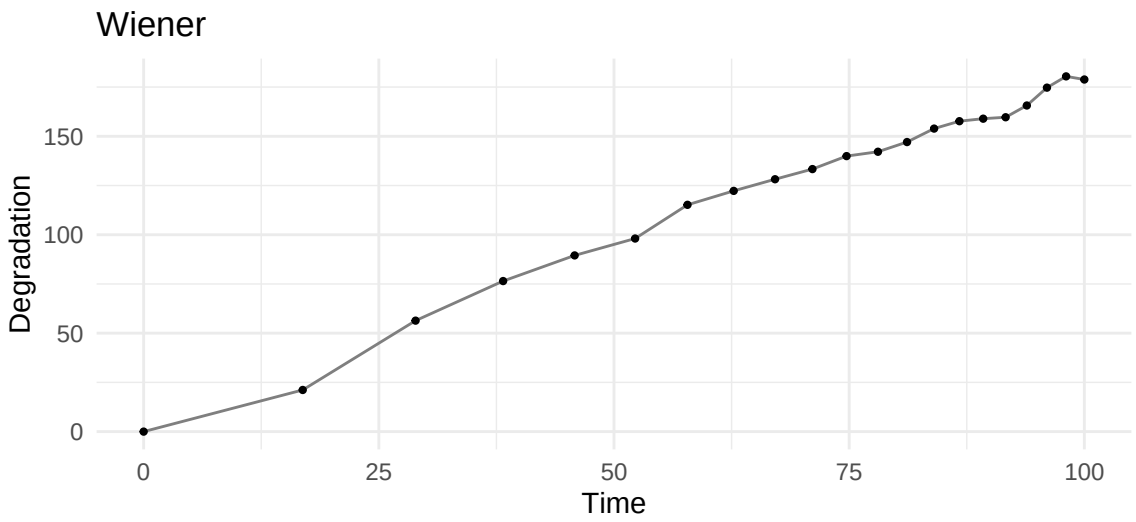
```
set.seed(22)
x <- sim_wiener(t2, mu = 1.5, sigma = 2)
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 21.15190
## [3,] 28.90648 56.36371
## [4,] 38.21229 76.47126
## [5,] 45.81569 89.49119
## [6,] 52.24427 98.07444
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 21.152 35.212 20.108 13.020  8.583 17.123  7.075  5.908  5.165  6.588
## [11]  2.216  4.924  6.837  3.762  1.271  0.728  5.970  9.068  5.724 -1.591
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

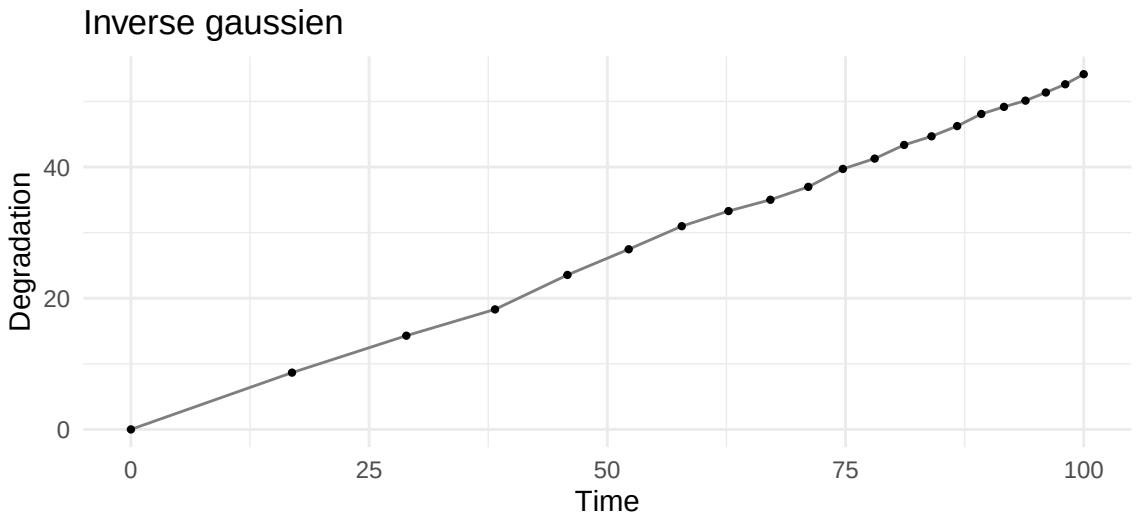
```
set.seed(23)
x <- sim_ig(t2, mu = 0.5, eta = 2)
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  8.655577
## [3,] 28.90648 14.289435
## [4,] 38.21229 18.296030
## [5,] 45.81569 23.551265
## [6,] 52.24427 27.462389
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 8.656 5.634 4.007 5.255 3.911 3.519 2.307 1.724 1.965 2.736 1.572 2.079
## [13] 1.321 1.558 1.849 1.079 0.948 1.244 1.255 1.538
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

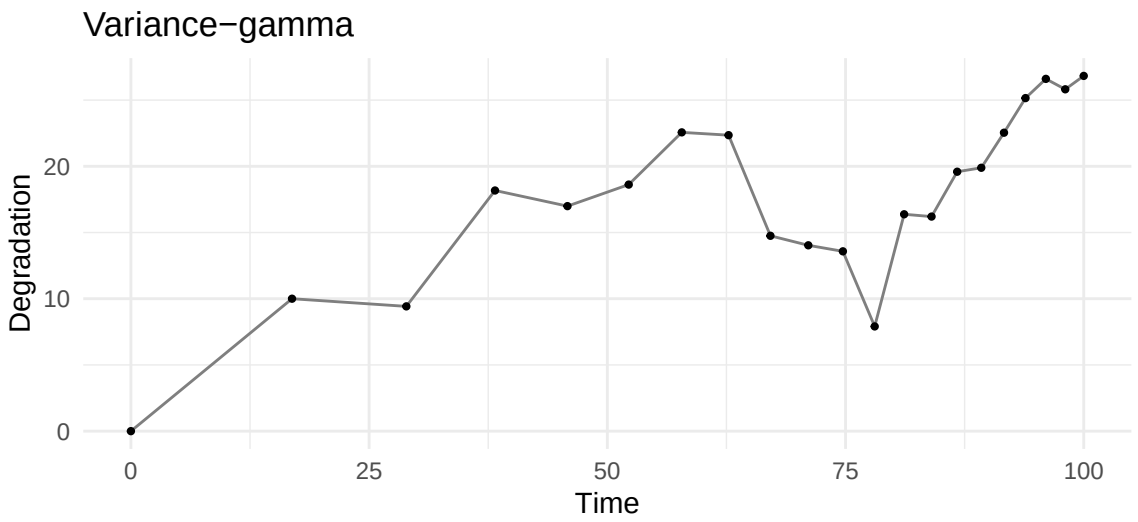
```
set.seed(24)
x <- sim_vg(t2, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 10.00188
## [3,] 28.90648  9.42331
## [4,] 38.21229 18.17461
## [5,] 45.81569 16.99351
## [6,] 52.24427 18.62047
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 10.002 -0.579  8.751 -1.181  1.627  3.944 -0.213 -7.600 -0.718 -0.455
## [11] -5.672  8.471 -0.174  3.388  0.298  2.649  2.613  1.452 -0.788  1.012
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

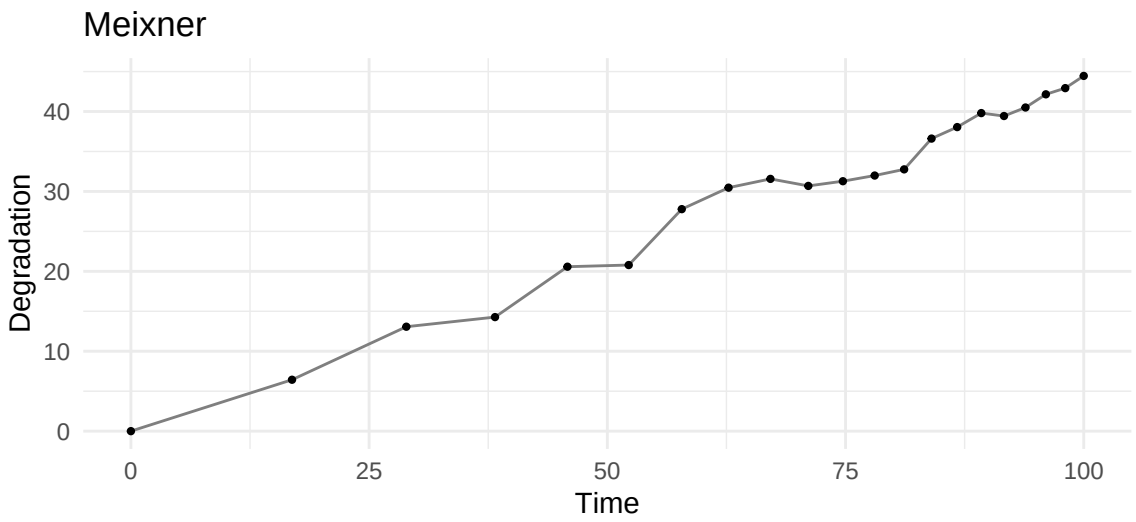
```
set.seed(25)
x <- sim_meixner(t2, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  6.431984
## [3,] 28.90648 13.068405
## [4,] 38.21229 14.274293
## [5,] 45.81569 20.578044
## [6,] 52.24427 20.786662
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  6.432  6.636  1.206  6.304  0.209  6.996  2.688  1.098 -0.879  0.589
## [11]  0.708  0.773  3.853  1.438  1.752 -0.366  1.064  1.650  0.775  1.531
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

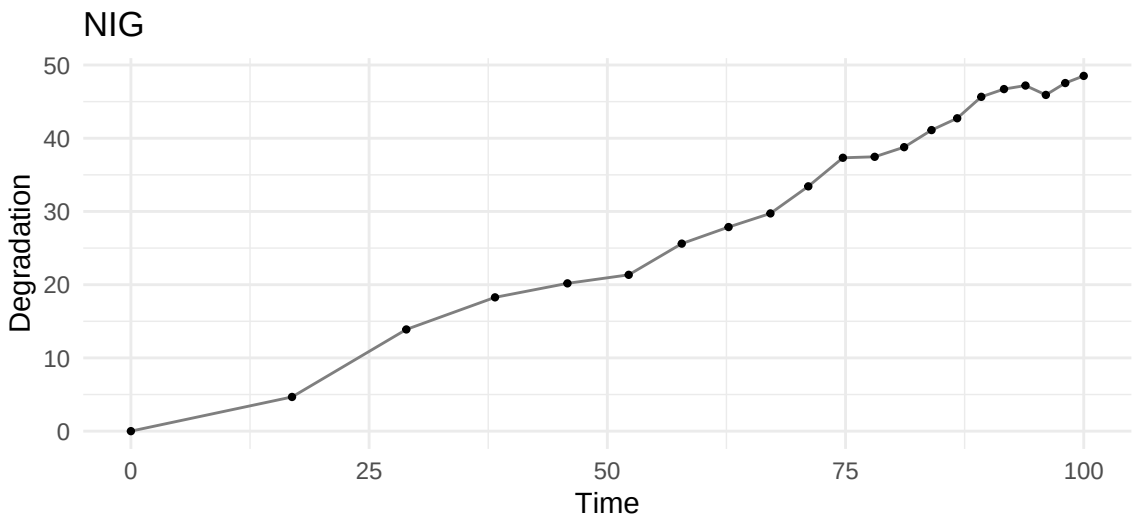
```
set.seed(26)
x <- sim_nig(t2, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  4.668869
## [3,] 28.90648 13.887229
## [4,] 38.21229 18.270337
## [5,] 45.81569 20.180364
## [6,] 52.24427 21.346607
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  4.669  9.218  4.383  1.910  1.166  4.261  2.262  1.864  3.696  3.902
## [11]  0.132  1.316  2.332  1.614  2.929  1.057  0.483 -1.279  1.621  0.976
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

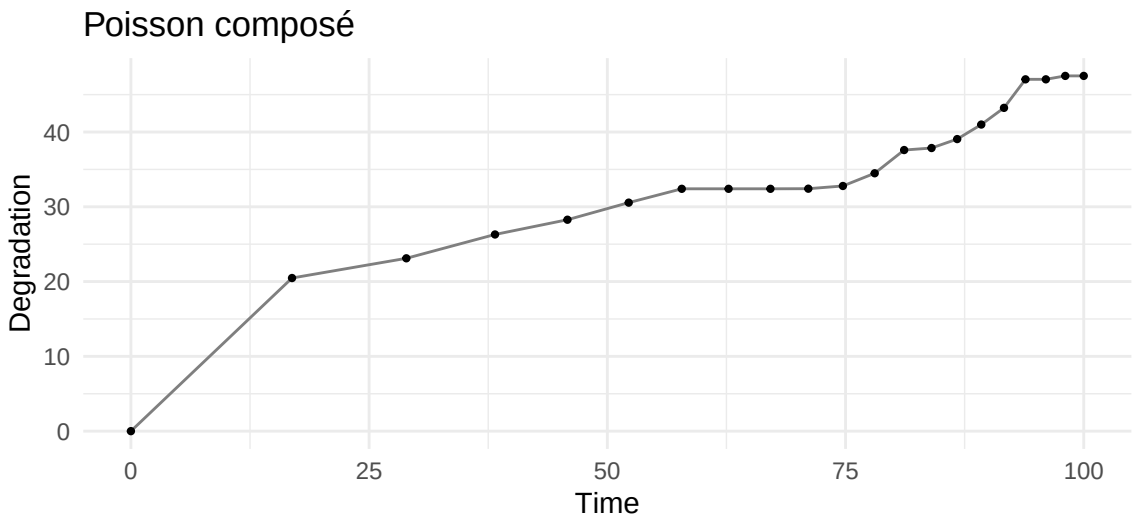
```
set.seed(27)
x <- sim_cpoisson(t2, lambda = 0.5)
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 20.47580
## [3,] 28.90648 23.11248
## [4,] 38.21229 26.29892
## [5,] 45.81569 28.27167
## [6,] 52.24427 30.55641
```

```
round(diff(x[, "X1"]), 3)
```

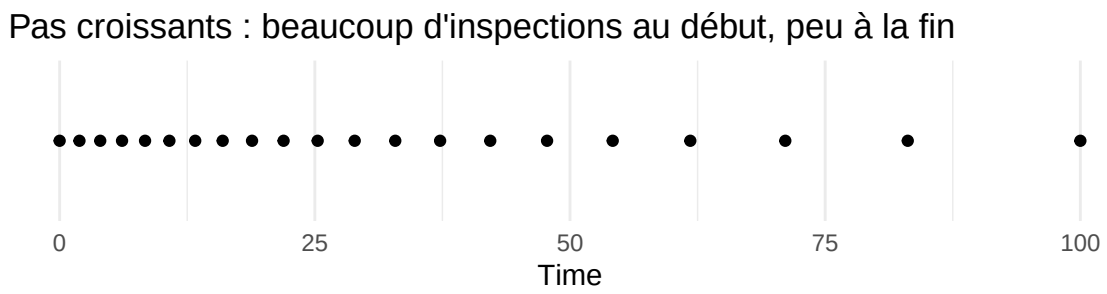
```
##  [1] 20.476  2.637  3.186  1.973  2.285  1.850  0.000  0.000  0.015  0.363
## [11]  1.700  3.106  0.269  1.196  1.943  2.232  3.814  0.000  0.459  0.000
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
0
10
20
30
40
0
25
50
75
100
Time

Cas 3 : pas croissants

```
t3 <- inspection_times("increasing", n = 20, tmax = 100)$time
plot_plan(t3, "Pas croissants : beaucoup d'inspections au début, peu à la fin")
```



Modèle 1 : gamma

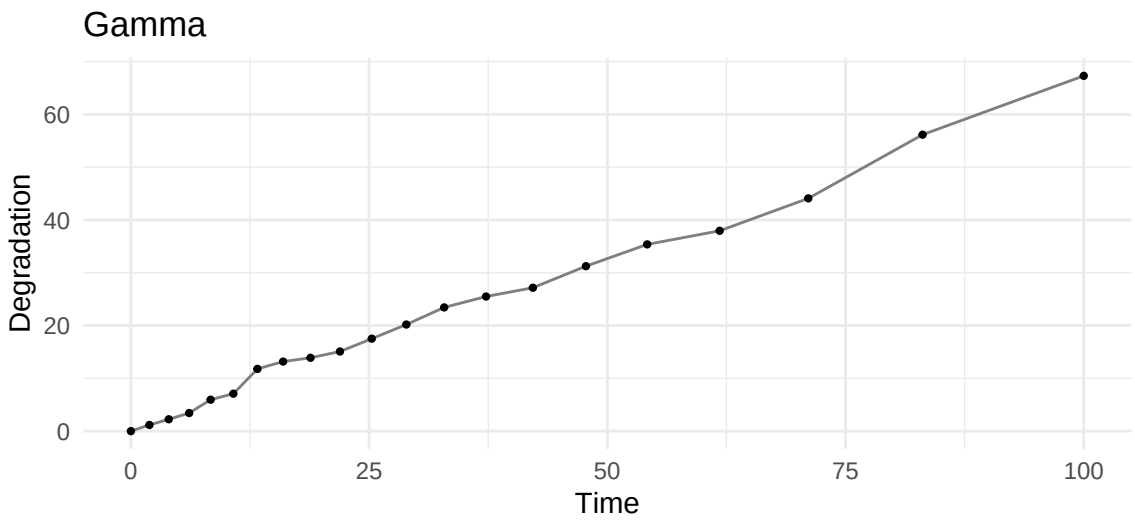
```
set.seed(31)
x <- sim_gamma(t3, a = 2, b = 3)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  1.940035 1.161003
## [3,]  3.974743 2.237610
## [4,]  6.113840 3.420305
## [5,]  8.368618 5.951271
## [6,] 10.752309 7.083982
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.161  1.077  1.183  2.531  1.133  4.699  1.389  0.714  1.189  2.432
## [11]  2.680  3.245  2.064  1.658  4.087  4.129  2.576  6.145 12.071 11.140
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

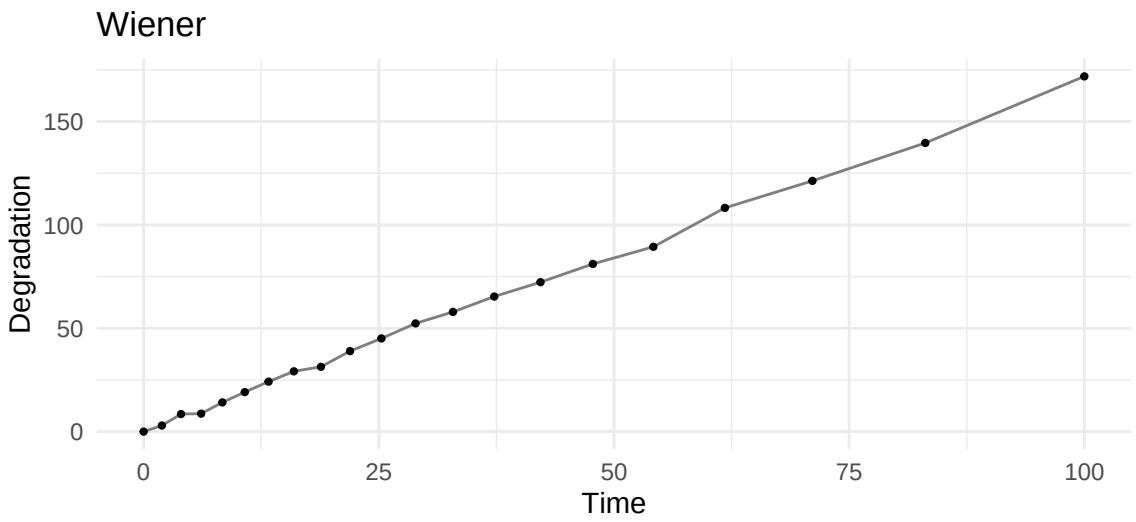
```
set.seed(32)
x <- sim_wiener(t3, mu = 1.5, sigma = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  1.940035  2.950837
## [3,]  3.974743  8.494273
## [4,]  6.113840  8.696041
## [5,]  8.368618 14.137385
## [6,] 10.752309 19.100712
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.951  5.543  0.202  5.441  4.963  5.087  4.971  2.198  7.588  6.144
## [11]  7.254  5.563  7.444  6.951  8.826  8.283 18.828 13.065 18.341 32.221
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

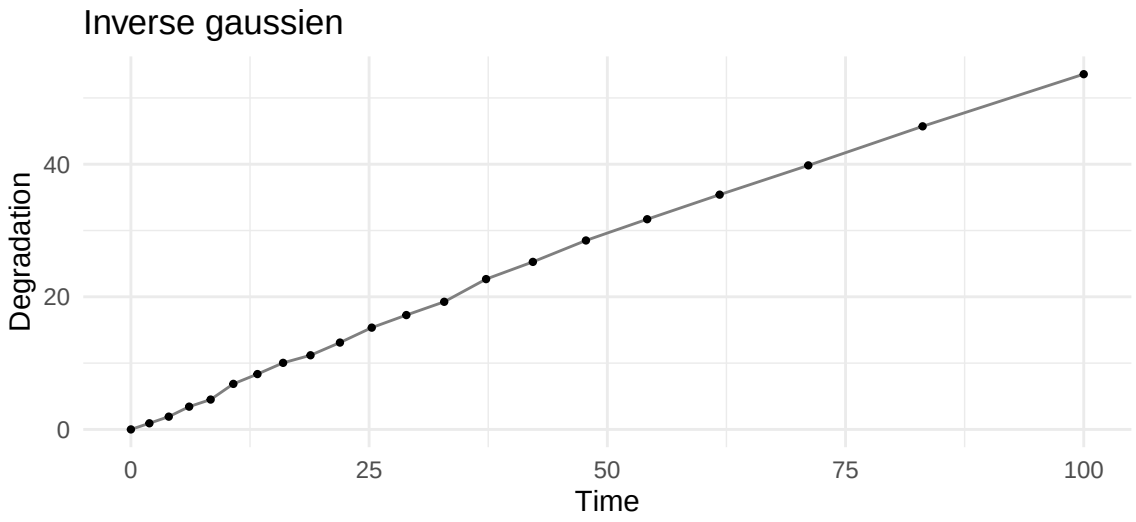
```
set.seed(33)
x <- sim_ig(t3, mu = 0.5, eta = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  1.940035 0.9238276
## [3,]  3.974743 1.9267367
## [4,]  6.113840 3.4350758
## [5,]  8.368618 4.5045984
## [6,] 10.752309 6.8688533
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.924 1.003 1.508 1.070 2.364 1.478 1.693 1.144 1.914 2.252 1.896 2.002
## [13] 3.424 2.595 3.231 3.205 3.704 4.415 5.894 7.865
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

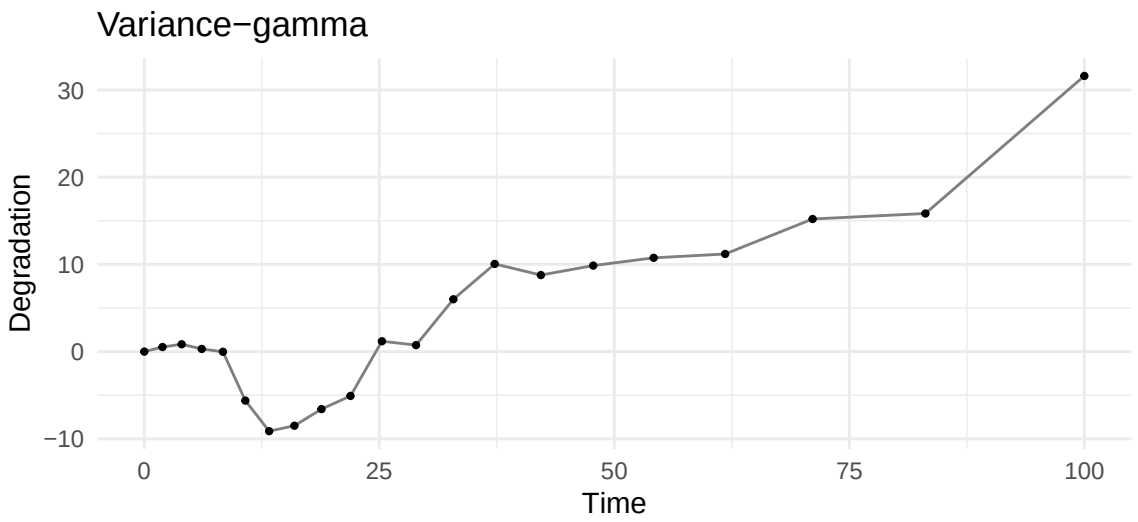
```
set.seed(34)
x <- sim_vg(t3, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t          X1
## [1,]  0.000000  0.00000000
## [2,]  1.940035  0.52692453
## [3,]  3.974743  0.84233589
## [4,]  6.113840  0.30912827
## [5,]  8.368618 -0.01932868
## [6,] 10.752309 -5.61595766
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.527  0.315 -0.533 -0.328 -5.597 -3.498  0.620  1.903  1.513  6.269
## [11] -0.446  5.257  4.048 -1.275  1.086  0.890  0.440  4.011  0.630 15.771
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

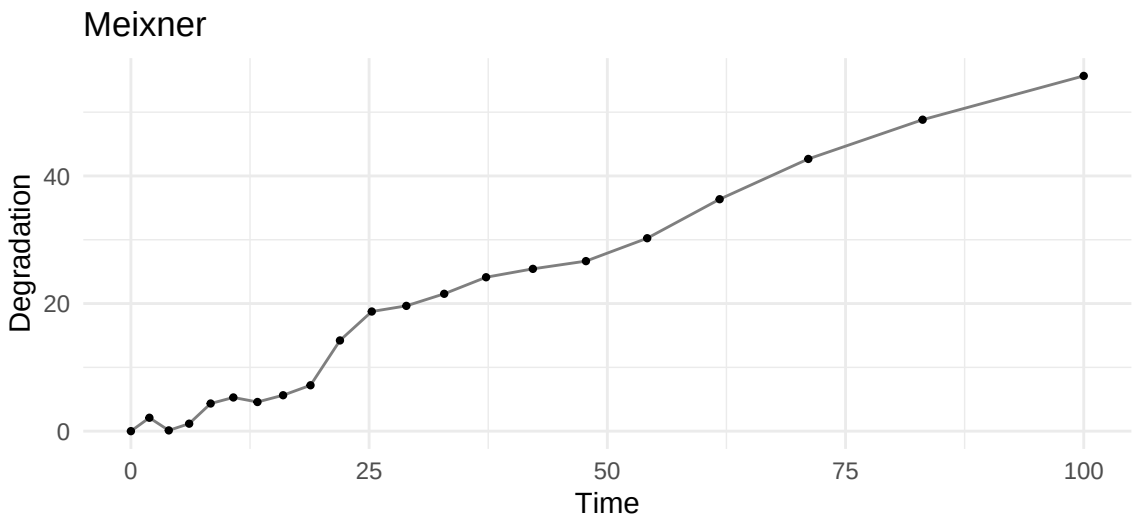
```
set.seed(35)
x <- sim_meixner(t3, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  1.940035 2.0981428
## [3,]  3.974743 0.1163328
## [4,]  6.113840 1.1689796
## [5,]  8.368618 4.3318852
## [6,] 10.752309 5.2801270
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.098 -1.982  1.053  3.163  0.948 -0.713  1.061  1.563  7.025  4.535
## [11]  0.886  1.892  2.595  1.315  1.213  3.589  6.116  6.314  6.149  6.868
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

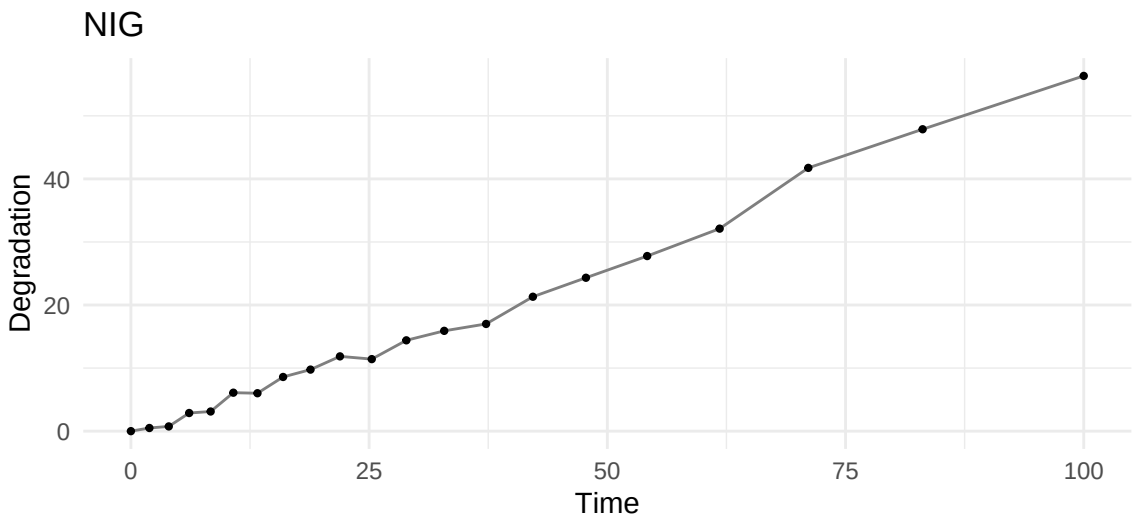
```
set.seed(36)
x <- sim_nig(t3, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  1.940035 0.4992997
## [3,]  3.974743 0.7488852
## [4,]  6.113840 2.8742631
## [5,]  8.368618 3.1134810
## [6,] 10.752309 6.0980221
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.499  0.250  2.125  0.239  2.985 -0.086  2.579  1.169  2.098 -0.438
## [11]  2.981  1.503  1.094  4.313  3.011  3.444  4.345  9.634  6.128  8.460
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

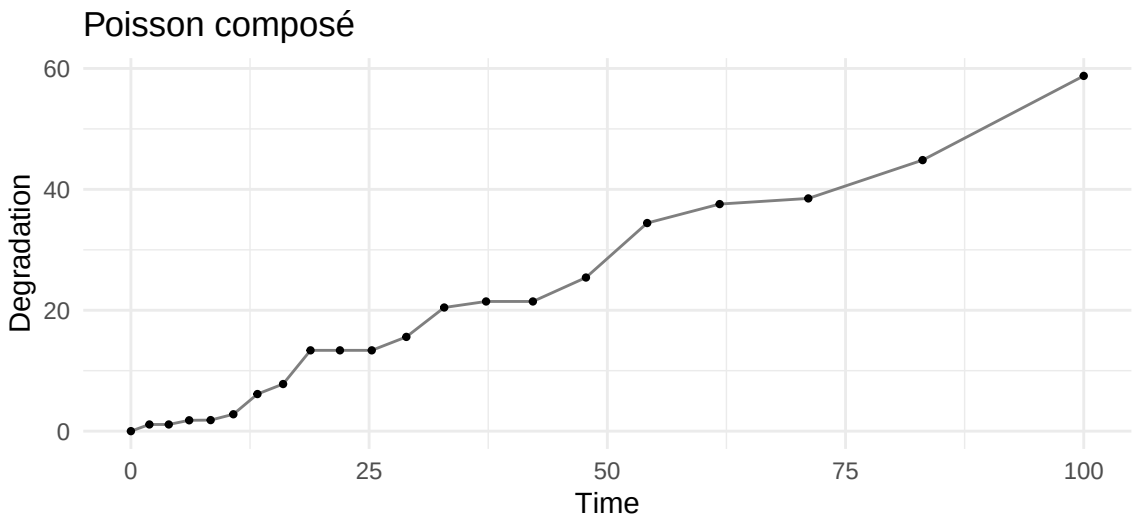
```
set.seed(37)
x <- sim_cpoisson(t3, lambda = 0.5)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  1.940035 1.104451
## [3,]  3.974743 1.104451
## [4,]  6.113840 1.796155
## [5,]  8.368618 1.827021
## [6,] 10.752309 2.792182
```

```
round(diff(x[, "X1"]), 3)
```

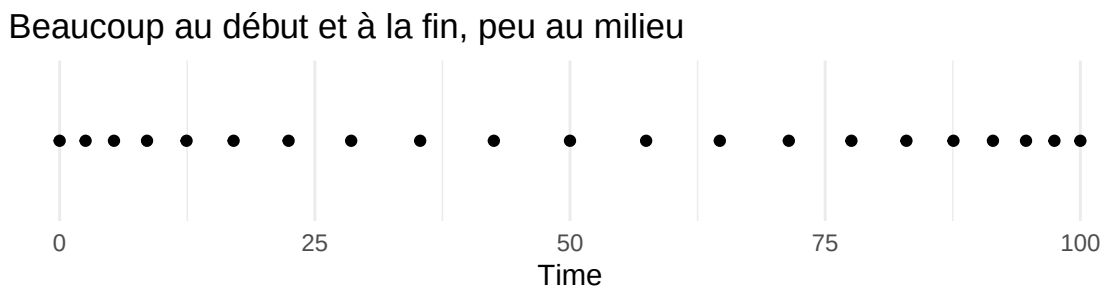
```
##  [1]  1.104  0.000  0.692  0.031  0.965  3.344  1.657  5.574  0.000  0.000
## [11]  2.224  4.862  0.999  0.000  3.960  9.014  3.136  0.937  6.341 13.930
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 4 : mixtes, beaucoup au début et à la fin

```
t4 <- inspection_times("inc-dec", n = 20, tmax = 100)$time
plot_plan(t4, "Beaucoup au début et à la fin, peu au milieu")
```



Modèle 1 : gamma

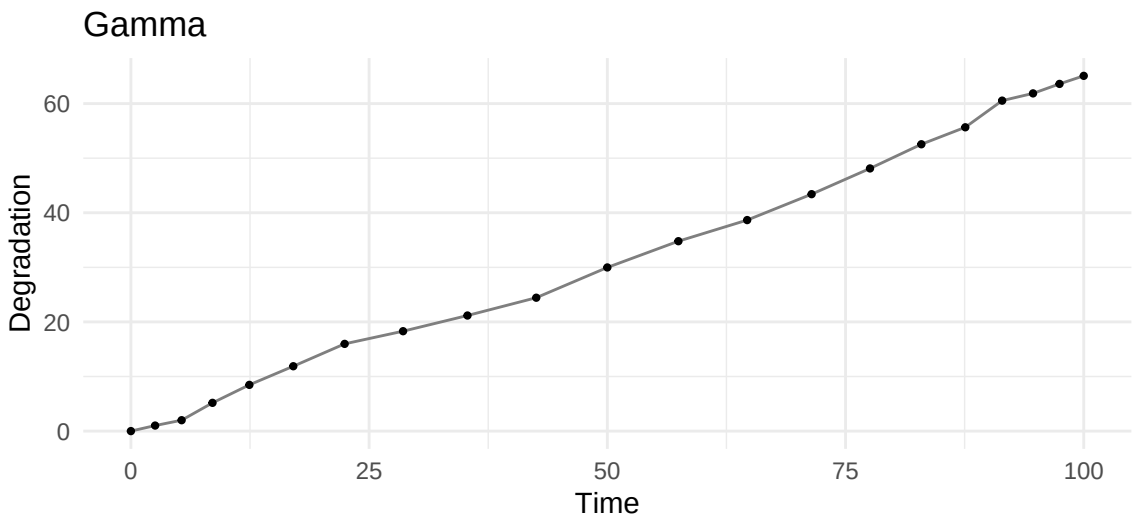
```
set.seed(41)
x <- sim_gamma(t4, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  2.540921  1.013078
## [3,]  5.322554  1.993097
## [4,]  8.562047  5.174992
## [5,] 12.431733  8.468824
## [6,] 17.042253 11.882103
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.013 0.980 3.182 3.294 3.413 4.097 2.322 2.872 3.263 5.544 4.811 3.869
## [13] 4.746 4.715 4.429 3.110 4.884 1.327 1.738 1.476
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

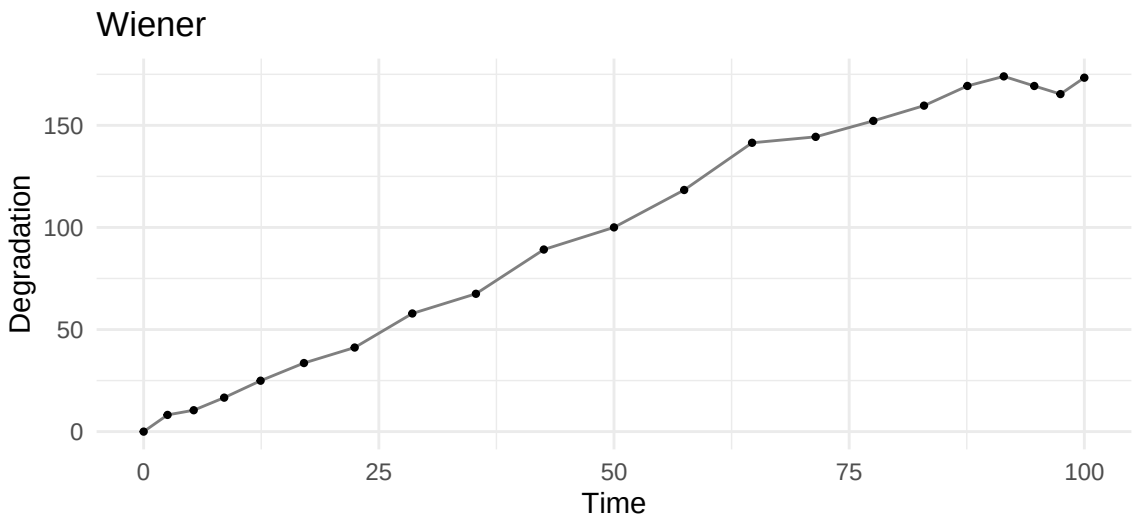
```
set.seed(42)
x <- sim_wiener(t4, mu = 1.5, sigma = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000  0.00000
## [2,]  2.540921  8.18207
## [3,]  5.322554 10.47089
## [4,]  8.562047 16.63729
## [5,] 12.431733 24.93169
## [6,] 17.042253 33.58357
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  8.182  2.289  6.166  8.294  8.652  7.591 16.680  9.649 21.673 10.846
## [11] 18.316 23.115  2.918  7.815  7.465  9.647  4.686 -4.703 -3.968  8.020
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

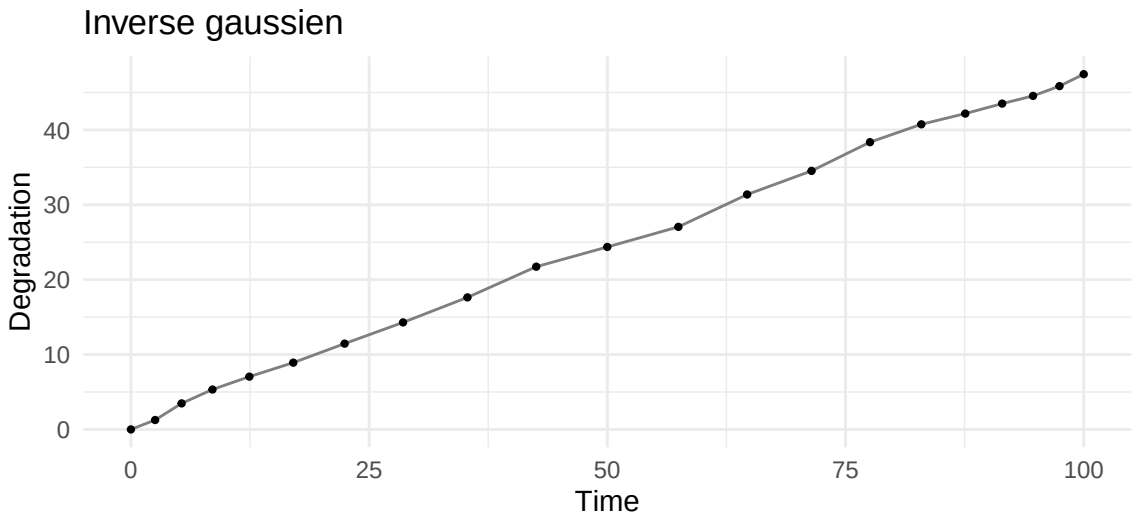
```
set.seed(43)
x <- sim_ig(t4, mu = 0.5, eta = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.540921 1.255599
## [3,]  5.322554 3.475956
## [4,]  8.562047 5.329629
## [5,] 12.431733 7.048824
## [6,] 17.042253 8.917167
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.256 2.220 1.854 1.719 1.868 2.538 2.835 3.341 4.100 2.635 2.685 4.320
## [13] 3.158 3.831 2.385 1.432 1.335 1.028 1.309 1.592
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

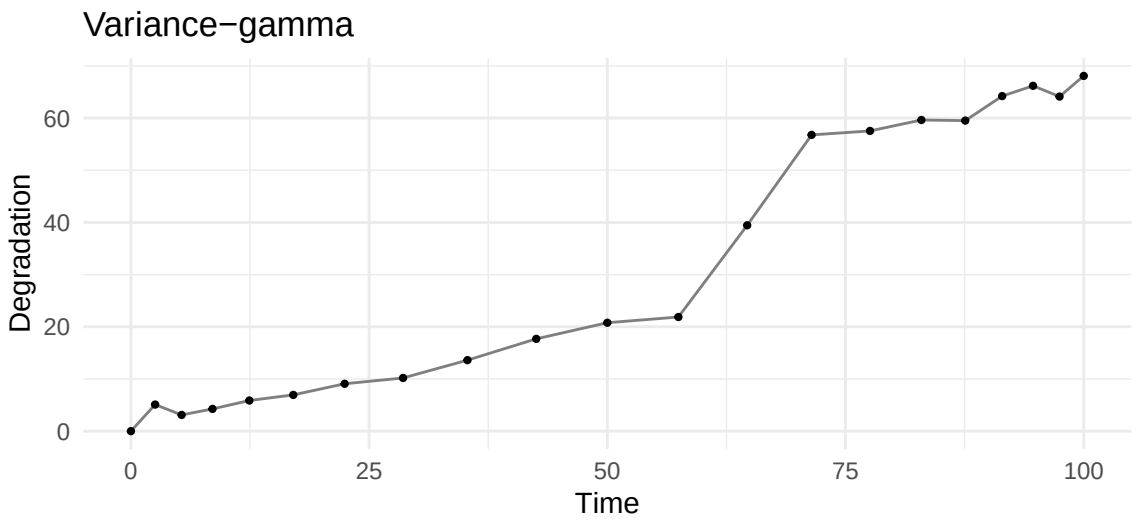
```
set.seed(44)
x <- sim_vg(t4, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.540921 5.095509
## [3,]  5.322554 3.104549
## [4,]  8.562047 4.258139
## [5,] 12.431733 5.869560
## [6,] 17.042253 6.941949
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.096 -1.991  1.154  1.611  1.072  2.137  1.112  3.414  4.080  3.092
## [11]  1.093 17.575 17.326  0.764  2.093 -0.115  4.696  1.969 -2.059  3.960
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

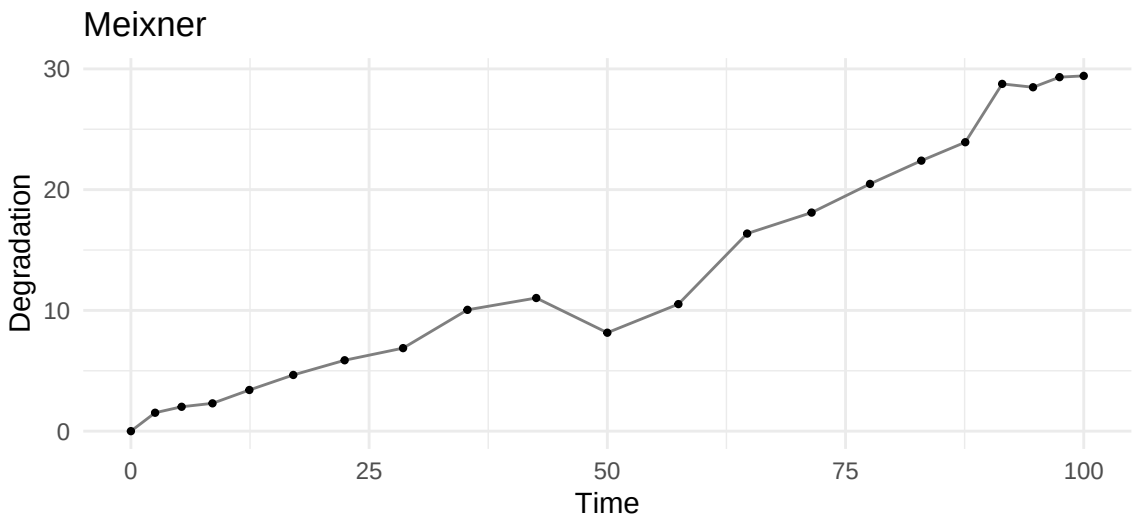
```
set.seed(45)
x <- sim_meixner(t4, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.540921 1.523821
## [3,]  5.322554 2.017361
## [4,]  8.562047 2.301981
## [5,] 12.431733 3.408298
## [6,] 17.042253 4.652765
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.524  0.494  0.285  1.106  1.244  1.217  1.005  3.171  0.980 -2.874
## [11]  2.368  5.843  1.737  2.374  1.926  1.526  4.830 -0.278  0.837  0.103
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

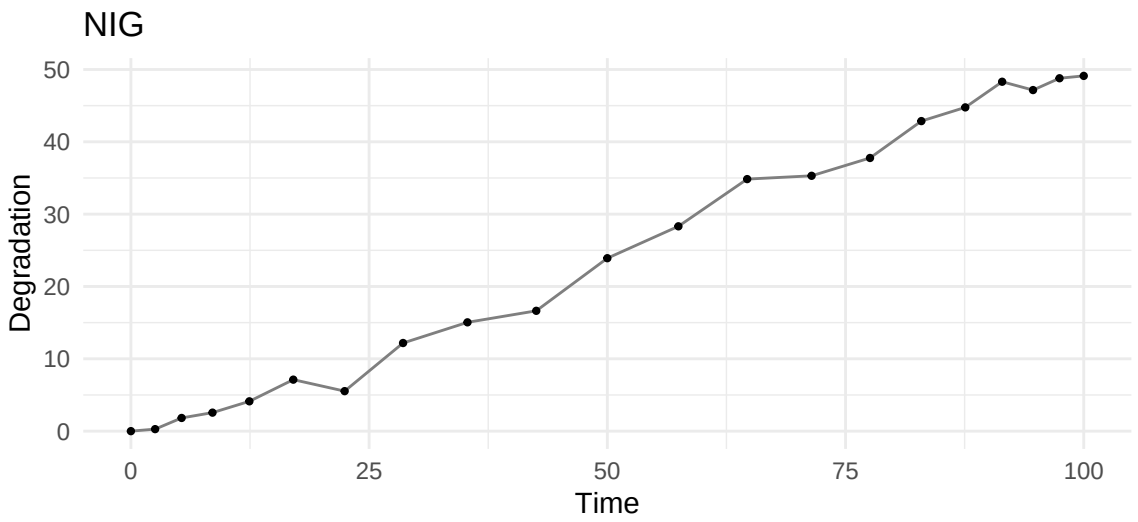
```
set.seed(46)
x <- sim_nig(t4, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  2.540921 0.2773168
## [3,]  5.322554 1.8228331
## [4,]  8.562047 2.5582928
## [5,] 12.431733 4.1226725
## [6,] 17.042253 7.1190832
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.277  1.546  0.735  1.564  2.996 -1.589  6.654  2.858  1.593  7.274
## [11]  4.408  6.528  0.456  2.463  5.101  1.895  3.545 -1.151  1.634  0.325
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

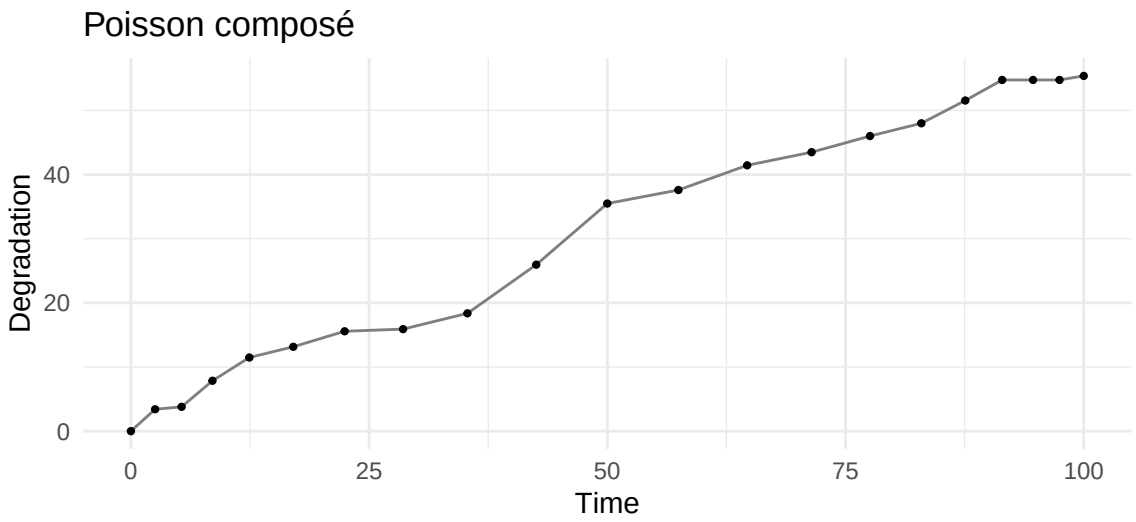
```
set.seed(47)
x <- sim_cpoisson(t4, lambda = 0.5)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  2.540921  3.410377
## [3,]  5.322554  3.788944
## [4,]  8.562047  7.855023
## [5,] 12.431733 11.480214
## [6,] 17.042253 13.146896
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.410 0.379 4.066 3.625 1.667 2.421 0.337 2.461 7.588 9.529 2.107 3.847
## [13] 2.046 2.519 1.991 3.545 3.210 0.000 0.000 0.631
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
0
20
40
0
25
50
75
100
Time

Cas 5 : mixtes, beaucoup au milieu

```
t5 <- inspection_times("dec-inc", n = 20, tmax = 100)$time
plot_plan(t5, "Peu au début et à la fin, beaucoup au milieu")
```



Modèle 1 : gamma

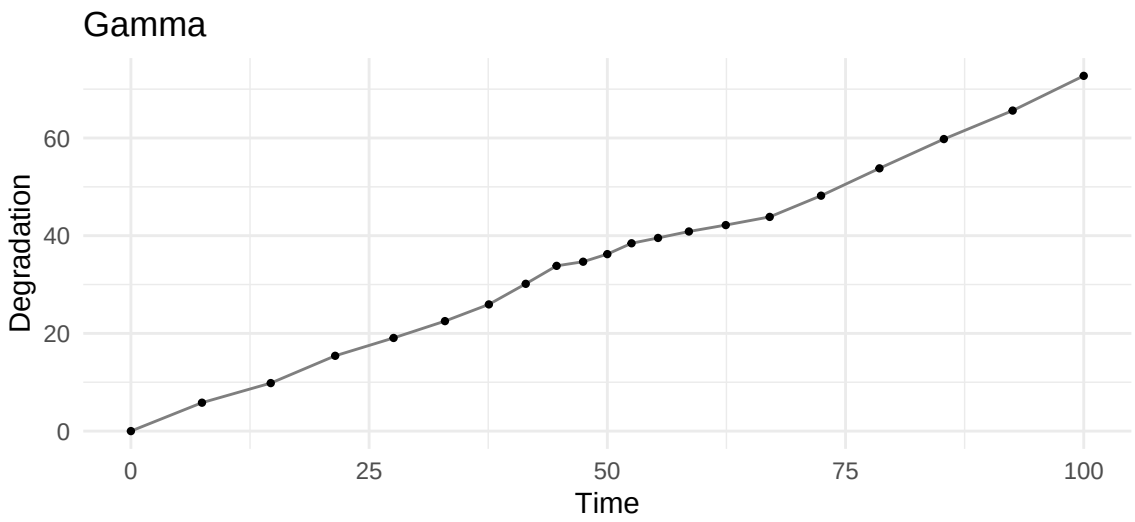
```
set.seed(51)
x <- sim_gamma(t5, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  5.814275
## [3,] 14.677446  9.824488
## [4,] 21.437953 15.412420
## [5,] 27.568267 19.065485
## [6,] 32.957747 22.511943
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 5.814 4.010 5.588 3.653 3.446 3.420 4.216 3.677 0.845 1.554 2.221 1.091
## [13] 1.322 1.314 1.673 4.344 5.600 5.990 5.831 7.110
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

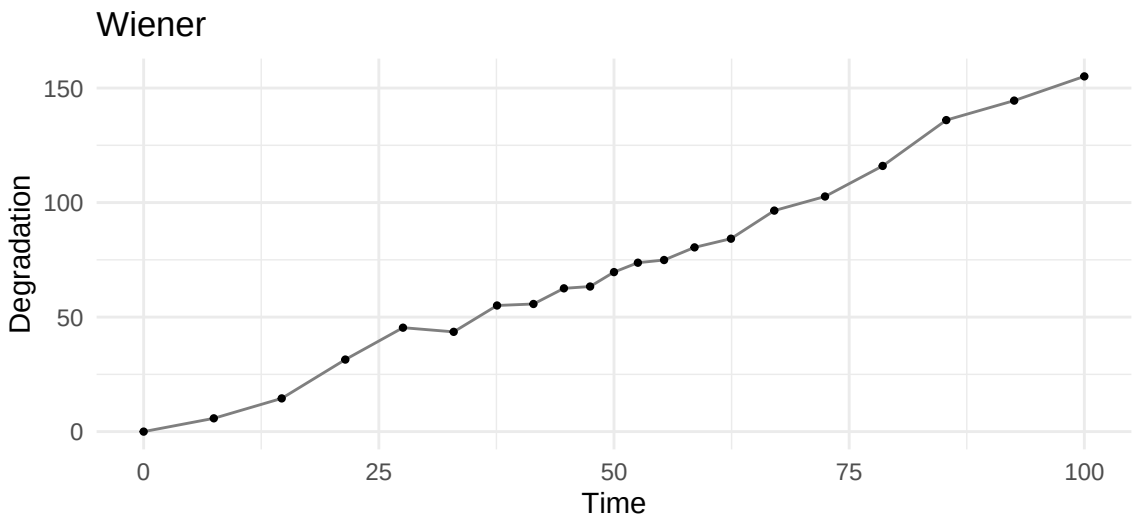
```
set.seed(52)
x <- sim_wiener(t5, mu = 1.5, sigma = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  5.801785
## [3,] 14.677446 14.507619
## [4,] 21.437953 31.452913
## [5,] 27.568267 45.376256
## [6,] 32.957747 43.556482
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.802  8.706 16.945 13.923 -1.820 11.506  0.661  6.829  0.794  6.298
## [11]  4.099  1.176  5.517  3.794 12.261  6.155 13.353 20.002  8.512 10.582
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

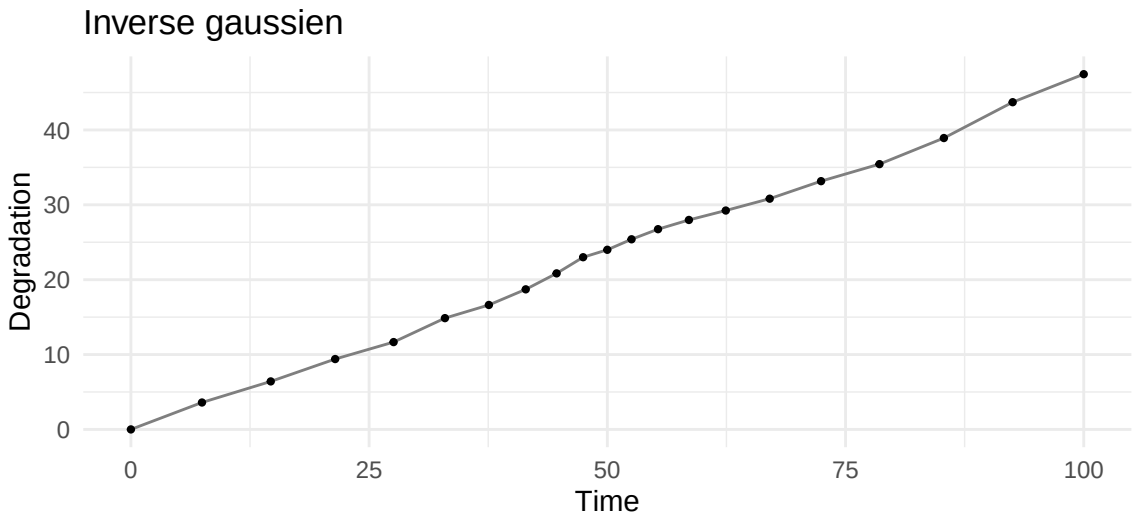
```
set.seed(53)
x <- sim_ig(t5, mu = 0.5, eta = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  3.595579
## [3,] 14.677446  6.412297
## [4,] 21.437953  9.399752
## [5,] 27.568267 11.663964
## [6,] 32.957747 14.863754
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.596 2.817 2.987 2.264 3.200 1.759 2.088 2.132 2.158 0.992 1.403 1.350
## [13] 1.232 1.260 1.581 2.342 2.275 3.485 4.787 3.742
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

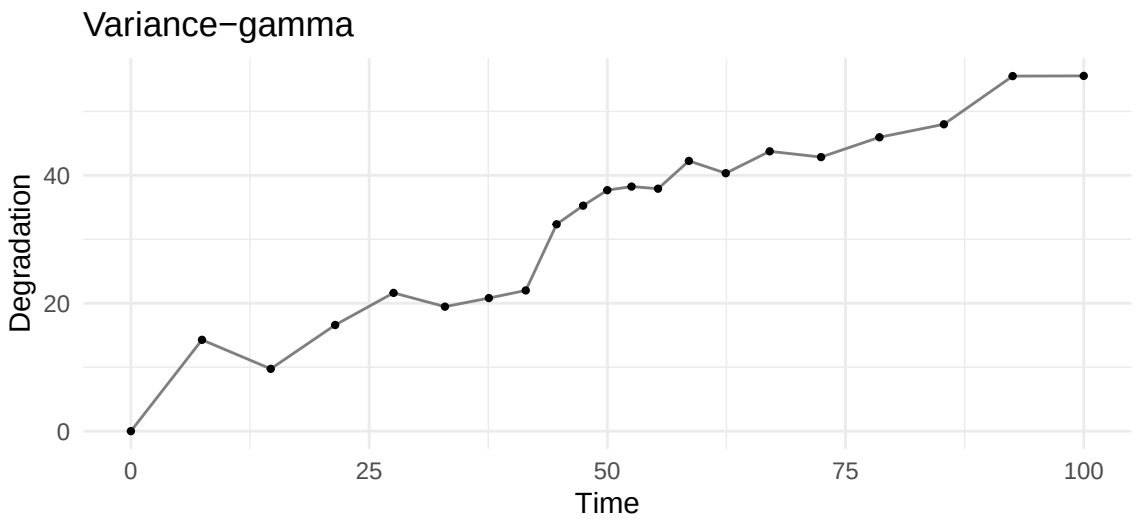
```
set.seed(54)
x <- sim_vg(t5, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079 14.277776
## [3,] 14.677446  9.742506
## [4,] 21.437953 16.599526
## [5,] 27.568267 21.619767
## [6,] 32.957747 19.463350
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 14.278 -4.535  6.857  5.020 -2.156  1.346  1.193 10.348  2.910  2.420
## [11]  0.563 -0.331  4.342 -1.921  3.421 -0.900  3.103  2.027  7.536  0.031
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

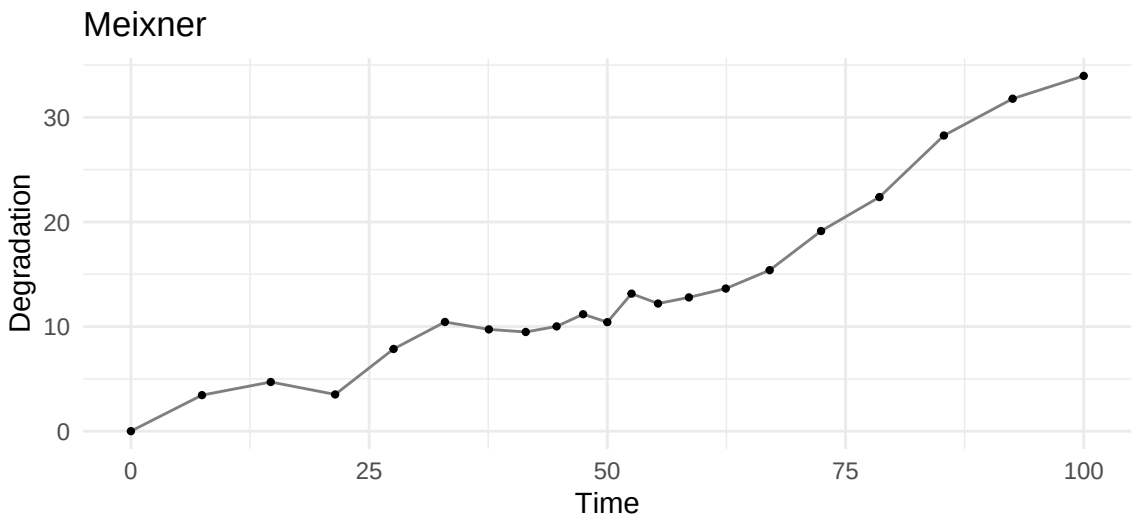
```
set.seed(55)
x <- sim_meixner(t5, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  3.442434
## [3,] 14.677446  4.697757
## [4,] 21.437953  3.507490
## [5,] 27.568267  7.857074
## [6,] 32.957747 10.436038
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.442  1.255 -1.190  4.350  2.579 -0.708 -0.247  0.531  1.165 -0.758
## [11]  2.727 -0.946  0.586  0.846  1.759  3.740  3.247  5.884  3.518  2.182
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

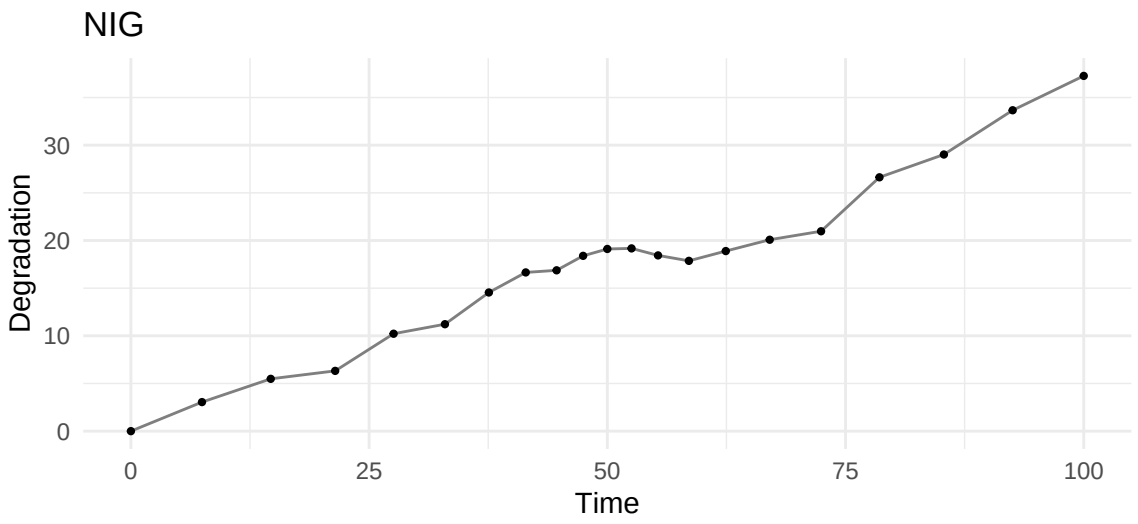
```
set.seed(56)
x <- sim_nig(t5, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  3.047164
## [3,] 14.677446  5.487323
## [4,] 21.437953  6.322425
## [5,] 27.568267 10.219941
## [6,] 32.957747 11.218525
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.047  2.440  0.835  3.898  0.999  3.328  2.105  0.220  1.519  0.720
## [11]  0.061 -0.735 -0.571  1.027  1.187  0.890  5.661  2.391  4.643  3.604
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

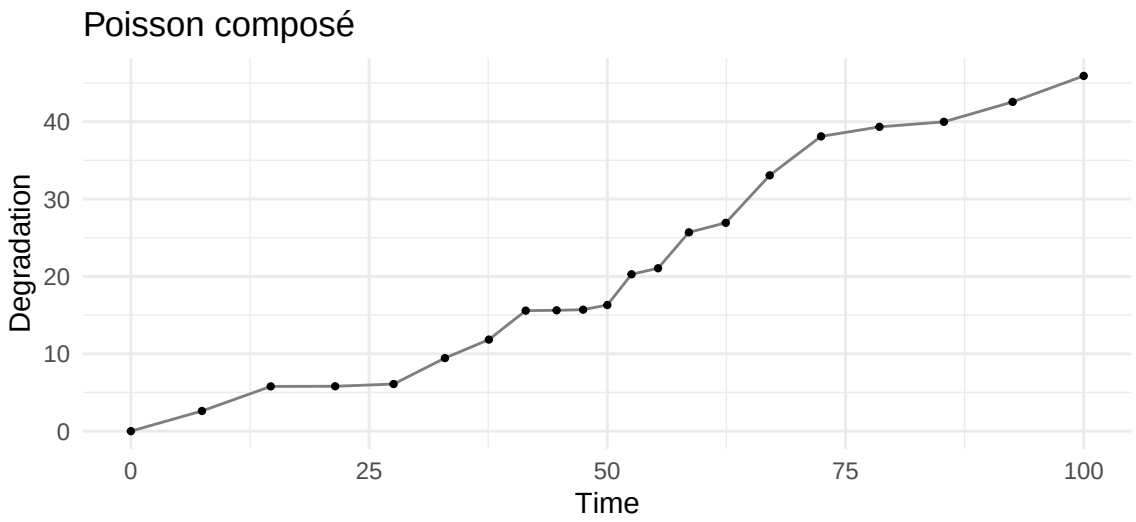
```
set.seed(57)
x <- sim_cpoisson(t5, lambda = 0.5)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  7.459079 2.606789
## [3,] 14.677446 5.781682
## [4,] 21.437953 5.802607
## [5,] 27.568267 6.088267
## [6,] 32.957747 9.442441
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.607 3.175 0.021 0.286 3.354 2.392 3.743 0.043 0.080 0.602 3.977 0.774
## [13] 4.646 1.231 6.136 5.036 1.230 0.651 2.577 3.361
```

```
plot_traj(x, "Poisson composé")
```

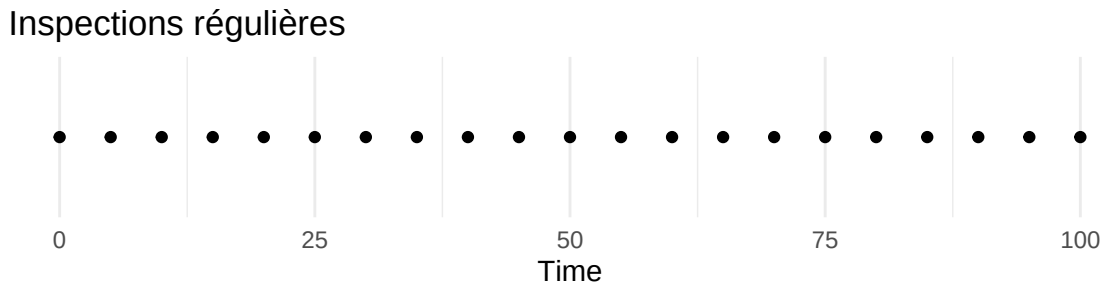

Poisson composé
Degradation
0
10
20
30
40
0
25
50
75
100
Time

II - Plusieurs trajectoires

5 trajectoires, chacune inspectée 20 fois entre 0 et 100, aux mêmes instants. `x` est une matrice : une colonne `t`, puis `X1`, ..., `X5`. On affiche les premières lignes des trois premières trajectoires.

Cas 1 : inspections régulières

```
p1 <- inspection_times("regular", n = 20, n_traj = 5, tmax = 100)
plot_plan(p1$time[p1$id == 1], "Inspections régulières")
```



Modèle 1 : gamma

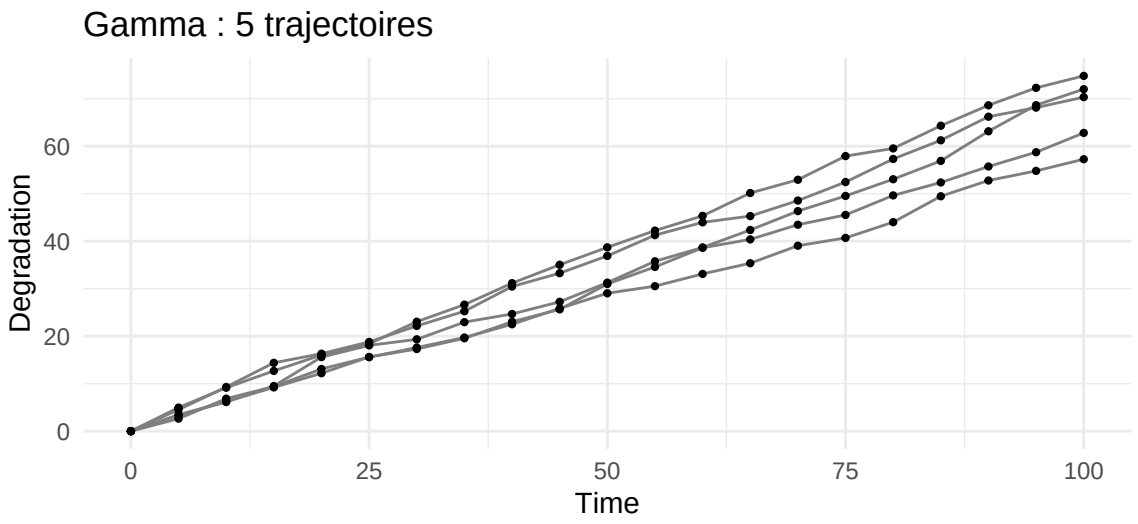
```
set.seed(111)
x <- sim_gamma(p1, a = 2, b = 3)
head(x[, 1:4])
```

```
##       t        X1        X2        X3
## [1,]  0  0.000000  0.000000  0.000000
## [2,]  5  3.412944  3.391215  4.962620
## [3,] 10  6.248927  6.129922  9.138235
## [4,] 15  9.207337  9.496121 12.684072
## [5,] 20 12.200879 15.598724 16.033187
## [6,] 25 15.606267 18.057525 18.389191
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.413 2.836 2.958 2.994 3.405 1.684 2.267 3.473 2.611 5.347 3.584 4.039
## [13] 1.762 3.088 2.059 4.112 2.727 3.365 3.005 4.065
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

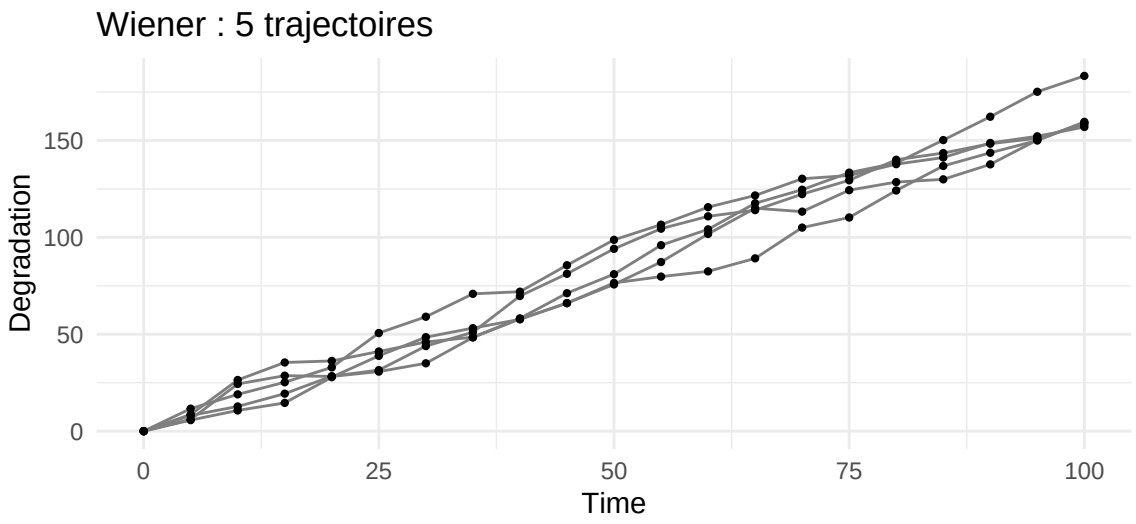
```
set.seed(112)
x <- sim_wiener(p1, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##       t        X1        X2       X3
## [1,]  0  0.000000  0.000000  0.00000
## [2,]  5  6.094713  8.613025 11.60806
## [3,] 10 24.342933 26.400095 18.97625
## [4,] 15 28.630759 35.438839 25.23648
## [5,] 20 28.257068 36.240624 32.90835
## [6,] 25 30.724657 41.112867 50.63640
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  6.095 18.248  4.288 -0.374  2.468  4.282 13.359  9.521  8.181 10.422
## [11]  3.263  2.710  6.713 15.872  5.218 13.935 12.657  6.808  6.338  9.608
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

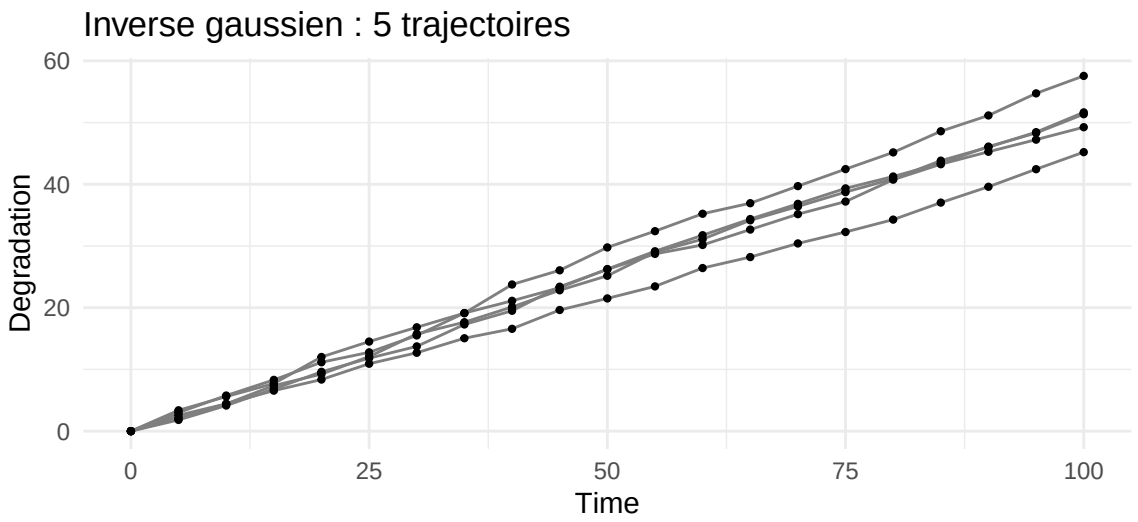
```
set.seed(113)
x <- sim_ig(p1, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##       t        X1        X2        X3
## [1,]  0  0.000000  0.000000  0.000000
## [2,]  5  2.575667  1.816995  3.380367
## [3,] 10  4.416063  4.148258  5.600797
## [4,] 15  7.371107  6.834152  7.721204
## [5,] 20  9.244923  9.608309 12.018158
## [6,] 25 12.077408 11.820806 14.513946
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.576 1.840 2.955 1.874 2.832 3.674 1.901 2.465 2.708 2.349 3.822 2.106
## [13] 3.062 2.242 2.329 2.141 2.951 2.186 2.432 3.209
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

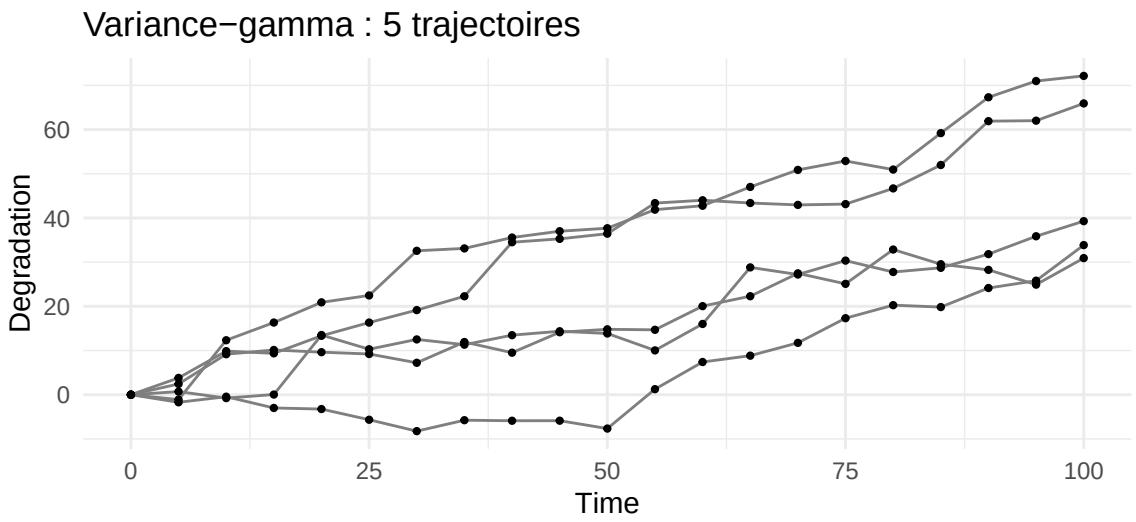
```
set.seed(114)
x <- sim_vg(p1, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##       t        X1          X2        X3
## [1,]  0  0.000000  0.00000000  0.000000
## [2,]  5  2.436627  0.70296417 -1.107265
## [3,] 10  9.162189 -0.77011502 12.323899
## [4,] 15 10.122663  0.03664116 16.329080
## [5,] 20  9.624831 13.50917123 20.901106
## [6,] 25  9.211656 10.28751843 22.462301
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.437  6.726  0.960 -0.498 -0.413 -1.990  4.699 -2.413  4.626  0.661
## [11] -0.109  5.361  2.242  5.178 -2.377  7.766 -3.320 -1.284 -3.340  6.006
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

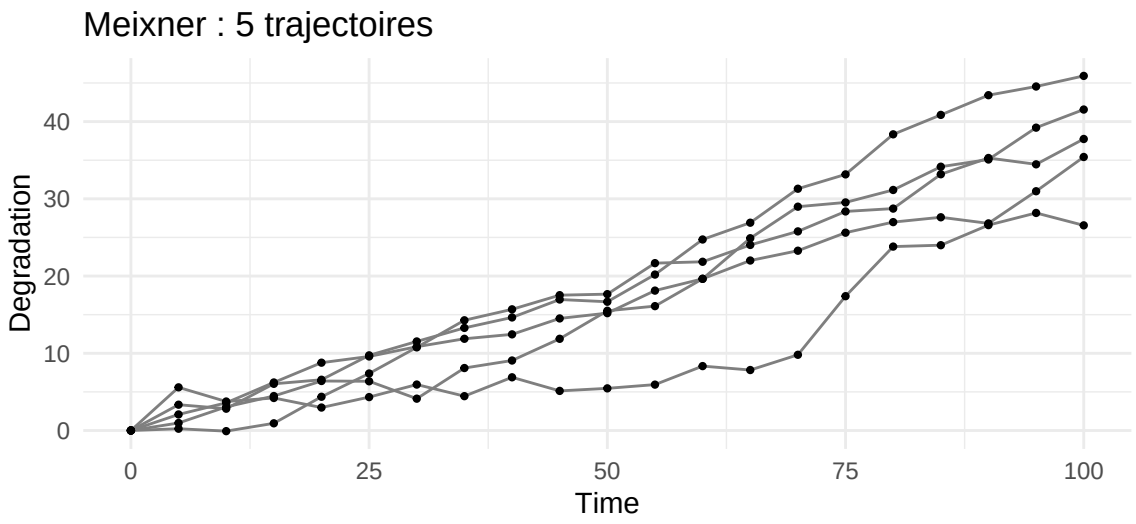
```
set.seed(115)
x <- sim_meixner(p1, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##       t       X1       X2          X3
## [1,]  0 0.000000 0.000000  0.00000000
## [2,]  5 3.339643 2.078708  0.22800433
## [3,] 10 2.823348 3.582343 -0.08350747
## [4,] 15 6.044890 6.217424  0.93099359
## [5,] 20 6.571270 8.782078  4.36570414
## [6,] 25 9.730140 9.586900  7.38028101
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.340 -0.516  3.222  0.526  3.159  1.800  1.755  1.354  2.323 -0.290
## [11]  3.520  4.541  2.174  4.399  1.858  5.186  2.518  2.551  1.127  1.372
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

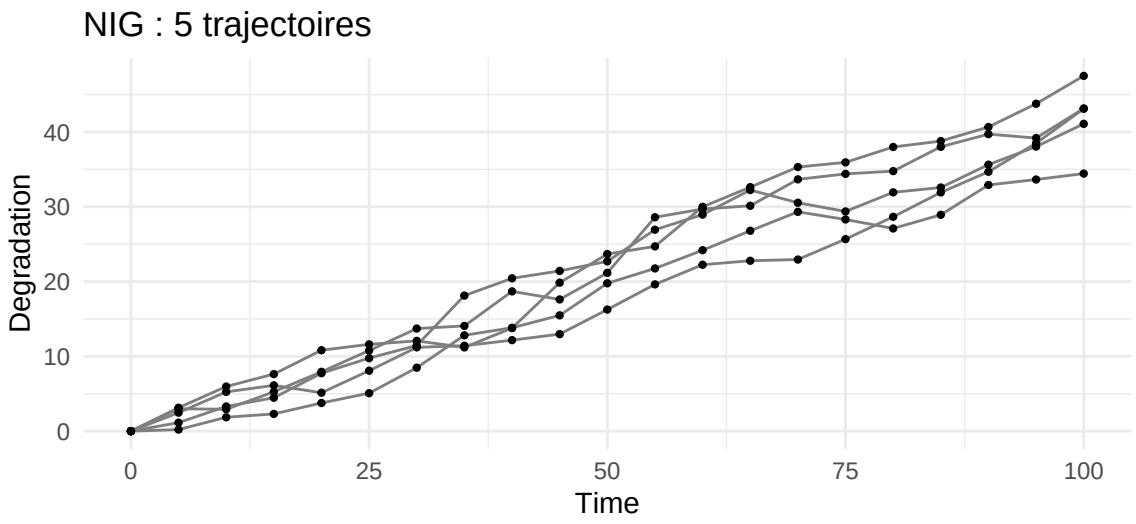
```
set.seed(116)
x <- sim_nig(p1, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##       t       X1        X2        X3
## [1,]  0 0.000000  0.000000 0.0000000
## [2,]  5 2.468533  2.990295 0.2074206
## [3,] 10 5.246575  2.923195 1.8555426
## [4,] 15 6.122326  5.263634 2.3019116
## [5,] 20 5.136418  7.929152 3.7644948
## [6,] 25 8.077892 10.777178 5.0749732
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.469  2.778  0.876 -0.986  2.941  3.123  0.221  0.743  0.804  3.282
## [11]  3.372  2.628  0.529  0.166  2.722  2.988  3.257  2.767  3.812  4.617
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

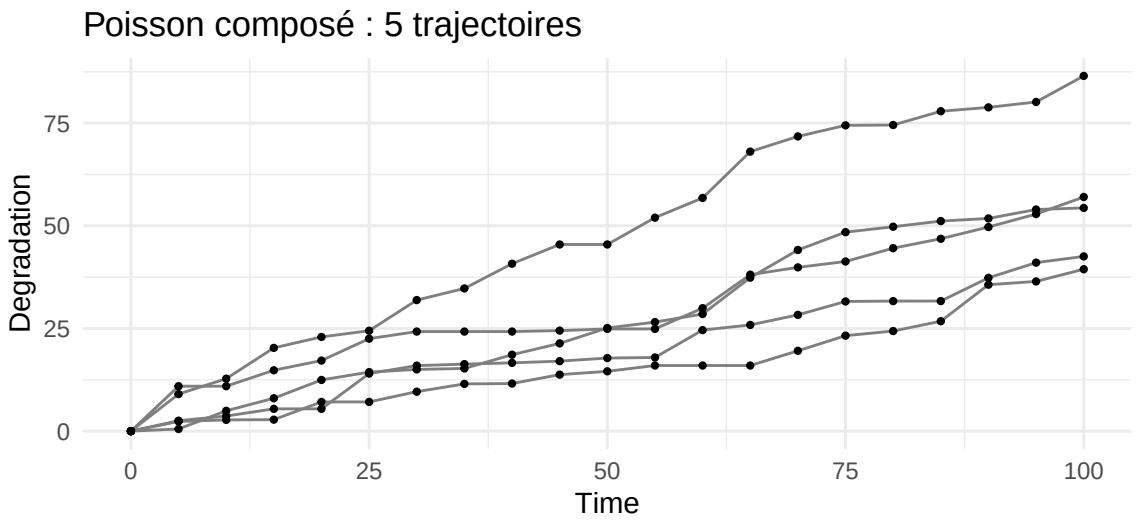
```
set.seed(117)
x <- sim_cpoisson(p1, lambda = 0.5)
head(x[, 1:4])
```

```
##       t       X1       X2        X3
## [1,]  0  0.00000  0.00000  0.000000
## [2,]  5 10.93951  9.02175  2.547582
## [3,] 10 10.93951 12.77248  3.677322
## [4,] 15 14.81978 20.27313  5.428941
## [5,] 20 17.18333 22.93611  5.428941
## [6,] 25 22.51752 24.44392 14.017366
```

```
round(diff(x[, "X1"]), 3)
```

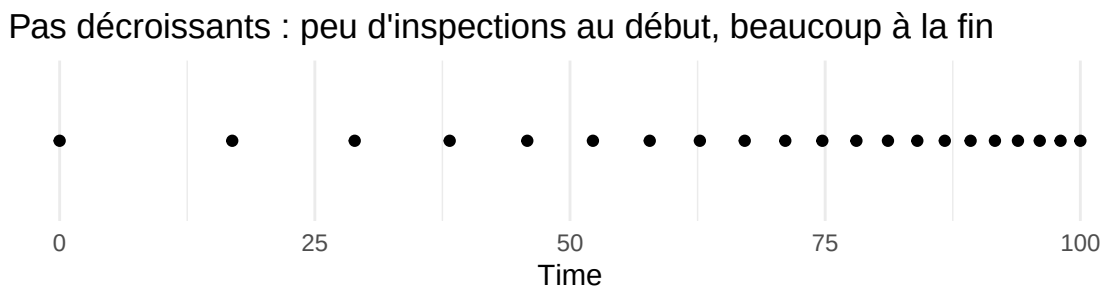
```
##  [1] 10.940  0.000  3.880  2.364  5.334  1.725  0.000  0.000  0.238  0.407
## [11]  0.000  5.049  8.156  1.785  1.432  3.239  2.295  2.852  3.165  4.153
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
0
25
50
75
0
25
50
75
100
Time

Cas 2 : pas décroissants

```
p2 <- inspection_times("decreasing", n = 20, n_traj = 5, tmax = 100)
plot_plan(p2$time[p2$id == 1], "Pas décroissants : peu d'inspections au début, beaucoup à la fin")
```



Modèle 1 : gamma

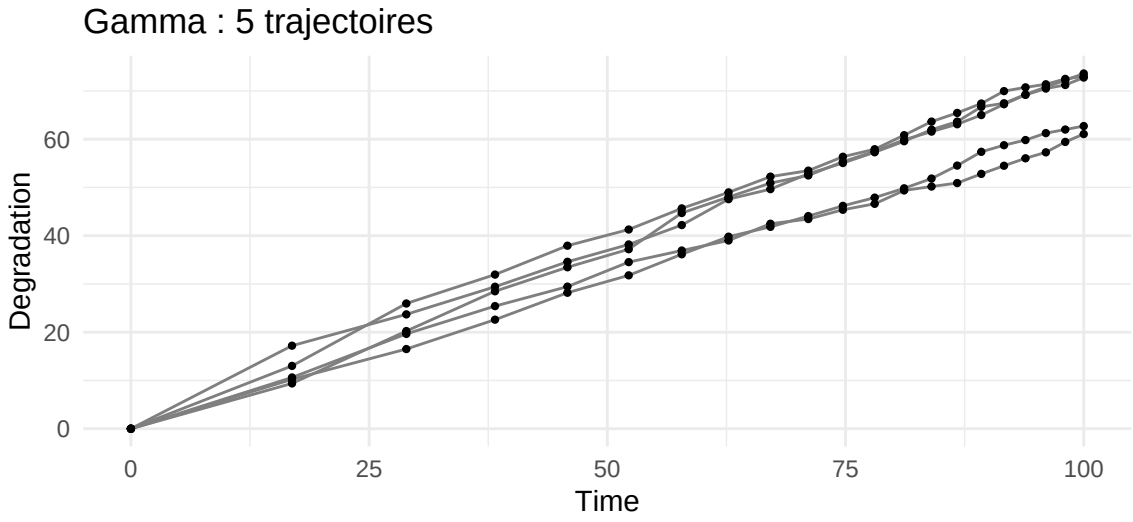
```
set.seed(121)
x <- sim_gamma(p2, a = 2, b = 3)
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.000000
## [2,] 16.90921 10.62024 17.20485  9.386244
## [3,] 28.90648 19.65015 23.69326 20.236937
## [4,] 38.21229 25.40798 29.41898 28.486875
## [5,] 45.81569 29.44038 34.59453 33.456933
## [6,] 52.24427 34.52621 38.18268 37.192447
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 10.620  9.030  5.758  4.032  5.086  2.398  2.089  3.466  0.972  1.918
## [11]  1.235  2.772  0.809  0.720  1.897  1.689  1.545  1.233  2.161  1.626
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

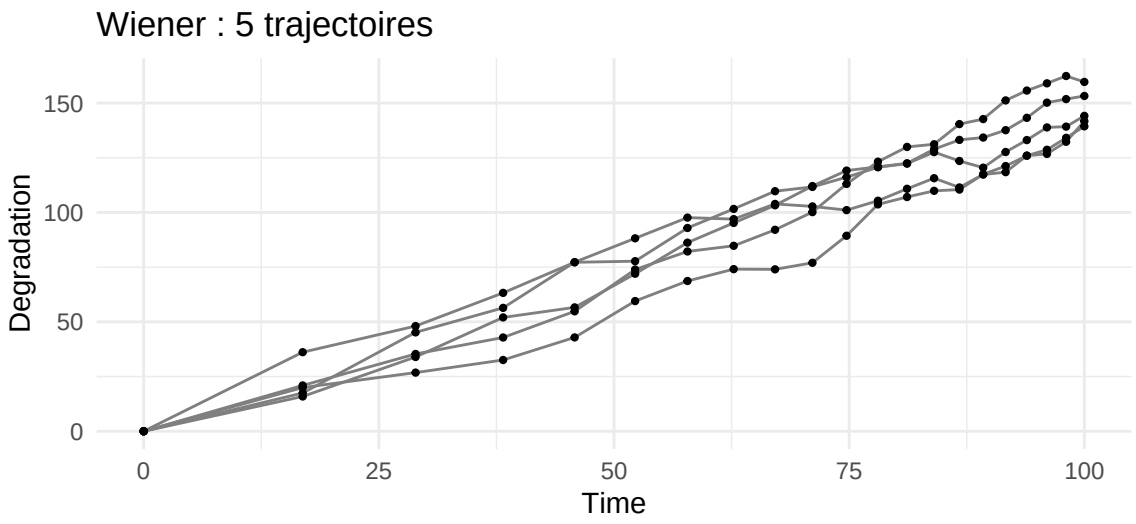
```
set.seed(122)
x <- sim_wiener(p2, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.00000
## [2,] 16.90921 36.14323 20.94065 19.95223
## [3,] 28.90648 48.07174 35.31596 26.80538
## [4,] 38.21229 63.24777 42.85070 32.56368
## [5,] 45.81569 77.22253 54.81566 42.88415
## [6,] 52.24427 77.72730 73.90821 59.50291
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 36.143 11.929 15.176 13.975  0.505 15.191  8.693  8.108  1.947  4.464
## [11]  4.524  1.677  5.246 -4.043 -3.056  7.155  5.454  5.783  0.361  4.913
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

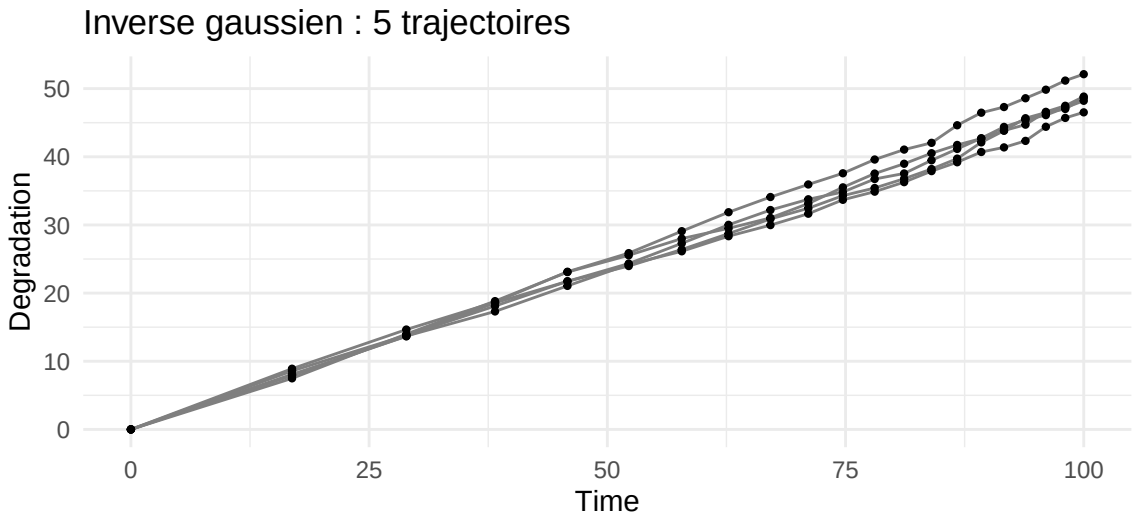
```
set.seed(123)
x <- sim_ig(p2, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##             t        X1        X2        X3
## [1,]  0.00000  0.000000  0.000000  0.000000
## [2,] 16.90921  7.897723  8.904532  8.073231
## [3,] 28.90648 13.700327 14.653043 13.652377
## [4,] 38.21229 17.306694 18.671536 18.566432
## [5,] 45.81569 21.060098 23.128694 21.727826
## [6,] 52.24427 24.193476 25.862694 24.330757
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 7.898 5.803 3.606 3.753 3.133 1.940 2.214 1.627 1.673 2.039 1.196 1.395
## [13] 1.619 1.301 1.505 0.673 0.955 2.070 1.300 0.819
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

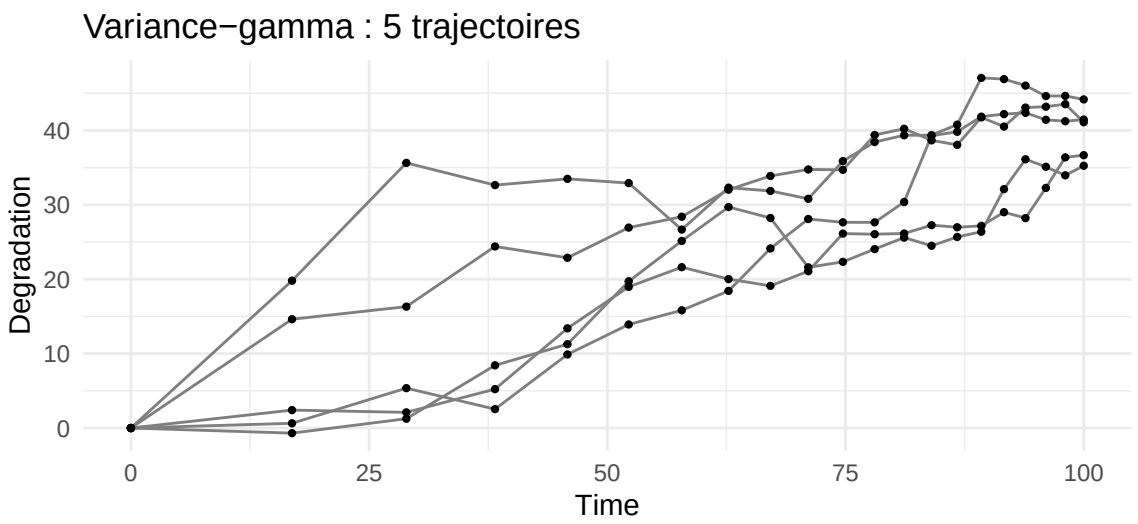
```
set.seed(124)
x <- sim_vg(p2, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##             t        X1        X2       X3
## [1,]  0.00000  0.000000  0.000000  0.00000
## [2,] 16.90921  0.637251  2.409797 19.80395
## [3,] 28.90648  5.363706  2.105576 35.63543
## [4,] 38.21229  2.540566  5.230256 32.65205
## [5,] 45.81569  9.891368 13.399796 33.49435
## [6,] 52.24427 13.917200 18.976941 32.92566
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.637  4.726 -2.823  7.351  4.026  1.908  2.606  5.707  3.958 -0.445
## [11] -0.007  2.737  8.907  0.533  2.031  0.344  0.170 -0.934 -0.195  0.226
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

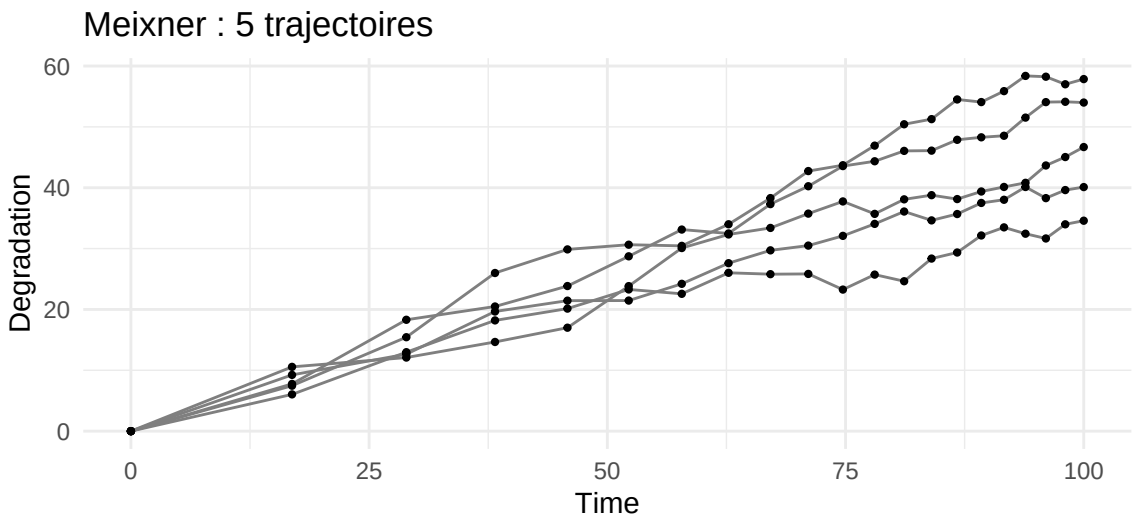
```
set.seed(125)
x <- sim_meixner(p2, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##             t       X1        X2       X3
## [1,]  0.00000  0.00000  0.000000  0.00000
## [2,] 16.90921 10.56185  6.039943  9.24308
## [3,] 28.90648 12.09210 12.976338 12.64901
## [4,] 38.21229 14.65336 18.195877 19.65971
## [5,] 45.81569 16.99743 20.136227 21.44034
## [6,] 52.24427 23.83157 23.285497 21.45917
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 10.562  1.530  2.561  2.344  6.834  6.259  2.230  1.072  2.353  2.001
## [11] -2.044  2.387  0.686 -0.641  1.247  0.744  0.680  2.850  1.385  1.643
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

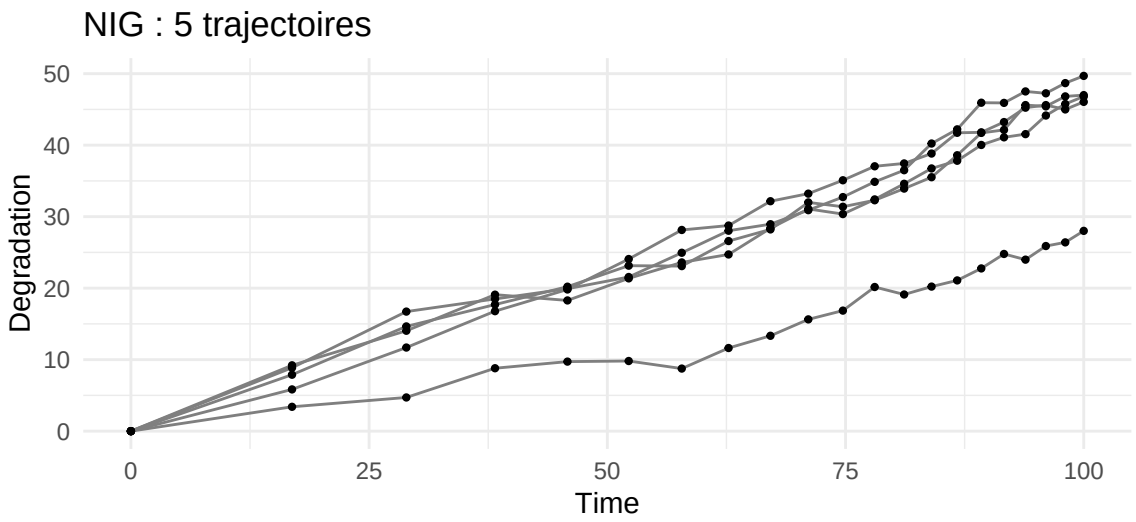
```
set.seed(126)
x <- sim_nig(p2, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##             t       X1        X2        X3
## [1,]  0.00000 0.000000  0.000000  0.000000
## [2,] 16.90921 3.403669  9.210465  8.863915
## [3,] 28.90648 4.711353 14.040403 16.720638
## [4,] 38.21229 8.797905 19.090496 18.494623
## [5,] 45.81569 9.725299 18.282901 19.921991
## [6,] 52.24427 9.809557 21.367375 21.559535
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.404  1.308  4.087  0.927  0.084 -1.058  2.875  1.712  2.294  1.226
## [11]  3.301 -1.046  1.104  0.866  1.675  2.024 -0.795  1.903  0.514  1.608
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

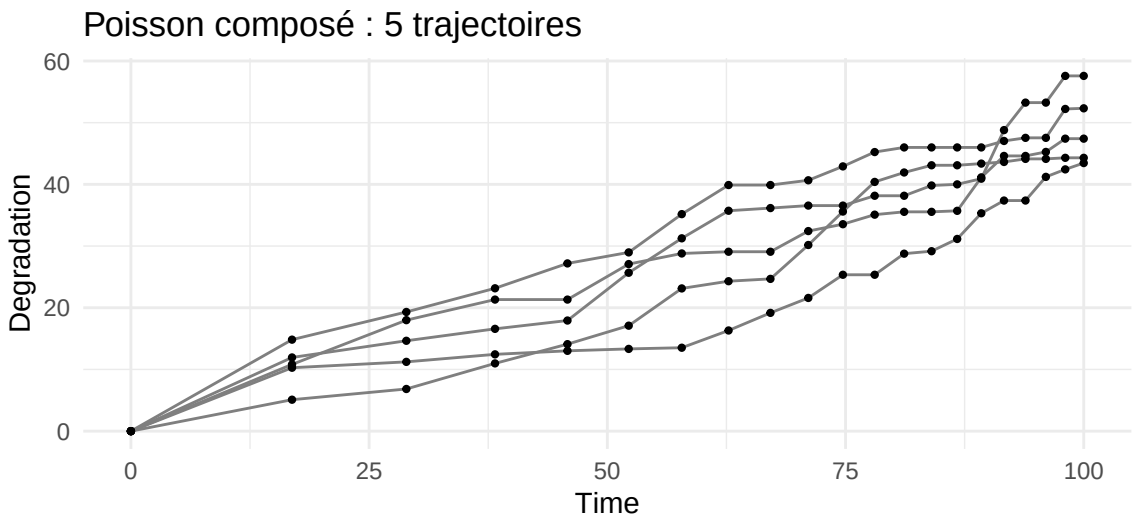
```
set.seed(127)
x <- sim_cpoisson(p2, lambda = 0.5)
head(x[, 1:4])
```

```
##             t        X1       X2       X3
## [1,]  0.00000  0.000000  0.00000  0.00000
## [2,] 16.90921  5.094637 10.27714 14.82955
## [3,] 28.90648  6.832984 11.22789 19.31514
## [4,] 38.21229 10.979469 12.44772 23.15368
## [5,] 45.81569 14.098460 13.01757 27.18597
## [6,] 52.24427 17.094252 13.32985 28.97341
```

```
round(diff(x[, "X1"]), 3)
```

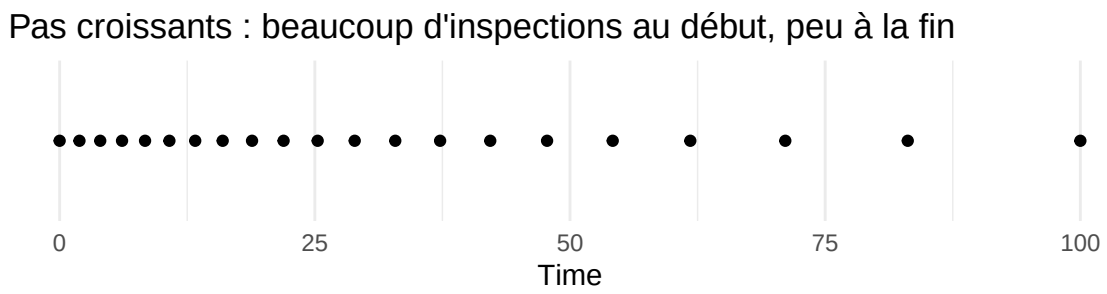
```
##  [1] 5.095 1.738 4.146 3.119 2.996 6.038 1.170 0.377 5.492 5.420 4.804 1.528
## [13] 1.169 0.000 0.274 0.266 0.499 0.000 0.170 0.000
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 3 : pas croissants

```
p3 <- inspection_times("increasing", n = 20, n_traj = 5, tmax = 100)
plot_plan(p3$time[p3$id == 1], "Pas croissants : beaucoup d'inspections au début, peu à la fin")
```



Modèle 1 : gamma

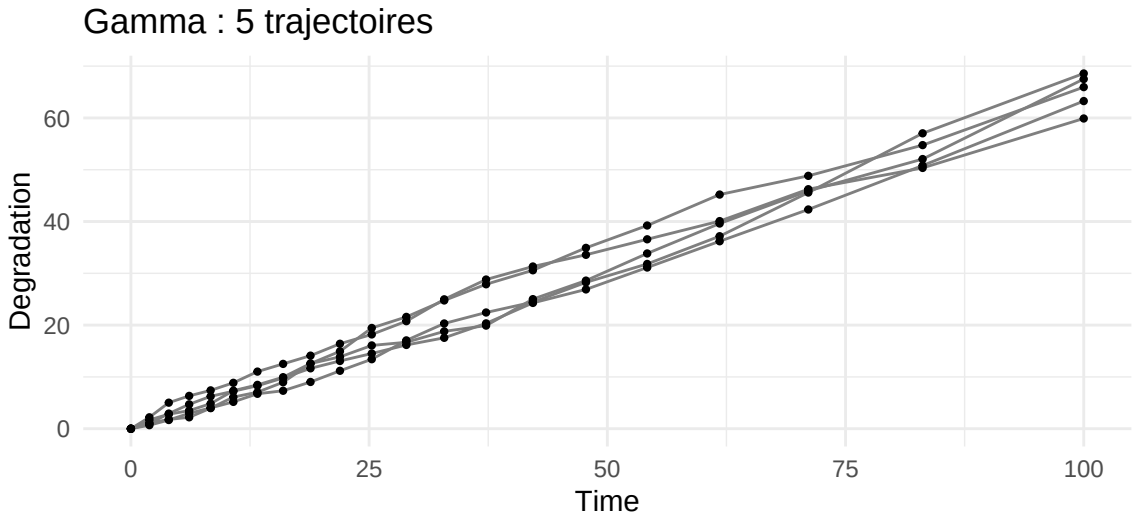
```
set.seed(131)
x <- sim_gamma(p3, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1       X2       X3
## [1,]  0.000000 0.0000000 0.000000 0.000000
## [2,]  1.940035 0.6807504 1.060241 2.206014
## [3,]  3.974743 1.6795349 1.791835 5.027377
## [4,]  6.113840 2.9693747 2.202258 6.336759
## [5,]  8.368618 4.0223494 3.975035 7.405013
## [6,] 10.752309 5.1828148 6.103820 8.862926
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.681  0.999  1.290  1.053  1.160  1.560  0.585  1.699  2.156  2.235
## [11]  3.633  3.264  2.122  1.999  3.815  3.560  5.353  8.435 11.436 11.550
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

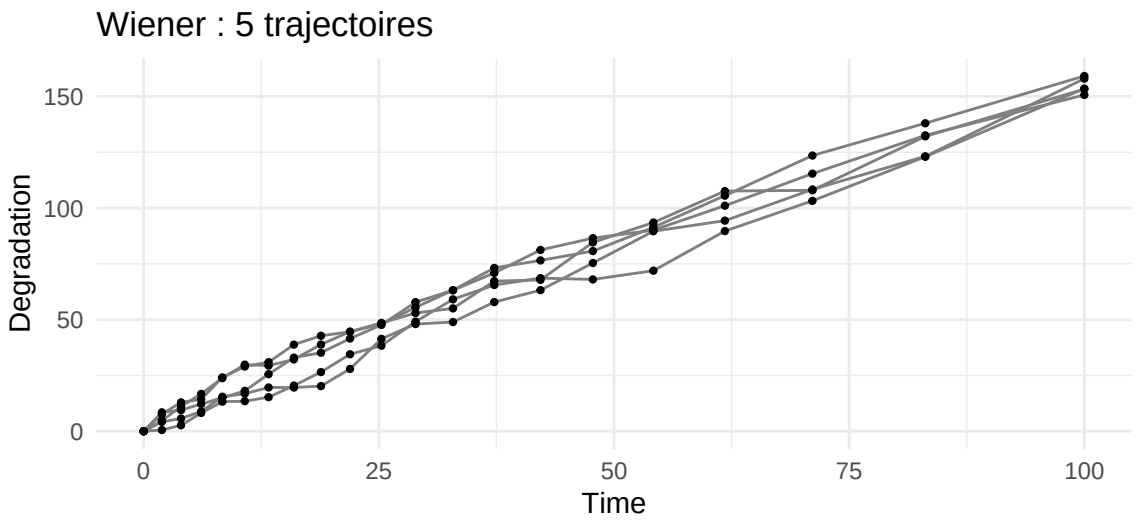
```
set.seed(132)
x <- sim_wiener(p3, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##              t        X1         X2        X3
## [1,]  0.000000  0.000000  0.0000000  0.000000
## [2,]  1.940035  4.230787  0.5691491  8.489047
## [3,]  3.974743  5.699667  2.6729907  9.454022
## [4,]  6.113840  8.879185  8.1968896 12.194740
## [5,]  8.368618 15.465729 13.2503463 14.987435
## [6,] 10.752309 16.842018 13.4664873 18.080960
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  4.231  1.469  3.180  6.587  1.376  2.773 -0.025  0.635  7.691 13.470
## [11]  6.643  0.893  8.950  5.330 12.177 14.230  4.739 13.966 14.832 34.817
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

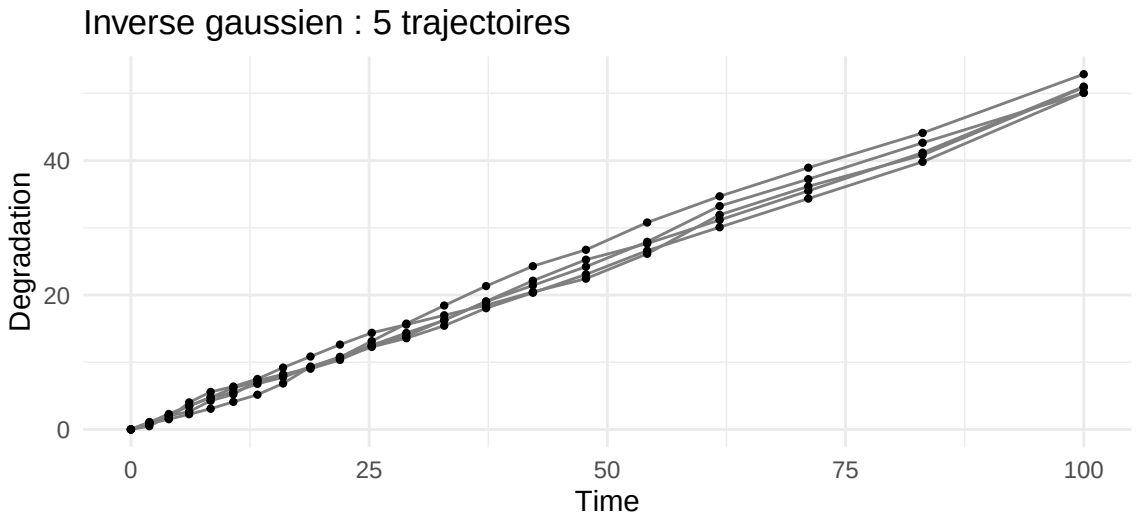
```
set.seed(133)
x <- sim_ig(p3, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1        X2       X3
## [1,]  0.000000 0.0000000 0.0000000 0.000000
## [2,]  1.940035 0.9390493 0.4884332 1.088656
## [3,]  3.974743 2.0880034 2.0127334 1.694220
## [4,]  6.113840 3.4899909 2.6631953 4.018758
## [5,]  8.368618 4.7335340 4.2683565 5.584057
## [6,] 10.752309 5.5338600 5.2210800 6.378180
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.939 1.149 1.402 1.244 0.800 1.245 0.987 1.482 1.473 2.417 2.603 2.697
## [13] 2.891 2.963 2.438 4.059 3.901 4.255 5.170 8.749
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

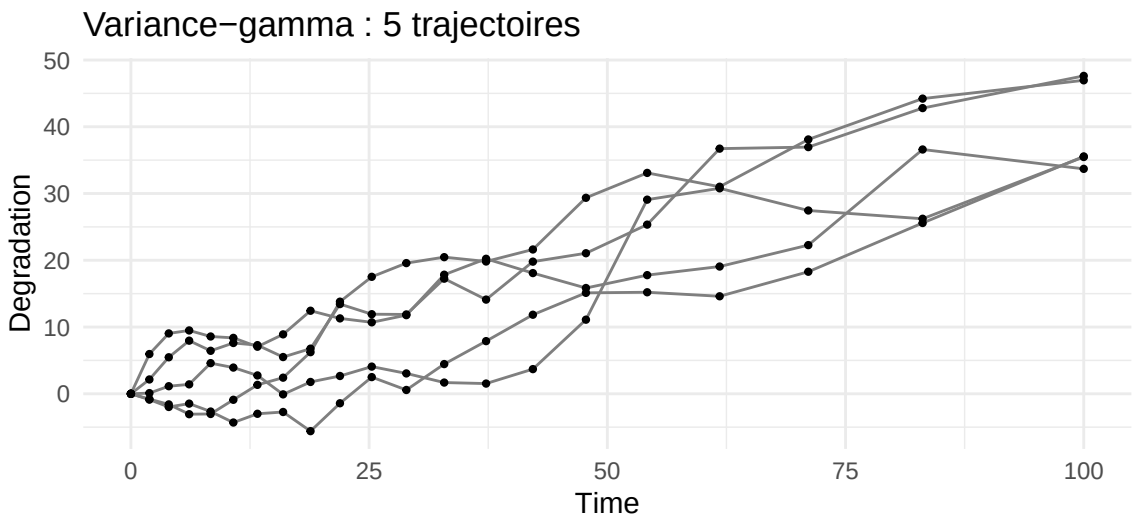
```
set.seed(134)
x <- sim_vg(p3, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t       X1        X2         X3
## [1,]  0.000000 0.000000 0.0000000  0.0000000
## [2,]  1.940035 5.953242 0.0991426 -0.8006998
## [3,]  3.974743 9.043546 1.1334917 -1.5900090
## [4,]  6.113840 9.487525 1.4108693 -3.0584259
## [5,]  8.368618 8.582190 4.5890451 -3.0247012
## [6,] 10.752309 8.374895 3.9255079 -0.8998503
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.953  3.090  0.444 -0.905 -0.207 -1.325  1.852  3.537 -1.146 -0.587
## [11]  1.073  6.059  2.383 -2.140 -2.239  1.929  1.291  3.208 14.324 -2.901
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

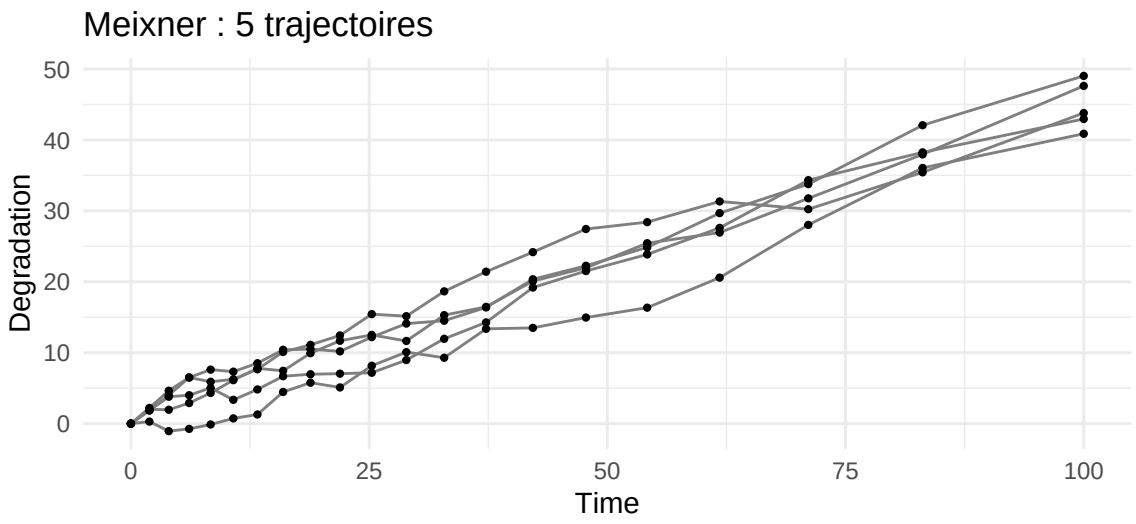
```
set.seed(135)
x <- sim_meixner(p3, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t         X1       X2       X3
## [1,]  0.000000  0.0000000 0.000000 0.000000
## [2,]  1.940035  0.2866694 2.024976 1.842790
## [3,]  3.974743 -1.0583616 1.958050 3.793688
## [4,]  6.113840 -0.7404516 2.908604 4.004078
## [5,]  8.368618 -0.1171312 4.326521 5.041161
## [6,] 10.752309  0.7393661 6.128973 3.356289
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.287 -1.345  0.318  0.623  0.856  0.548  3.176  1.307 -0.667  3.054
## [11]  1.925 -0.796  4.077  0.133  1.458  1.393  4.235  7.435  8.031  4.820
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

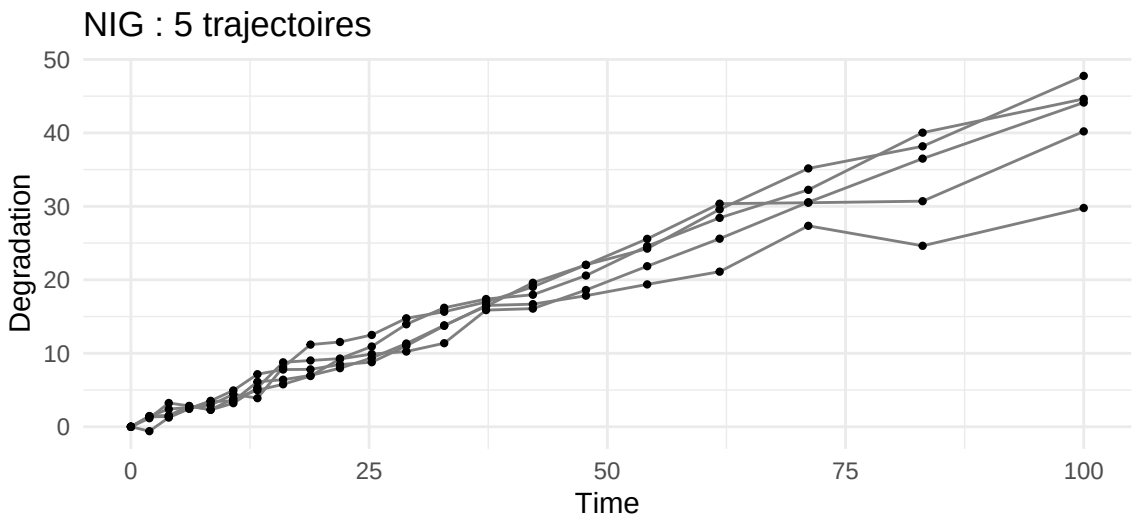
```
set.seed(136)
x <- sim_nig(p3, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t       X1       X2         X3
## [1,]  0.000000 0.000000 0.000000  0.0000000
## [2,]  1.940035 1.416675 1.163461 -0.5974616
## [3,]  3.974743 2.398329 3.233627  1.2486142
## [4,]  6.113840 2.639810 2.835734  2.4510788
## [5,]  8.368618 2.962988 2.302184  3.4433145
## [6,] 10.752309 4.418144 3.207621  3.5021879
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.417  0.982  0.241  0.323  1.455 -0.528  4.205  3.088  0.357  0.954
## [11]  2.269  0.891  1.326  2.062  2.999  2.201  5.348  5.570  3.014  9.580
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

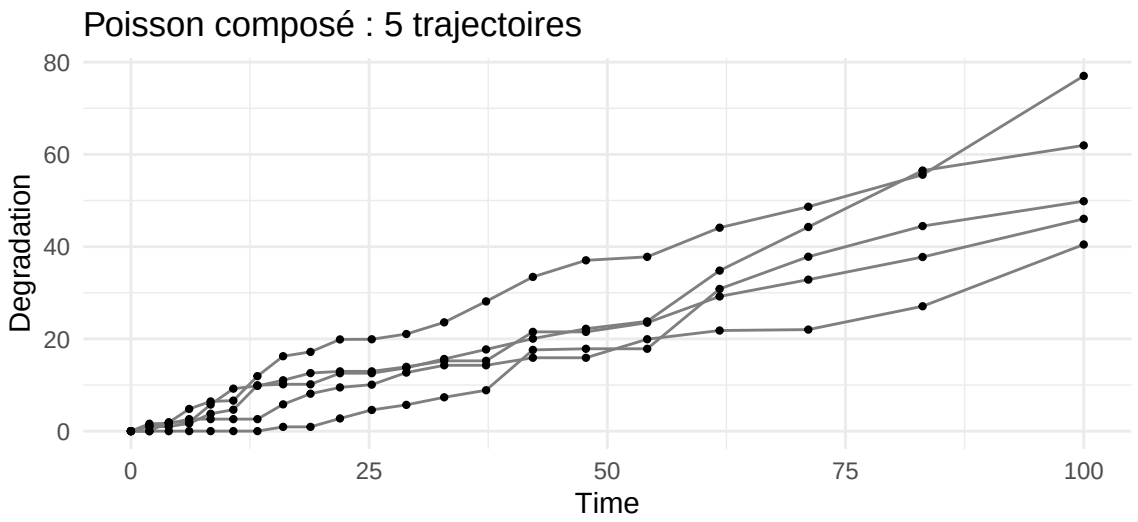
```
set.seed(137)
x <- sim_cpoisson(p3, lambda = 0.5)
head(x[, 1:4])
```

```
##              t       X1        X2       X3
## [1,]  0.000000 0.000000 0.0000000 0.000000
## [2,]  1.940035 1.126505 0.9923666 0.000000
## [3,]  3.974743 1.856240 0.9923666 1.922071
## [4,]  6.113840 4.813572 1.6737306 1.922071
## [5,]  8.368618 6.442241 3.7726864 5.759813
## [6,] 10.752309 6.619919 4.6405605 9.211759
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.127  0.730  2.957  1.629  0.178  5.304  4.321  0.932  2.706  0.037
## [11]  1.134  2.535  4.539  5.310  3.599  0.752  6.321  4.543  6.941 21.428
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
Time
0
20
40
60
80
0
25
50
75
100

Cas 4 : mixtes, beaucoup au début et à la fin

```
p4 <- inspection_times("inc-dec", n = 20, n_traj = 5, tmax = 100)
plot_plan(p4$time[p4$id == 1], "Beaucoup au début et à la fin, peu au milieu")
```



Modèle 1 : gamma

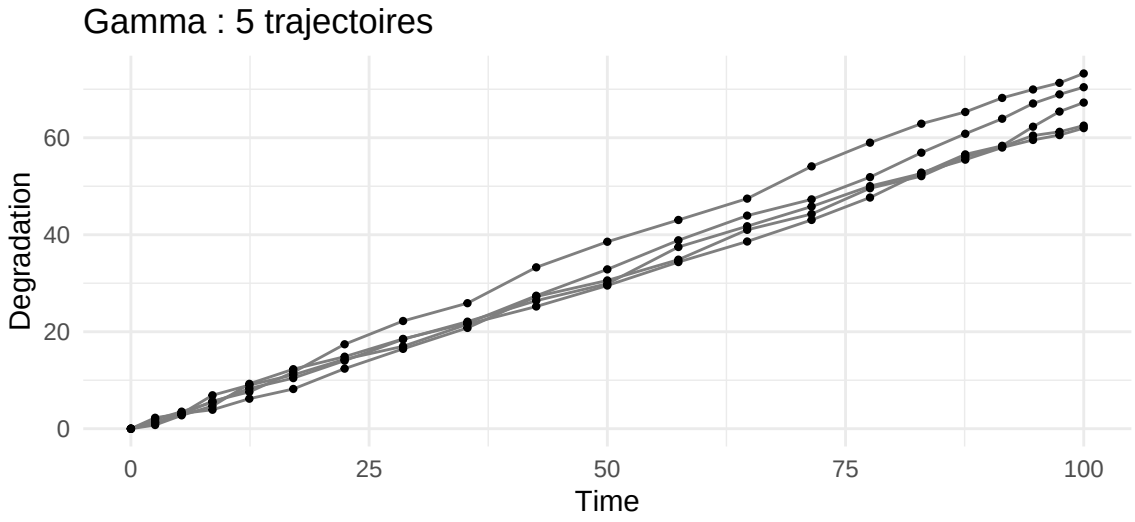
```
set.seed(141)
x <- sim_gamma(p4, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000  0.000000 0.000000  0.000000
## [2,]  2.540921  1.916327 1.174257  2.256682
## [3,]  5.322554  3.520553 3.137588  3.099718
## [4,]  8.562047  5.407939 3.909497  6.897871
## [5,] 12.431733  8.292020 6.188806  8.993691
## [6,] 17.042253 10.434968 8.191329 11.098034
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.916 1.604 1.887 2.884 2.143 3.602 4.399 3.640 4.331 3.499 7.567 4.281
## [13] 4.062 4.209 2.591 3.954 1.772 3.936 3.120 1.856
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

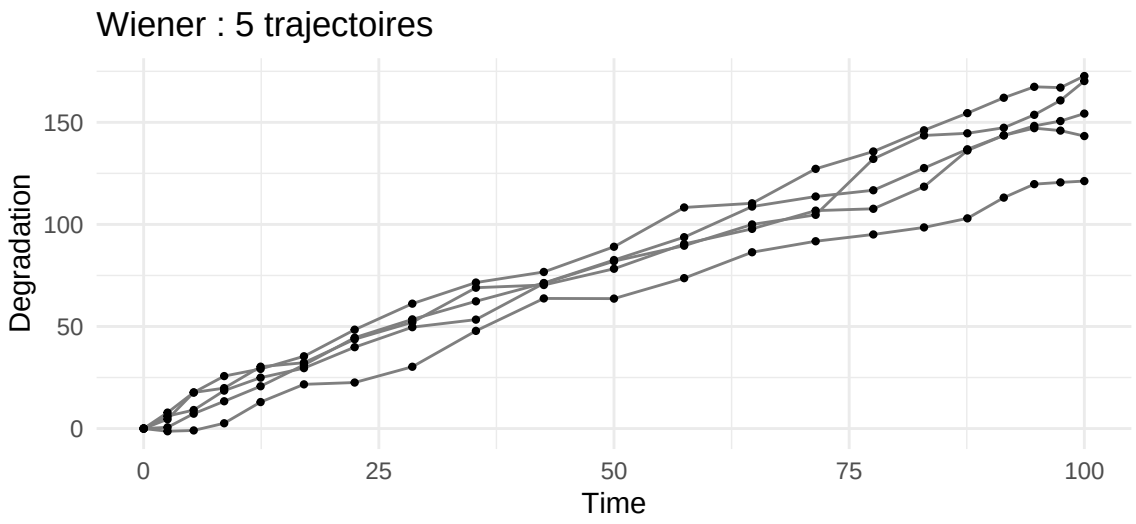
```
set.seed(142)
x <- sim_wiener(p4, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##              t        X1         X2        X3
## [1,]  0.000000  0.000000  0.0000000  0.000000
## [2,]  2.540921  7.811498 -1.3223131  4.565071
## [3,]  5.322554 17.668172 -0.9279949 17.670392
## [4,]  8.562047 25.696328  2.5898009 19.801467
## [5,] 12.431733 29.212882 12.9757735 30.310208
## [6,] 17.042253 35.402095 21.6501585 32.251385
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  7.811  9.857  8.028  3.517  6.189 13.028 12.753 10.340  5.187 12.348
## [11] 19.250  2.036 16.878  8.513 10.390  8.398  7.528  5.360 -0.391  5.692
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

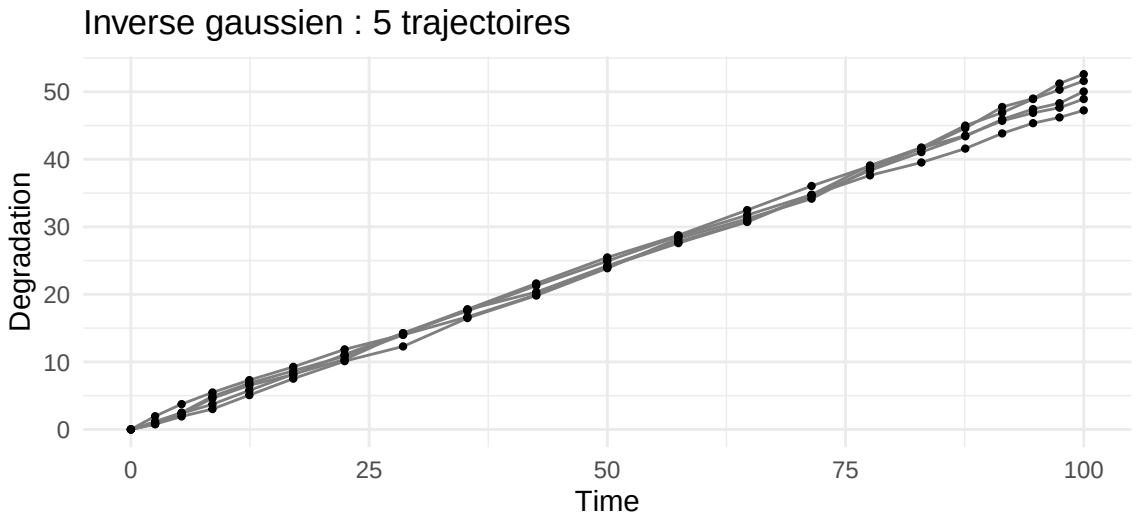
```
set.seed(143)
x <- sim_ig(p4, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000 0.0000000 0.000000 0.0000000
## [2,]  2.540921 0.7772242 1.956862 0.9506147
## [3,]  5.322554 1.9233357 3.745237 2.3784685
## [4,]  8.562047 3.0144983 5.472639 3.7276869
## [5,] 12.431733 5.0832297 7.300213 5.7749032
## [6,] 17.042253 7.5167033 9.259178 8.1877298
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.777 1.146 1.091 2.069 2.433 2.597 2.175 4.202 3.330 4.050 4.402 2.917
## [13] 2.977 4.438 2.984 3.016 3.141 1.239 2.242 1.378
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

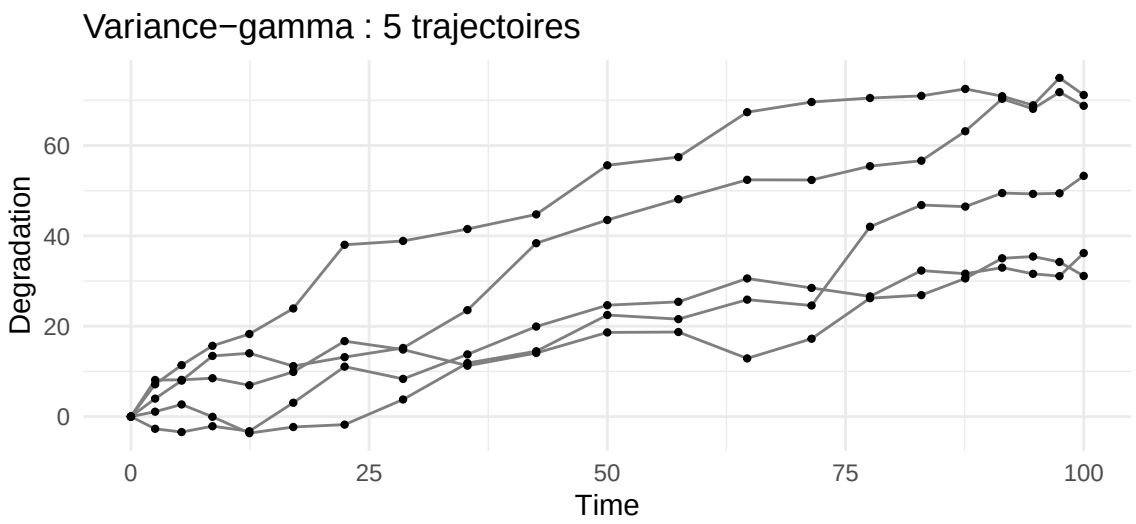
```
set.seed(144)
x <- sim_vg(p4, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t       X1        X2        X3
## [1,]  0.000000 0.000000  0.000000  0.000000
## [2,]  2.540921 8.101337  7.146385 -2.701680
## [3,]  5.322554 8.131089 11.386395 -3.421002
## [4,]  8.562047 8.493392 15.665319 -2.130697
## [5,] 12.431733 6.936565 18.279288 -3.241100
## [6,] 17.042253 9.887127 23.929366  3.061084
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  8.101  0.030  0.362 -1.557  2.951  6.814 -1.871 -3.565  2.809  4.555
## [11]  0.087 -5.843  4.361  8.967  0.692  3.686  4.459  0.392 -1.206 -3.099
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

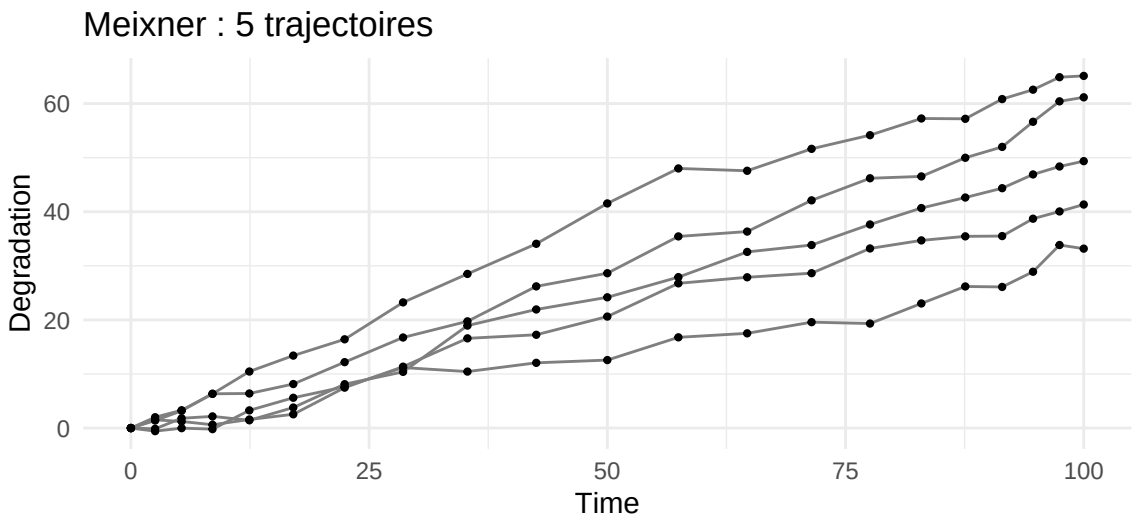
```
set.seed(145)
x <- sim_meixner(p4, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t        X1          X2        X3
## [1,]  0.000000  0.000000  0.00000000 0.0000000
## [2,]  2.540921  2.004889 -0.57081430 1.5262858
## [3,]  5.322554  3.288517 -0.01945328 1.2183414
## [4,]  8.562047  6.329637 -0.19546133 0.6123311
## [5,] 12.431733 10.456133  3.27044396 1.5840534
## [6,] 17.042253 13.399033  5.59940925 2.5455475
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.005  1.284  3.041  4.126  2.943  3.014  6.826  5.267  5.568  7.465
## [11]  6.469 -0.429  4.055  2.525  3.087 -0.058  3.650  1.735  2.309  0.241
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

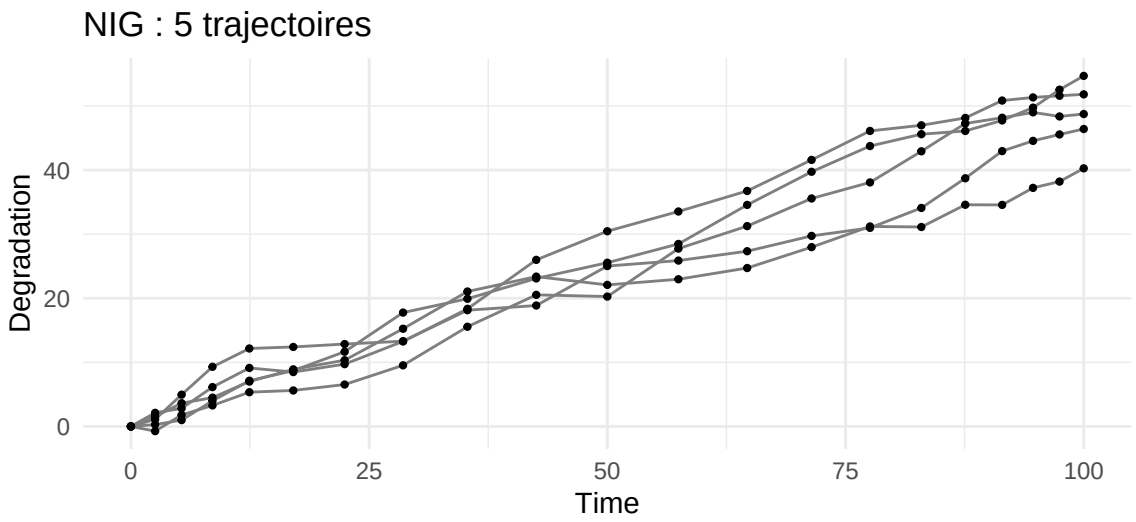
```
set.seed(146)
x <- sim_nig(p4, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t       X1         X2        X3
## [1,]  0.000000 0.000000  0.0000000 0.0000000
## [2,]  2.540921 1.561046 -0.7367223 0.2876020
## [3,]  5.322554 3.624618  1.7851754 0.9848451
## [4,]  8.562047 4.495267  3.2748986 3.9989290
## [5,] 12.431733 7.001454  5.3411814 7.1301402
## [6,] 17.042253 8.906618  5.6123716 8.7004438
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.561  2.064  0.871  2.506  1.905  1.425  4.912  5.809  2.327 -1.294
## [11]  0.883  1.751  3.257  3.229 -0.084  3.469 -0.022  2.659  0.981  2.066
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

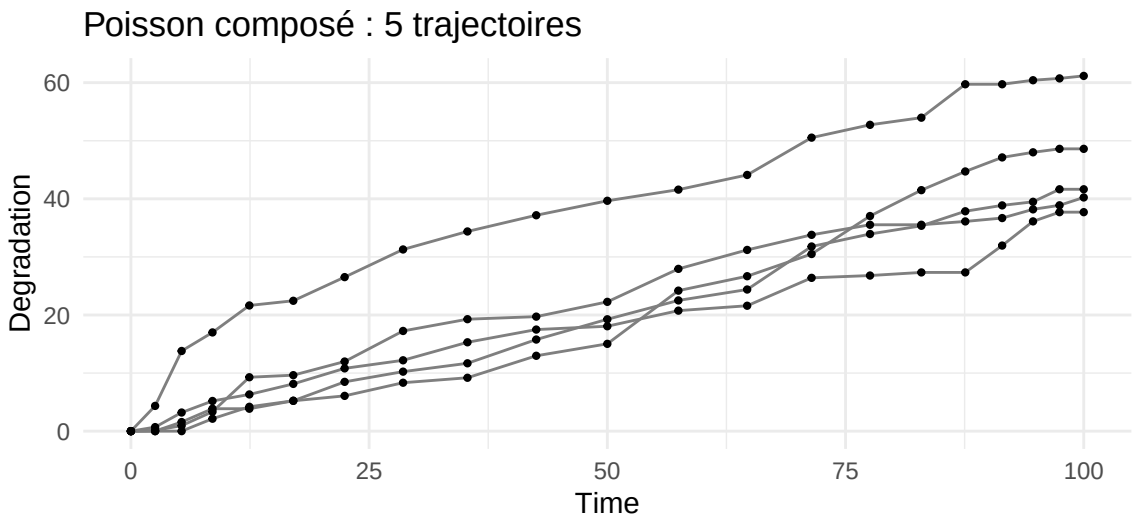
```
set.seed(147)
x <- sim_cpoisson(p4, lambda = 0.5)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000 0.0000000  0.000000 0.0000000
## [2,]  2.540921 0.1210714  4.345207 0.6949144
## [3,]  5.322554 0.9519051 13.802905 3.2070645
## [4,]  8.562047 3.3447454 16.996240 5.1935449
## [5,] 12.431733 9.2809750 21.638766 6.3193773
## [6,] 17.042253 9.6390935 22.445798 8.1355463
```

```
round(diff(x[, "X1"]), 3)
```

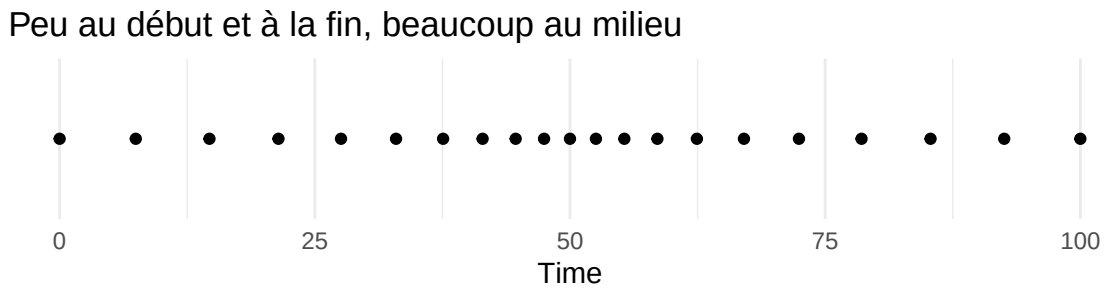
```
##  [1] 0.121 0.831 2.393 5.936 0.358 2.346 5.264 2.027 0.440 2.544 5.680 3.261
## [13] 2.598 1.733 0.000 0.576 0.580 1.499 0.698 1.351
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 5 : mixtes, beaucoup au milieu

```
p5 <- inspection_times("dec-inc", n = 20, n_traj = 5, tmax = 100)
plot_plan(p5$time[p5$id == 1], "Peu au début et à la fin, beaucoup au milieu")
```



Modèle 1 : gamma

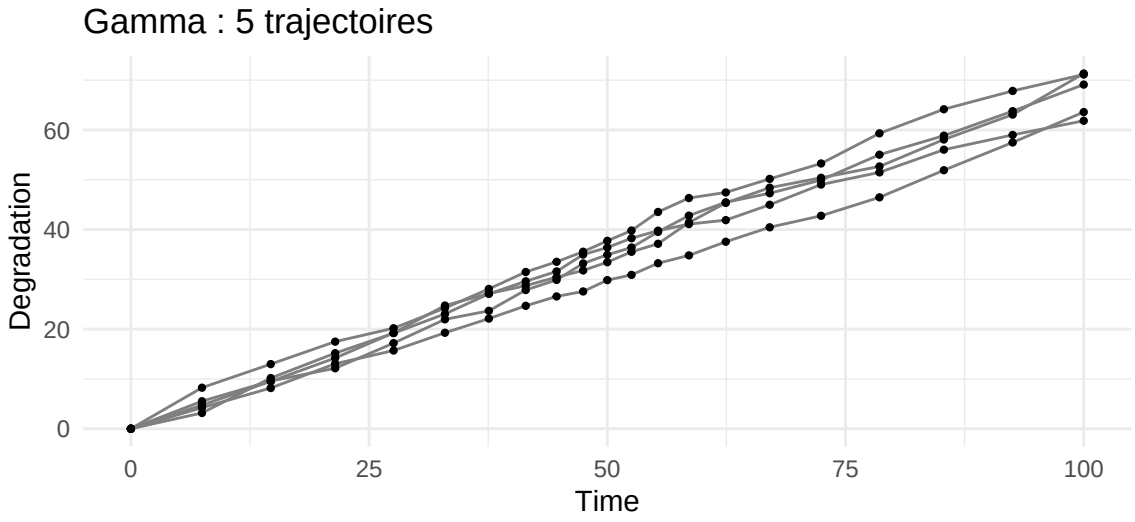
```
set.seed(151)
x <- sim_gamma(p5, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  7.459079  4.741041  5.521434  4.228339
## [3,] 14.677446  9.508096  9.575761  8.175986
## [4,] 21.437953 14.203284 12.146490 13.038912
## [5,] 27.568267 19.244251 17.192410 15.703649
## [6,] 32.957747 24.736728 21.962976 19.268742
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 4.741 4.767 4.695 5.041 5.492 2.407 1.586 1.710 1.344 1.661 2.085 1.614
## [13] 4.256 3.951 3.052 1.993 2.272 5.417 4.993 8.263
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

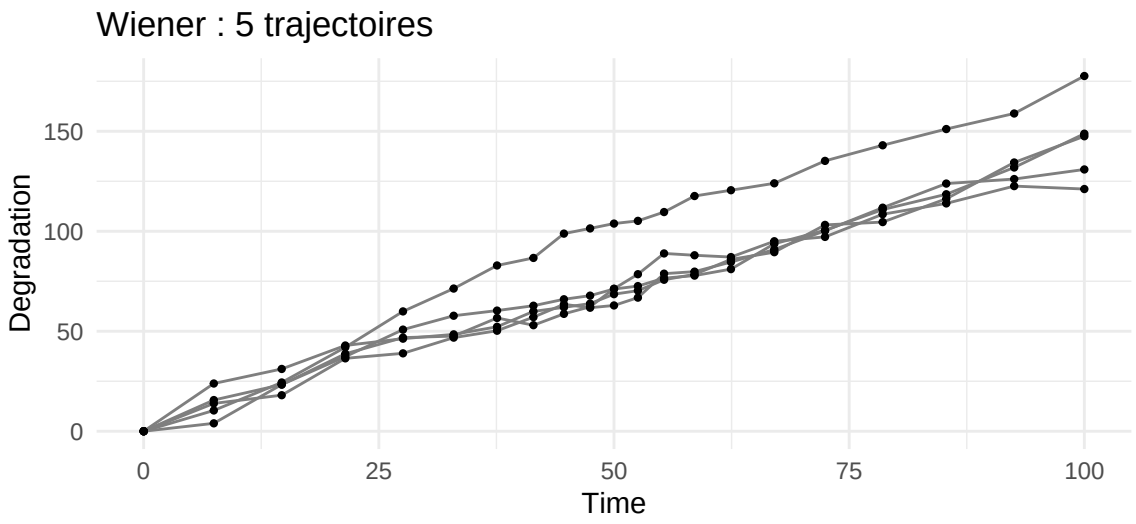
```
set.seed(152)
x <- sim_wiener(p5, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##              t       X1       X2       X3
## [1,]  0.000000  0.00000  0.00000  0.000000
## [2,]  7.459079 23.86975 15.59424  3.958588
## [3,] 14.677446 31.14916 23.32680 23.360176
## [4,] 21.437953 42.87639 38.71552 37.302685
## [5,] 27.568267 46.29445 46.82211 50.815925
## [6,] 32.957747 48.44725 47.57995 57.756556
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 23.870  7.279 11.727  3.418  2.153  3.777  7.718  2.015  1.883  4.704
## [11]  1.753  5.451  2.693  7.475  3.619 13.646  1.412 11.651 18.147 13.120
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

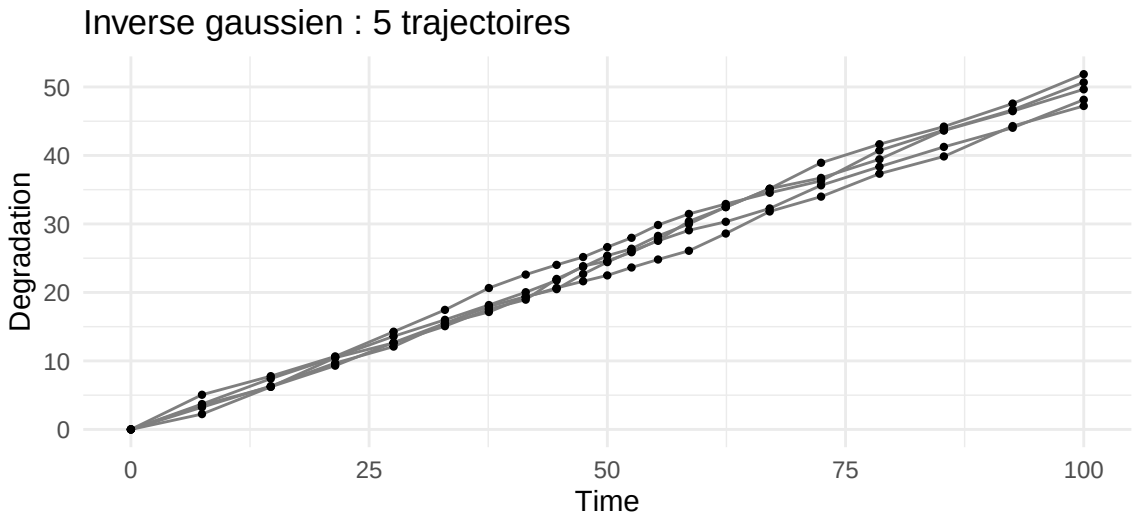
```
set.seed(153)
x <- sim_ig(p5, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  7.459079  2.238359  3.240561  3.557905
## [3,] 14.677446  6.217203  6.325747  6.219868
## [4,] 21.437953 10.468404  9.677937  9.307220
## [5,] 27.568267 12.628815 12.094830 12.680646
## [6,] 32.957747 15.074764 15.487383 15.478020
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.238 3.979 4.251 2.160 2.446 2.483 1.386 3.026 1.761 1.638 1.012 1.874
## [13] 1.749 2.454 2.684 3.788 2.710 2.572 3.362 4.294
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

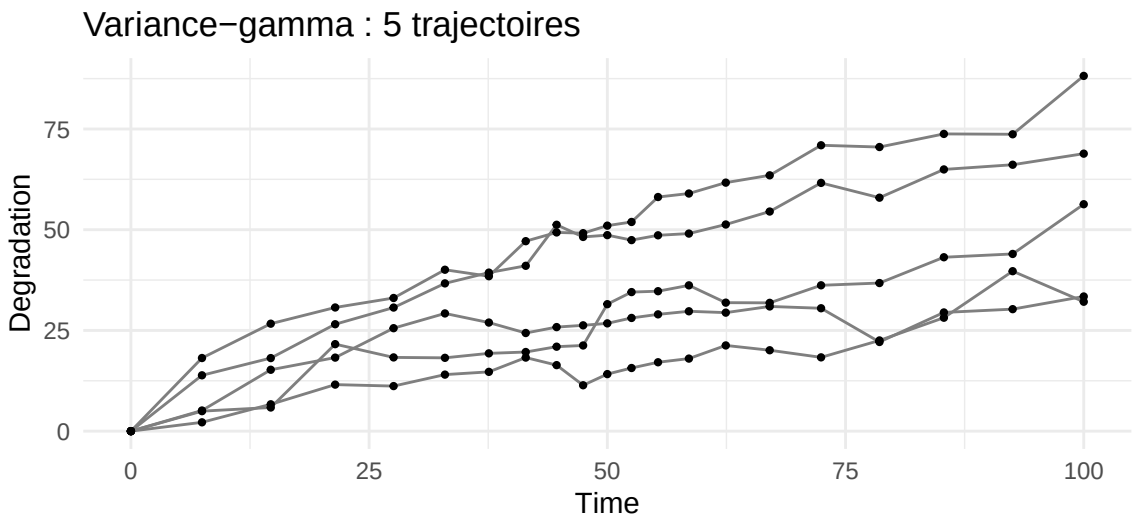
```
set.seed(154)
x <- sim_vg(p5, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t        X1        X2       X3
## [1,]  0.000000  0.000000  0.000000  0.00000
## [2,]  7.459079  2.190125  4.960023 13.85976
## [3,] 14.677446  6.667534  5.870367 18.13231
## [4,] 21.437953 11.550031 21.591089 26.53618
## [5,] 27.568267 11.182291 18.299595 30.67430
## [6,] 32.957747 14.030616 18.197226 36.67401
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.190  4.477  4.882 -0.368  2.848  0.685  3.542 -1.873 -4.993  2.766
## [11]  1.512  1.428  0.910  3.264 -1.194 -1.765  4.196  5.645 11.551 -7.583
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

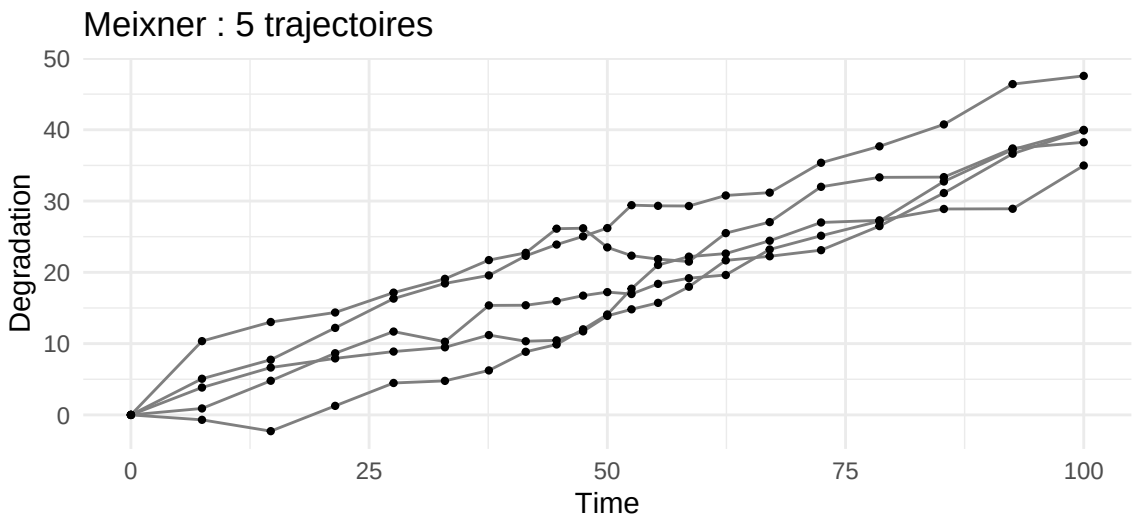
```
set.seed(155)
x <- sim_meixner(p5, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t        X1       X2       X3
## [1,]  0.000000  0.000000 0.000000  0.00000
## [2,]  7.459079  5.071196 3.839912 10.35035
## [3,] 14.677446  7.739227 6.643030 13.02948
## [4,] 21.437953 12.206749 7.927603 14.36807
## [5,] 27.568267 16.315425 8.886013 17.15169
## [6,] 32.957747 18.441474 9.482853 19.08732
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.071  2.668  4.468  4.109  2.126  1.128  2.741  1.584  1.152  1.161
## [11]  3.216 -0.093 -0.024  1.478  0.398  4.188  2.306  3.073  5.657  1.151
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

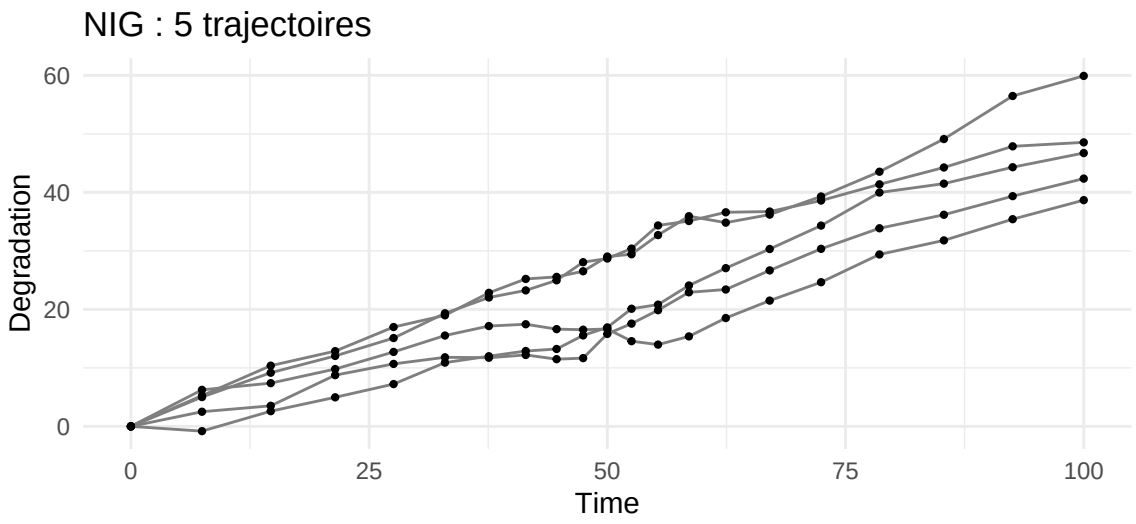
```
set.seed(156)
x <- sim_nig(p5, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t         X1        X2        X3
## [1,]  0.000000  0.0000000  0.000000  0.000000
## [2,]  7.459079 -0.8154205  6.241988  5.009794
## [3,] 14.677446  2.6002278  7.387626  9.169580
## [4,] 21.437953  4.9651138  9.796440 12.038473
## [5,] 27.568267  7.2308250 12.724568 15.099153
## [6,] 32.957747 10.8952491 15.527961 19.331089
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -0.815  3.416  2.365  2.266  3.664  1.096  0.888  0.346  2.322  1.370
## [11]  3.189  0.709  3.267  2.964  3.271  4.003  5.652  1.520  2.809  2.423
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

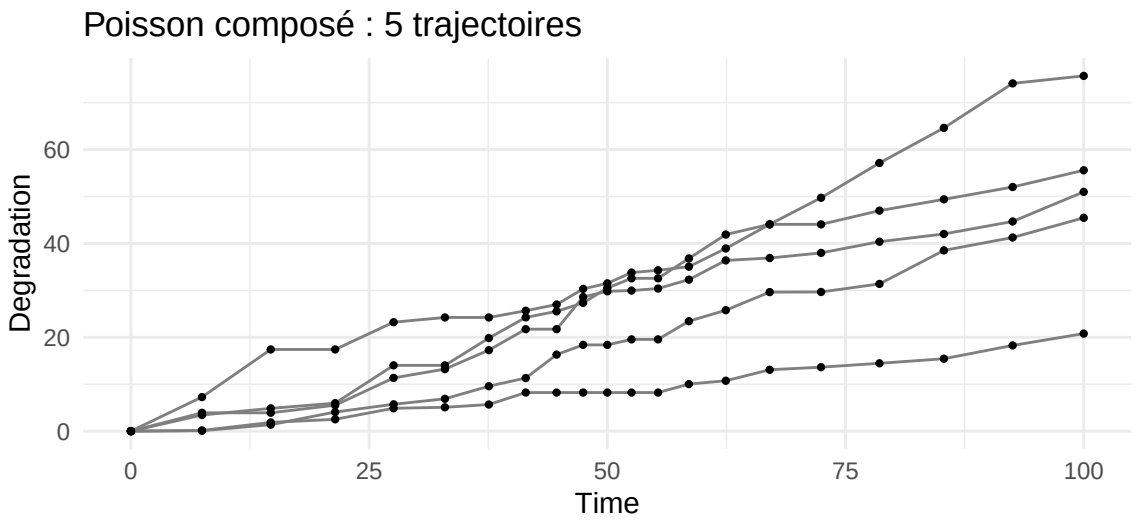
```
set.seed(157)
x <- sim_cpoisson(p5, lambda = 0.5)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000 0.0000000
## [2,]  7.459079  3.911296  3.438826 0.1384595
## [3,] 14.677446  3.911296  4.852867 1.8628205
## [4,] 21.437953  5.528340  5.961461 2.5230237
## [5,] 27.568267 11.344923 14.017337 4.8769255
## [6,] 32.957747 13.224447 14.017337 5.0783270
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.911 0.000 1.617 5.817 1.880 4.040 4.465 0.000 6.858 1.222 0.144 0.439
## [13] 1.896 4.099 0.505 1.094 2.385 1.647 2.663 6.315
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
0
20
40
60
0
25
50
75
100
Time

III - Processus de Lévy à échelle de temps

Une fonction d'échelle de temps A (fonction croissante, continue à droite avec des limites à gauche, $A(0) = 0$ et $A(t) \to +\infty$) donne le processus de Lévy à échelle de temps $X_t = Y_{A(t)}$, où Y est un processus de Lévy. Le paramètre que A remplace (identifiabilité) n'est alors pas donné :

Processus	Fonction	$X_t - X_s$, $\Delta A = A(t) - A(s)$	Avec $A(t) = at$ ou μt
Gamma	`sim_gamma(t, b, A)`	$\Gamma(\Delta A, b)$	gamma de forme a
Wiener	`sim_wiener(t, sigma, A)`	$\mathcal{N}(\Delta A, \sigma^2 \Delta A)$	Wiener $(\mu, \sigma\sqrt{\mu})$
Inverse gaussien	`sim_ig(t, eta, A)`	$IG(\Delta A, \eta \Delta A^2)$	IG $(\mu, \eta\mu^2)$

Pour le Wiener, $X_t = \mu t + \sigma B_{\mu t}$ a la loi de $\mu t + \sigma\sqrt{\mu}\, B_t$, car $(B_{\mu t})_{t\geq 0}$ a la loi de $(\sqrt{\mu}\, B_t)_{t\geq 0}$.

On prend $A(t) = 0.2\, t^{1.5}$, de sorte que $A(100) = 200$, et les 5 trajectoires du cas 2 de la partie II (pas décroissants).

Modèle 1 : gamma à échelle de temps (Cha et Mercier, 2019, exemple 2)

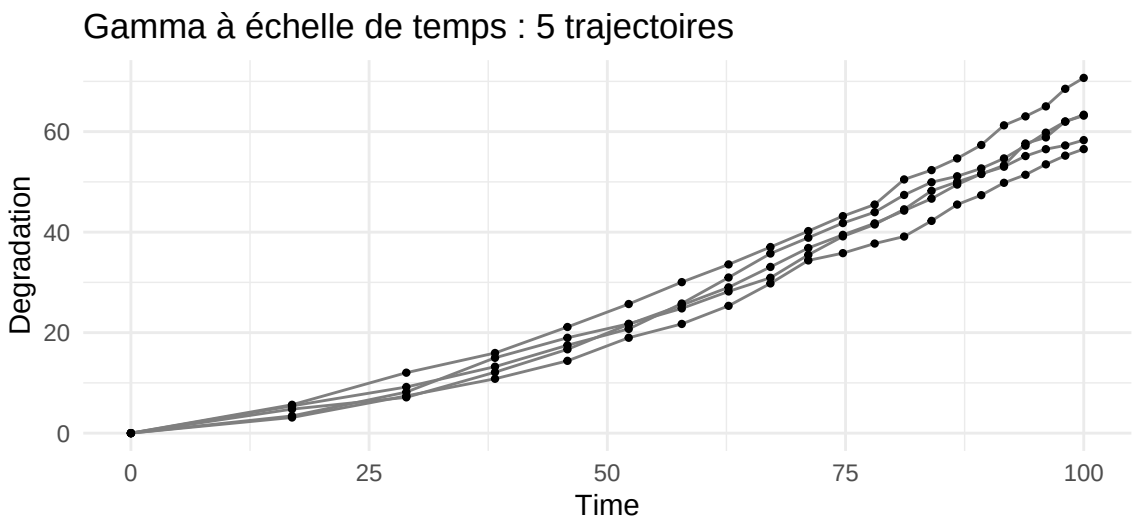
```
A <- function(t) 0.2 * t^1.5
set.seed(301)
x <- sim_gamma(p2, b = 3, A = A)
x
```

```
## 5 simulated trajectory(ies) of the time-scaled gamma process, same inspection times (matrix)
## parameters: b = 3
## simulation method: RWA
## time-scaling function: A = function (t)  0.2 * t^1.5
##
##             t        X1        X2        X3        X4        X5
## [1,]  0.00000  0.000000  0.000000  0.000000  0.000000  0.000000
## [2,] 16.90921  4.773915  5.643509  5.414966  3.115465  3.424173
## [3,] 28.90648  7.123770 12.030601  9.179291  7.408775  8.170816
## [4,] 38.21229 12.125653 15.944868 13.227631 10.815661 14.990427
## [5,] 45.81569 16.679852 21.122246 17.490494 14.387050 18.975245
## [6,] 52.24427 21.656789 25.693115 20.742016 18.980764 21.755772
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 4.774 2.350 5.002 4.554 4.977 3.818 3.535 4.066 3.771 2.608 2.292 2.531
## [13] 2.378 2.808 2.138 1.412 2.123 1.368 0.747 1.053
```

```
plot_traj(x, "Gamma à échelle de temps : 5 trajectoires")
```



Modèle 2 : Wiener à échelle de temps (Cha et Mercier, 2019, exemple 1)

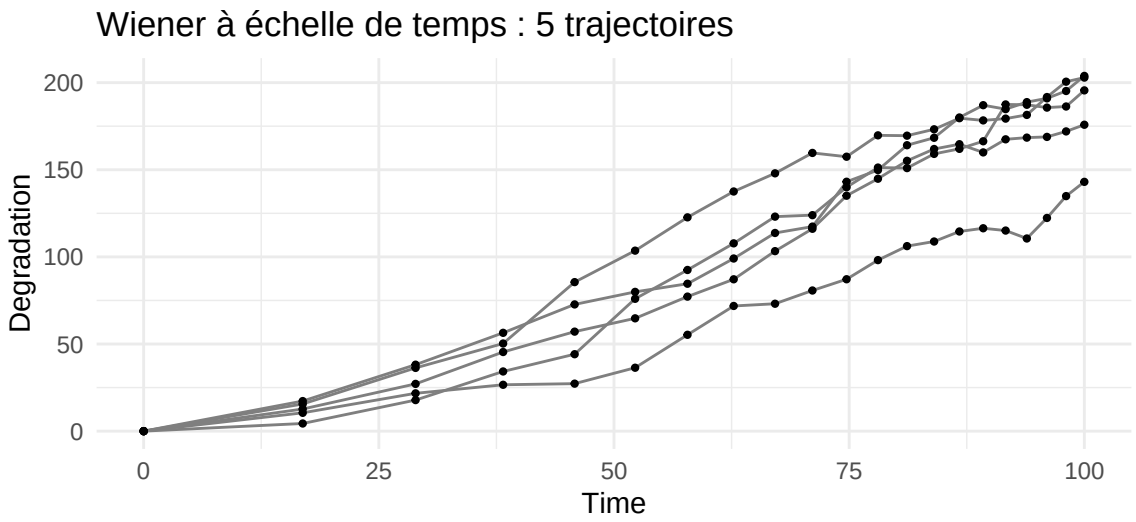
```
set.seed(302)
x <- sim_wiener(p2, sigma = 2, A = A)
head(x[, 1:4])
```

```
##             t        X1       X2       X3
## [1,]  0.00000   0.00000  0.00000  0.00000
## [2,] 16.90921  15.59075 17.25295 10.51275
## [3,] 28.90648  36.29836 38.13262 21.70767
## [4,] 38.21229  50.28924 56.42033 26.60491
## [5,] 45.81569  85.51731 72.71654 27.24921
## [6,] 52.24427 103.57496 79.91758 36.39190
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 15.591 20.708 13.991 35.228 18.058 19.114 14.837 10.431 11.701 -2.172
## [11] 12.259 -0.191  3.656  6.382 -1.274  1.034  2.118 10.247  8.816  2.342
```

```
plot_traj(x, "Wiener à échelle de temps : 5 trajectoires")
```



Modèle 3 : inverse gaussien à échelle de temps

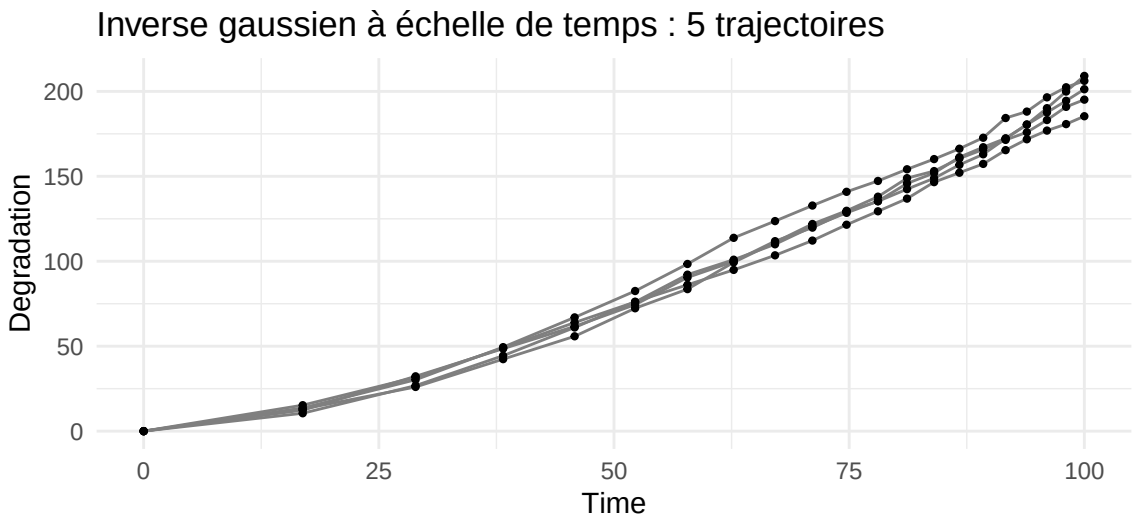
```
set.seed(303)
x <- sim_ig(p2, eta = 2, A = A)
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.00000
## [2,] 16.90921 15.27997 12.34472 13.01598
## [3,] 28.90648 31.65861 32.26962 26.01430
## [4,] 38.21229 48.94874 48.56201 42.37829
## [5,] 45.81569 63.91646 61.62486 55.83160
## [6,] 52.24427 75.99712 74.46398 72.33299
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 15.280 16.379 17.290 14.968 12.081 16.122  8.795  9.072 10.441  8.073
## [11]  6.860 10.427  6.183  9.291  5.868  5.212  8.070  7.190  6.849  6.818
```

```
plot_traj(x, "Inverse gaussien à échelle de temps : 5 trajectoires")
```



Vérification : avec $A(t) = at$, on retrouve le processus gamma de forme a

Avec la même graine, le gamma à échelle de temps avec $A(t) = 2t$ et le gamma $(a, b) = (2, 3)$ donnent les mêmes trajectoires :

```
set.seed(304)
y1 <- sim_gamma(p2, b = 3, A = function(t) 2 * t)
set.seed(304)
y2 <- sim_gamma(p2, a = 2, b = 3)
all.equal(unclass(y1)[, -1], unclass(y2)[, -1], check.attributes = FALSE)
```

```
## [1] TRUE
```

IV - Méthode Bridge

Les parties I à III utilisent la méthode RWA, qui cumule des accroissements indépendants. Avec `method = "Bridge"`, la valeur à la dernière inspection est tirée d'abord. Les autres valeurs sont ensuite tirées une à une, en prenant chaque fois la date du milieu de la liste, par les fonctions `gamma_bridge`, `wiener_bridge`, `ig_bridge`, `vg_bridge` et `meixner_bridge` du package `levyBridge`, installé avec `DegradSim`.

Les trajectoires ont la même loi qu'avec la méthode RWA. Avec une même graine, les valeurs obtenues sont différentes, et le calcul est plus lent. Le normal inverse gaussien et le Poisson composé n'ont pas de pont : pour eux, seule la méthode RWA est disponible.

Cas 1 : une trajectoire, inspections régulières

Les inspections sont celles du cas 1 de la partie I. L'affichage de `x` indique la méthode utilisée.

Modèle 1 : gamma

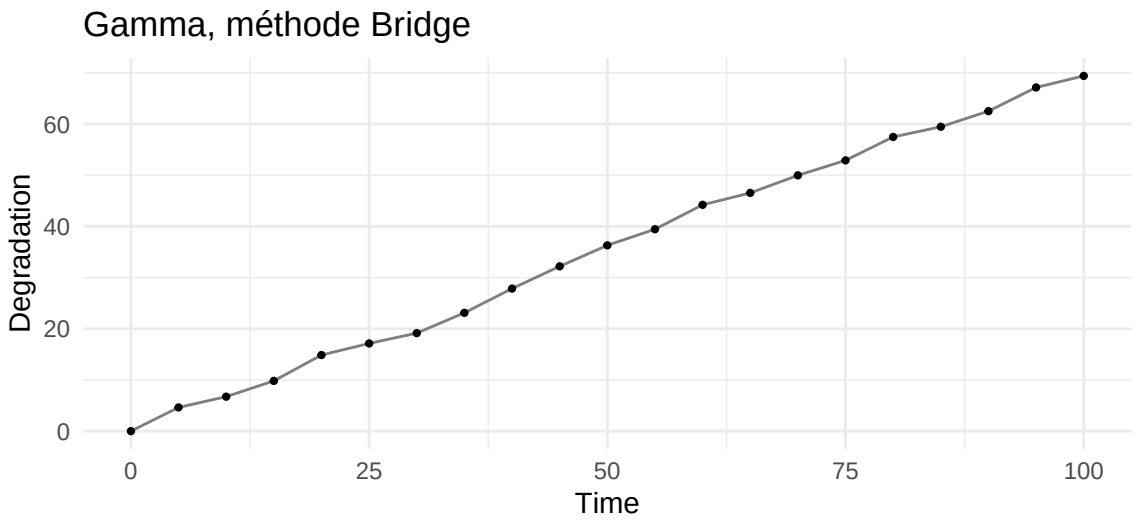
```
set.seed(411)
x <- sim_gamma(t1, a = 2, b = 3, method = "Bridge")
x
```

```
## 1 simulated trajectory(ies) of the gamma process, same inspection times (matrix)
## parameters: a = 2, b = 3
## simulation method: Bridge
##
##       t        X1
## [1,]  0  0.000000
## [2,]  5  4.629840
## [3,] 10  6.731952
## [4,] 15  9.816780
## [5,] 20 14.864175
## [6,] 25 17.131037
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 4.630 2.102 3.085 5.047 2.267 2.011 3.972 4.730 4.348 4.101 3.142 4.773
## [13] 2.320 3.435 2.926 4.576 1.989 3.044 4.632 2.258
```

```
plot_traj(x, "Gamma, méthode Bridge")
```



Modèle 2 : Wiener

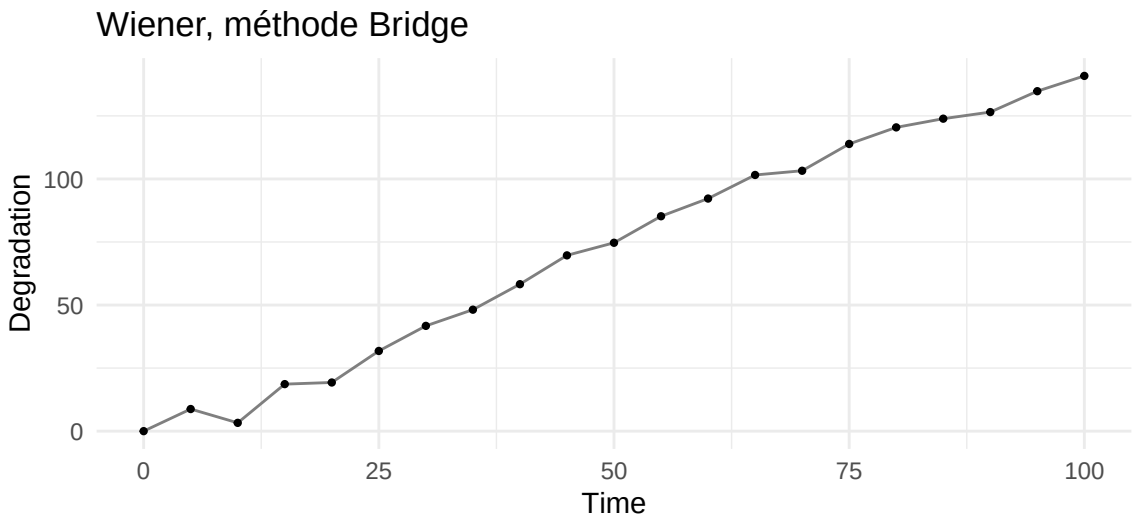
```
set.seed(412)
x <- sim_wiener(t1, mu = 1.5, sigma = 2, method = "Bridge")
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  8.766412
## [3,] 10  3.286023
## [4,] 15 18.642434
## [5,] 20 19.289664
## [6,] 25 31.793960
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  8.766 -5.480 15.356  0.647 12.504  9.942  6.423 10.111 11.424  4.991
## [11] 10.519  7.015  9.351  1.671 10.632  6.567  3.448  2.615  8.284  6.064
```

```
plot_traj(x, "Wiener, méthode Bridge")
```



Modèle 3 : inverse gaussien

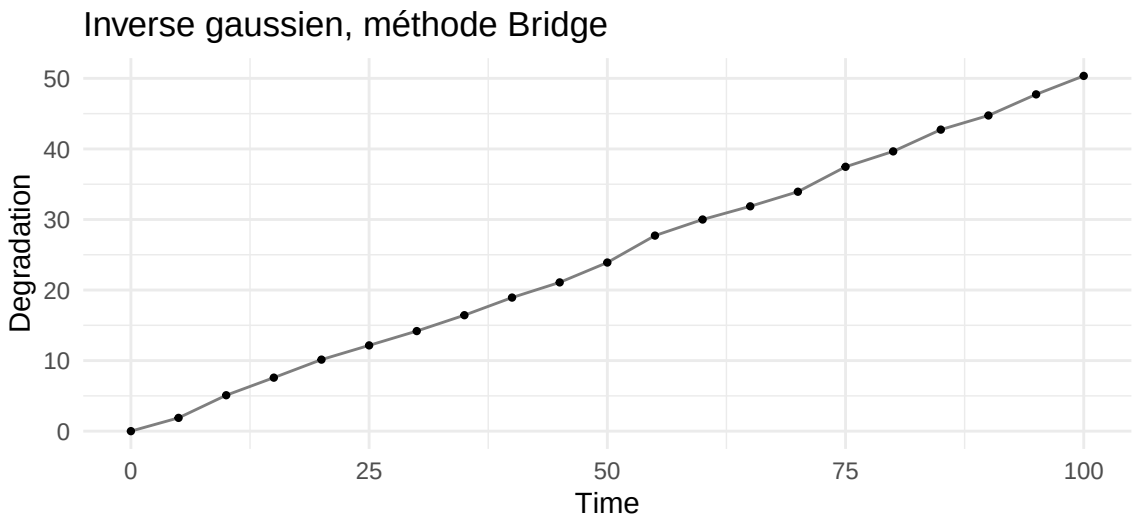
```
set.seed(413)
x <- sim_ig(t1, mu = 0.5, eta = 2, method = "Bridge")
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  1.880276
## [3,] 10  5.088247
## [4,] 15  7.581137
## [5,] 20 10.136199
## [6,] 25 12.154311
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.880 3.208 2.493 2.555 2.018 2.029 2.250 2.498 2.153 2.815 3.818 2.279
## [13] 1.873 2.065 3.533 2.186 3.083 2.005 2.992 2.618
```

```
plot_traj(x, "Inverse gaussien, méthode Bridge")
```



Modèle 4 : variance-gamma

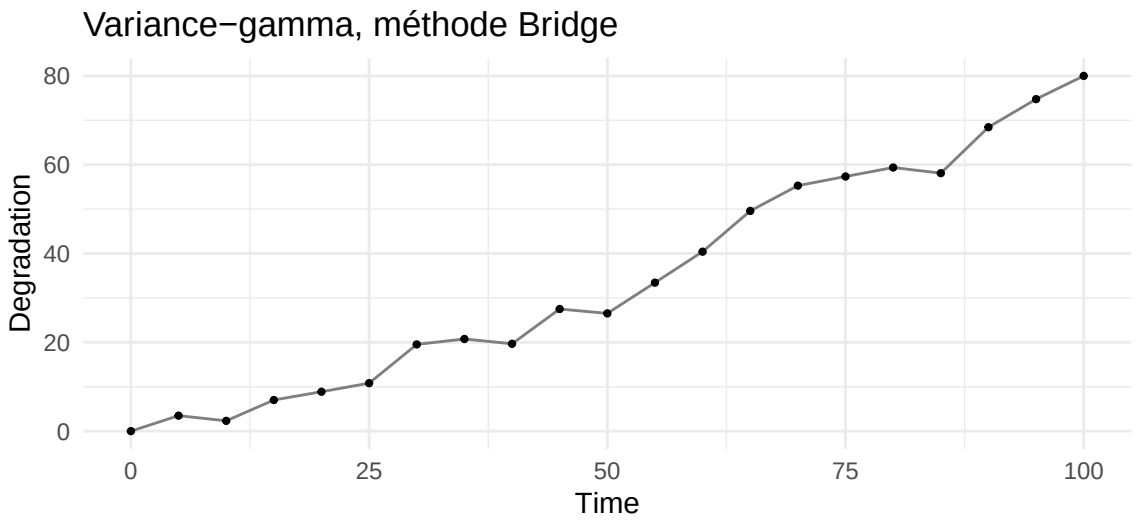
```
set.seed(414)
x <- sim_vg(t1, theta = 0.5, sigma = 1.5, nu = 2, method = "Bridge")
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  3.486393
## [3,] 10  2.317832
## [4,] 15  7.009930
## [5,] 20  8.872142
## [6,] 25 10.801312
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.486 -1.169  4.692  1.862  1.929  8.724  1.229 -1.087  7.833 -0.989
## [11]  6.926  6.978  9.177  5.685  2.061  2.022 -1.273 10.352  6.323  5.231
```

```
plot_traj(x, "Variance-gamma, méthode Bridge")
```



Modèle 5 : Meixner

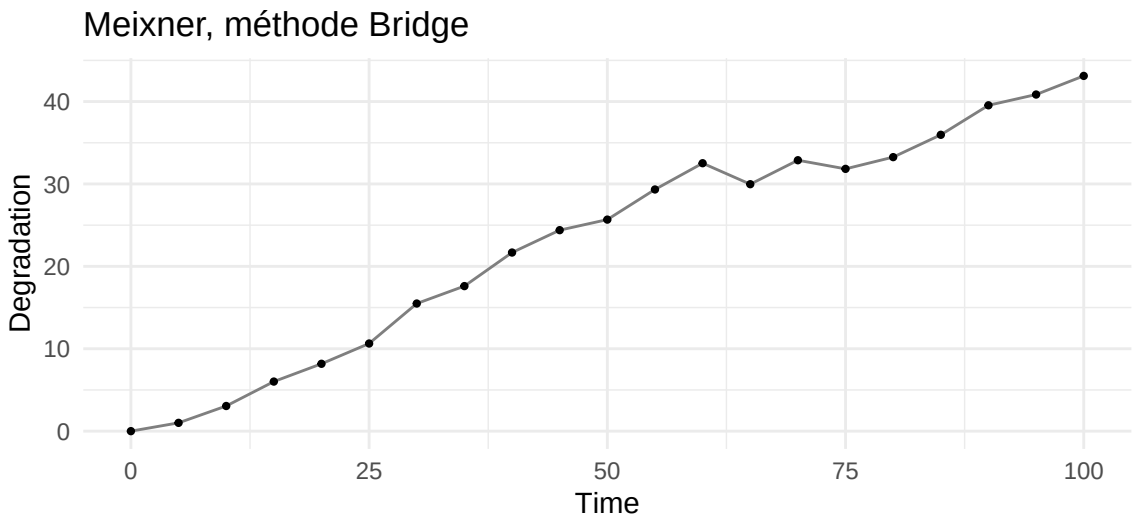
```
set.seed(415)
x <- sim_meixner(t1, a = 1, b = 0.3, d = 1.5, m = 0.2, method = "Bridge")
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  1.006599
## [3,] 10  3.051490
## [4,] 15  6.017266
## [5,] 20  8.174742
## [6,] 25 10.635094
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.007  2.045  2.966  2.157  2.460  4.853  2.115  4.076  2.716  1.280
## [11]  3.658  3.187 -2.547  2.909 -1.047  1.425  2.710  3.580  1.307  2.262
```

```
plot_traj(x, "Meixner, méthode Bridge")
```


Meixner, méthode Bridge
Degradation
0
10
20
30
40
0
25
50
75
100
Time

Cas 2 : une trajectoire, pas décroissants

Les inspections sont celles du cas 2 de la partie I.

Modèle 1 : gamma

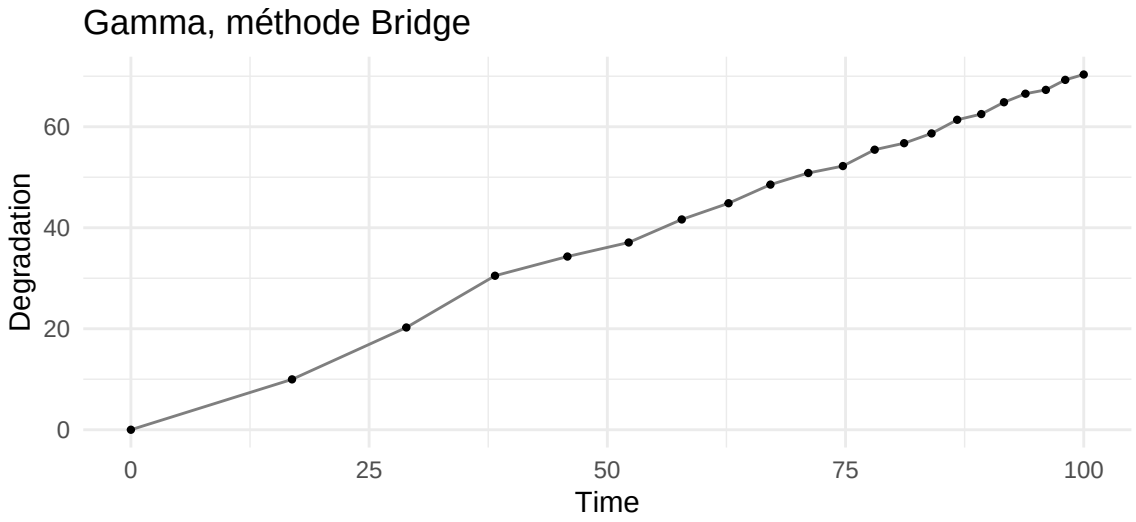
```
set.seed(421)
x <- sim_gamma(t2, a = 2, b = 3, method = "Bridge")
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  9.975064
## [3,] 28.90648 20.246261
## [4,] 38.21229 30.492266
## [5,] 45.81569 34.296333
## [6,] 52.24427 37.076793
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  9.975 10.271 10.246  3.804  2.780  4.564  3.213  3.690  2.295  1.376
## [11]  3.238  1.288  1.934  2.701  1.131  2.340  1.694  0.758  1.979  1.065
```

```
plot_traj(x, "Gamma, méthode Bridge")
```



Modèle 2 : Wiener

```
set.seed(422)
x <- sim_wiener(t2, mu = 1.5, sigma = 2, method = "Bridge")
head(x)
```

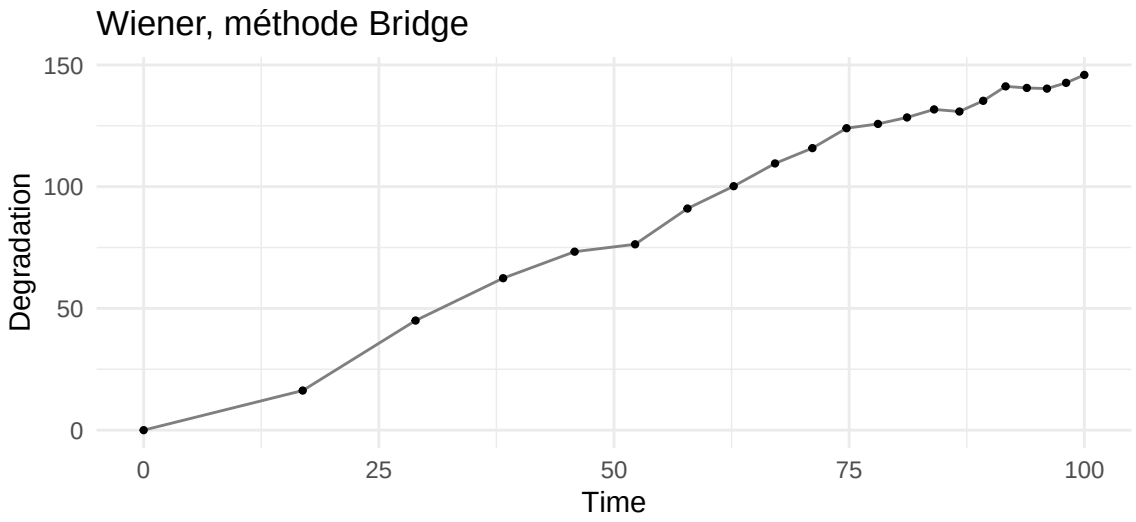
```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 16.26360
## [3,] 28.90648 45.01257
## [4,] 38.21229 62.39900
```

```
## [5,] 45.81569 73.29481
## [6,] 52.24427 76.32360
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 16.264 28.749 17.386 10.896  3.029 14.690  9.179  9.341  6.289  8.180
## [11]  1.762  2.659  3.295 -0.861  4.399  5.928 -0.647 -0.271  2.359  3.271
```

```
plot_traj(x, "Wiener, méthode Bridge")
```



Modèle 3 : inverse gaussien

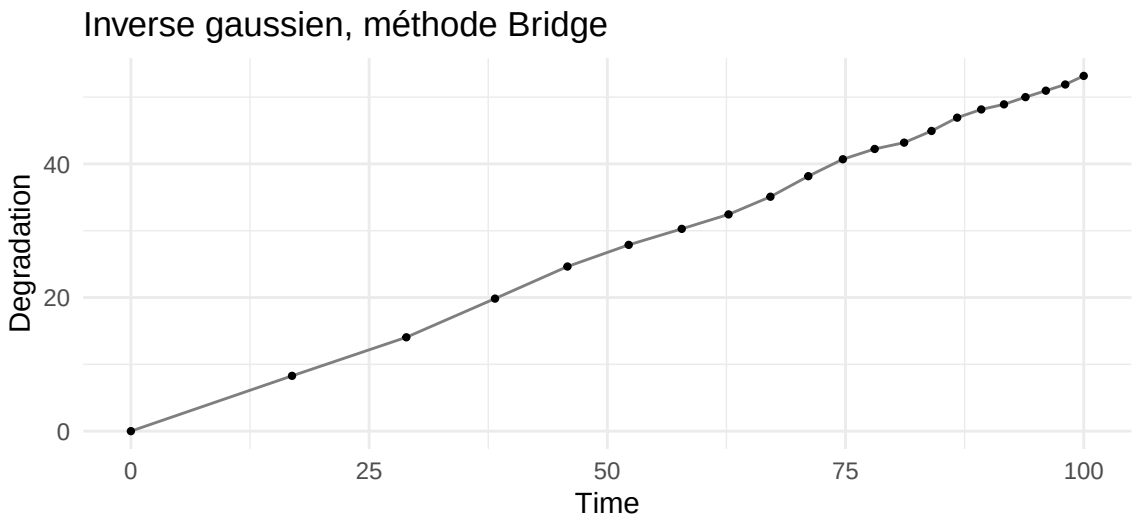
```
set.seed(423)
x <- sim_ig(t2, mu = 0.5, eta = 2, method = "Bridge")
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  8.275556
## [3,] 28.90648 14.049238
## [4,] 38.21229 19.837928
## [5,] 45.81569 24.641122
## [6,] 52.24427 27.882289
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 8.276 5.774 5.789 4.803 3.241 2.397 2.166 2.646 3.073 2.538 1.537 0.947
## [13] 1.749 1.990 1.229 0.762 1.084 0.960 0.930 1.285
```

```
plot_traj(x, "Inverse gaussien, méthode Bridge")
```



Modèle 4 : variance-gamma

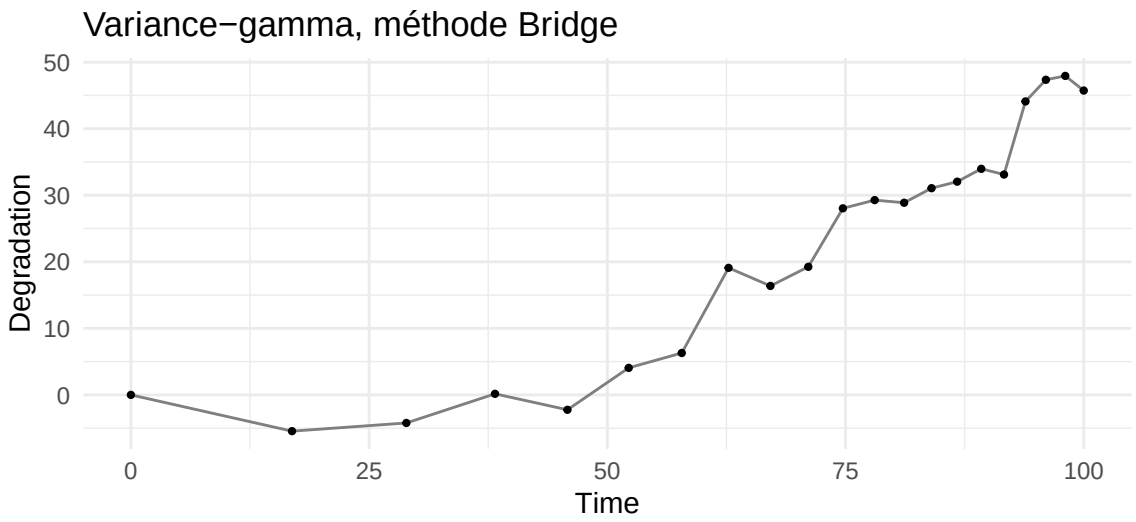
```
set.seed(424)
x <- sim_vg(t2, theta = 0.5, sigma = 1.5, nu = 2, method = "Bridge")
head(x)
```

```
##             t         X1
## [1,]  0.00000  0.0000000
## [2,] 16.90921 -5.4476867
## [3,] 28.90648 -4.2229991
## [4,] 38.21229  0.1595717
## [5,] 45.81569 -2.2396554
## [6,] 52.24427  4.0610719
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -5.448  1.225  4.383 -2.399  6.301  2.236 12.792 -2.716  2.876  8.786
## [11]  1.236 -0.397  2.188  0.980  1.931 -0.860 10.989  3.251  0.585 -2.202
```

```
plot_traj(x, "Variance-gamma, méthode Bridge")
```



Modèle 5 : Meixner

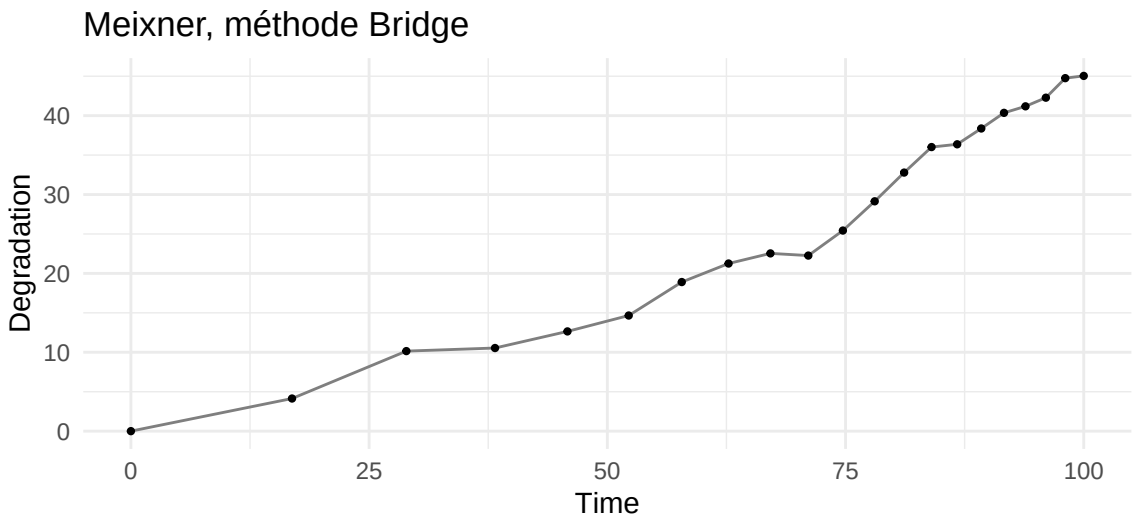
```
set.seed(425)
x <- sim_meixner(t2, a = 1, b = 0.3, d = 1.5, m = 0.2, method = "Bridge")
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921  4.13572
## [3,] 28.90648 10.14732
## [4,] 38.21229 10.53812
## [5,] 45.81569 12.64474
## [6,] 52.24427 14.65923
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  4.136  6.012  0.391  2.107  2.014  4.244  2.349  1.284 -0.276  3.173
## [11]  3.707  3.644  3.234  0.352  2.005  1.980  0.828  1.095  2.478  0.285
```

```
plot_traj(x, "Meixner, méthode Bridge")
```


Meixner, méthode Bridge
Degradation
0
10
20
30
40
0
25
50
75
100
Time

Cas 3 : une trajectoire, pas croissants

Les inspections sont celles du cas 3 de la partie I.

Modèle 1 : gamma

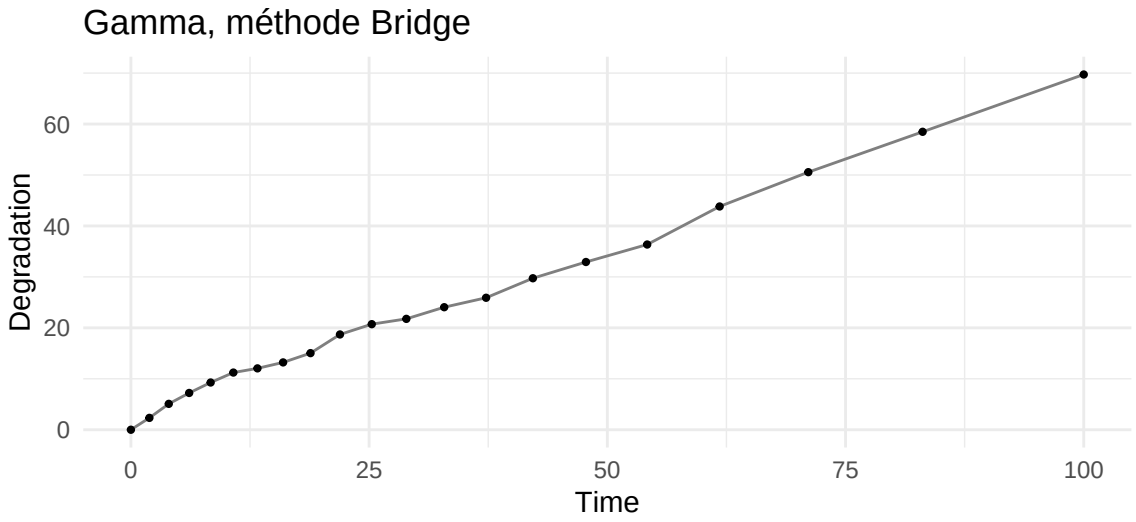
```
set.seed(431)
x <- sim_gamma(t3, a = 2, b = 3, method = "Bridge")
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  1.940035  2.329127
## [3,]  3.974743  5.081529
## [4,]  6.113840  7.219826
## [5,]  8.368618  9.271957
## [6,] 10.752309 11.223763
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.329  2.752  2.138  2.052  1.952  0.823  1.170  1.822  3.662  2.023
## [11]  1.039  2.290  1.852  3.832  3.189  3.432  7.467  6.736  7.922 11.246
```

```
plot_traj(x, "Gamma, méthode Bridge")
```



Modèle 2 : Wiener

```
set.seed(432)
x <- sim_wiener(t3, mu = 1.5, sigma = 2, method = "Bridge")
head(x)
```

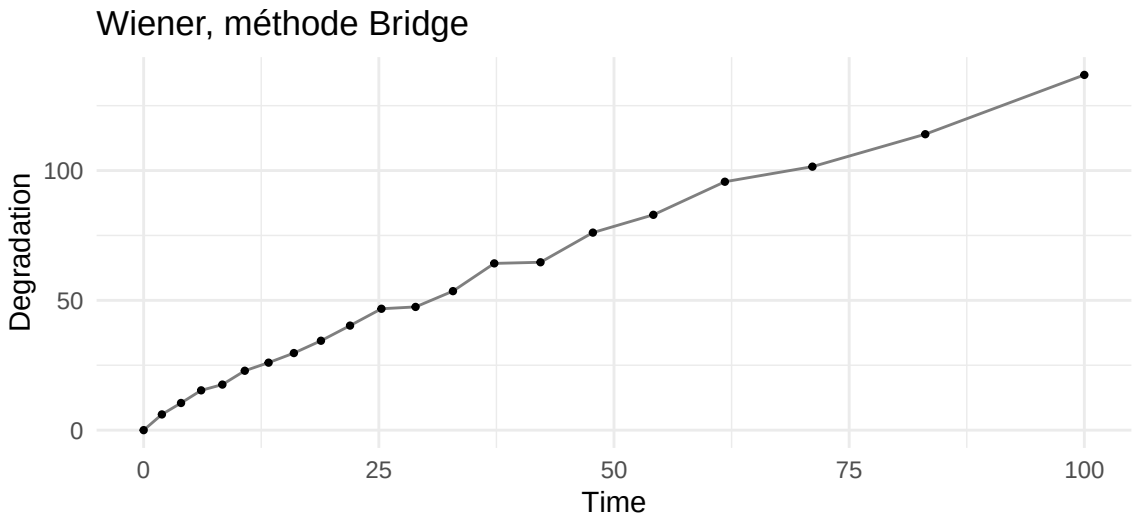
```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  1.940035  6.050583
## [3,]  3.974743 10.460403
## [4,]  6.113840 15.328770
```

```
## [5,]  8.368618 17.550983
## [6,] 10.752309 22.878592
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  6.051  4.410  4.868  2.222  5.328  3.117  3.696  4.745  5.835  6.486
## [11]  0.714  6.077 10.689  0.432 11.424  6.855 12.742  5.814 12.497 22.848
```

```
plot_traj(x, "Wiener, méthode Bridge")
```



Modèle 3 : inverse gaussien

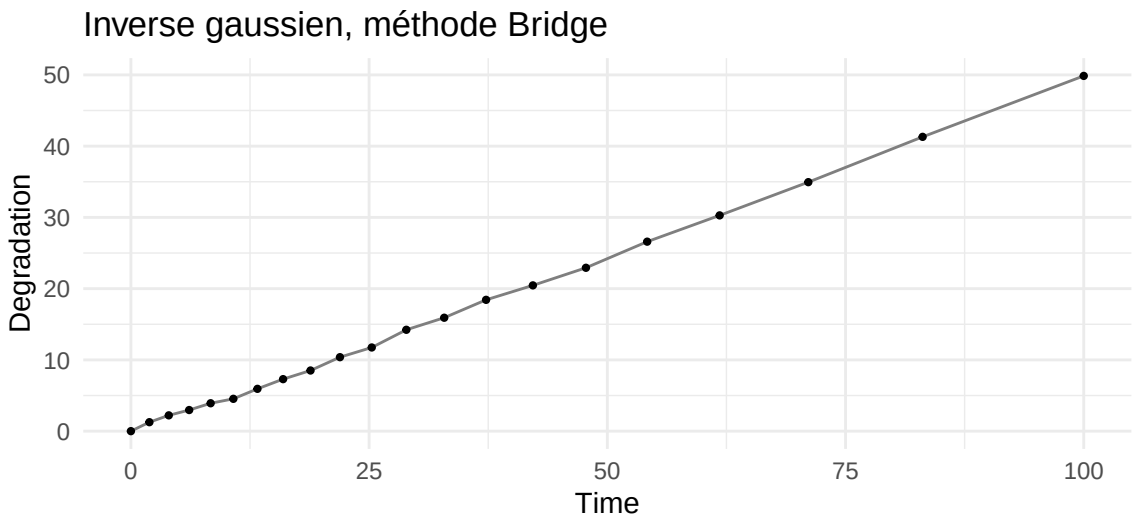
```
set.seed(433)
x <- sim_ig(t3, mu = 0.5, eta = 2, method = "Bridge")
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  1.940035 1.257262
## [3,]  3.974743 2.215204
## [4,]  6.113840 2.962286
## [5,]  8.368618 3.914138
## [6,] 10.752309 4.534792
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.257 0.958 0.747 0.952 0.621 1.402 1.360 1.211 1.873 1.359 2.484 1.690
## [13] 2.513 2.029 2.469 3.665 3.689 4.663 6.358 8.555
```

```
plot_traj(x, "Inverse gaussien, méthode Bridge")
```



Modèle 4 : variance-gamma

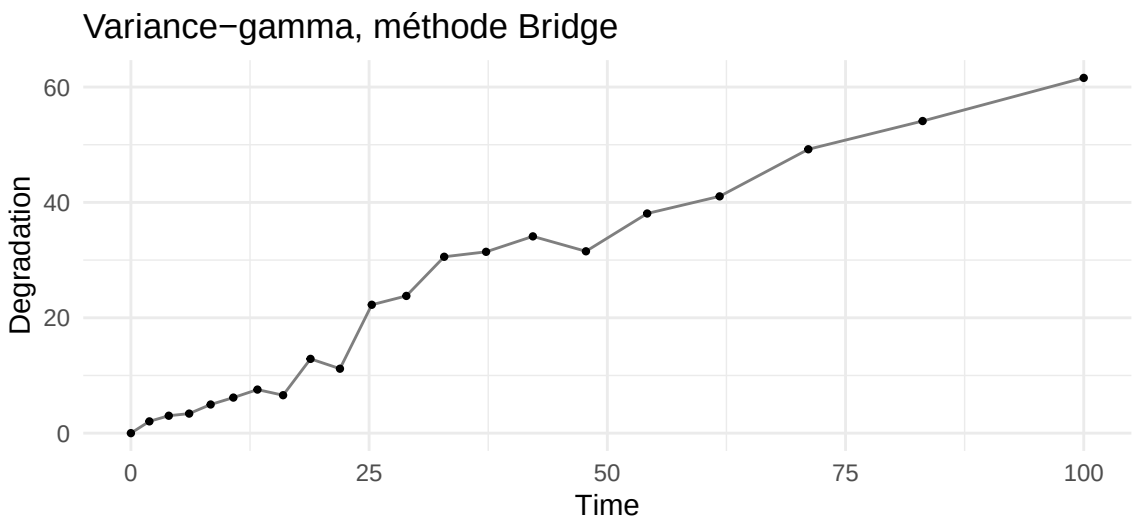
```
set.seed(434)
x <- sim_vg(t3, theta = 0.5, sigma = 1.5, nu = 2, method = "Bridge")
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  1.940035 2.041241
## [3,]  3.974743 3.017546
## [4,]  6.113840 3.387751
## [5,]  8.368618 4.965831
## [6,] 10.752309 6.163499
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.041  0.976  0.370  1.578  1.198  1.387 -0.976  6.299 -1.702 11.084
## [11]  1.534  6.787  0.856  2.679 -2.586  6.551  2.973  8.162  4.896  7.477
```

```
plot_traj(x, "Variance-gamma, méthode Bridge")
```



Modèle 5 : Meixner

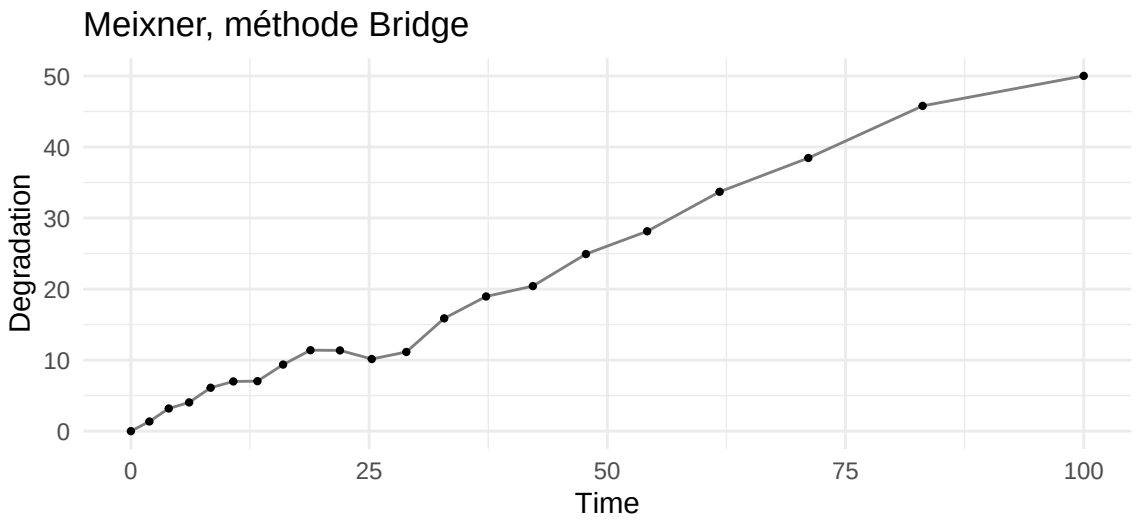
```
set.seed(435)
x <- sim_meixner(t3, a = 1, b = 0.3, d = 1.5, m = 0.2, method = "Bridge")
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  1.940035 1.360412
## [3,]  3.974743 3.189150
## [4,]  6.113840 4.042679
## [5,]  8.368618 6.110828
## [6,] 10.752309 6.997767
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.360  1.829  0.854  2.068  0.887  0.039  2.334  2.025 -0.029 -1.212
## [11]  0.991  4.732  3.081  1.464  4.509  3.206  5.561  4.756  7.338  4.222
```

```
plot_traj(x, "Meixner, méthode Bridge")
```


Meixner, méthode Bridge
Degradation
Time
0
10
20
30
40
50
0
25
50
75
100

Cas 4 : une trajectoire, mixtes, beaucoup au début et à la fin

Les inspections sont celles du cas 4 de la partie I.

Modèle 1 : gamma

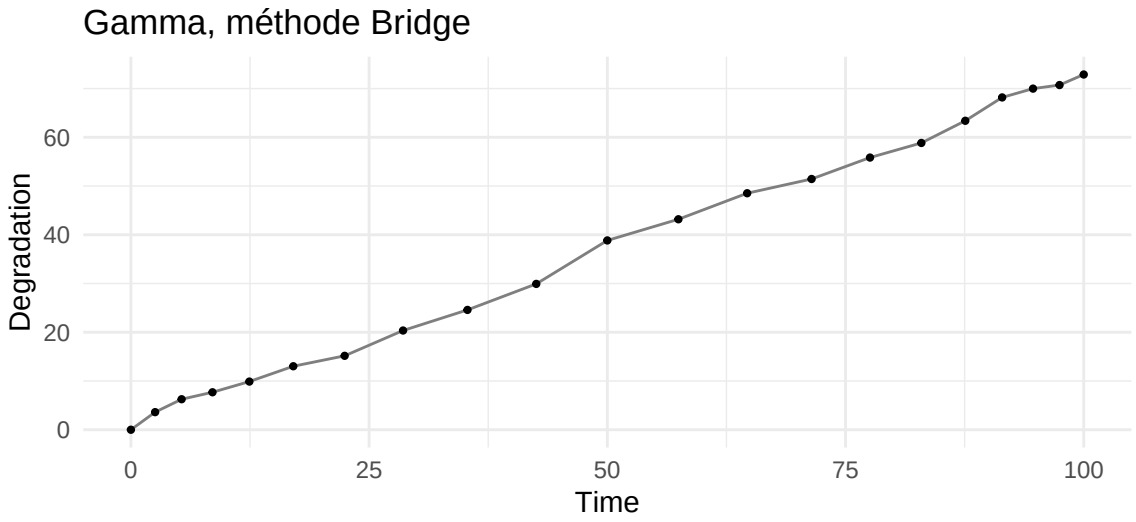
```
set.seed(441)
x <- sim_gamma(t4, a = 2, b = 3, method = "Bridge")
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  2.540921  3.611674
## [3,]  5.322554  6.260762
## [4,]  8.562047  7.690060
## [5,] 12.431733  9.883583
## [6,] 17.042253 13.026928
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.612 2.649 1.429 2.194 3.143 2.140 5.192 4.225 5.337 8.919 4.343 5.339
## [13] 2.924 4.398 2.998 4.547 4.792 1.806 0.735 2.164
```

```
plot_traj(x, "Gamma, méthode Bridge")
```



Modèle 2 : Wiener

```
set.seed(442)
x <- sim_wiener(t4, mu = 1.5, sigma = 2, method = "Bridge")
head(x)
```

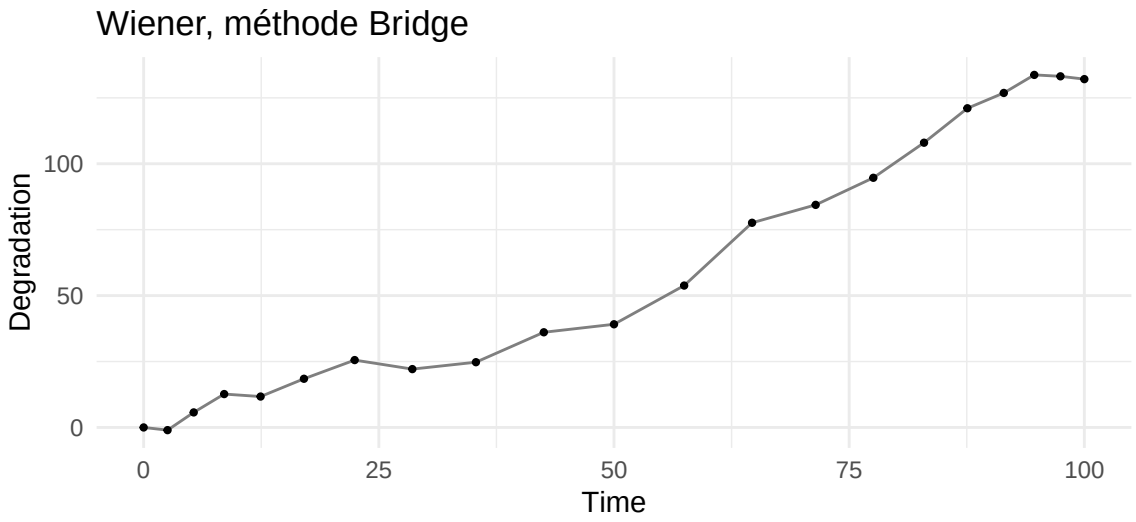
```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  2.540921 -1.022307
## [3,]  5.322554  5.699086
## [4,]  8.562047 12.658174
```

```
## [5,] 12.431733 11.718798
## [6,] 17.042253 18.459034
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -1.022  6.721  6.959 -0.939  6.740  7.090 -3.423  2.620 11.346  3.037
## [11] 14.681 23.811  6.795 10.269 13.305 13.061  5.809  6.845 -0.537 -1.065
```

```
plot_traj(x, "Wiener, méthode Bridge")
```



Modèle 3 : inverse gaussien

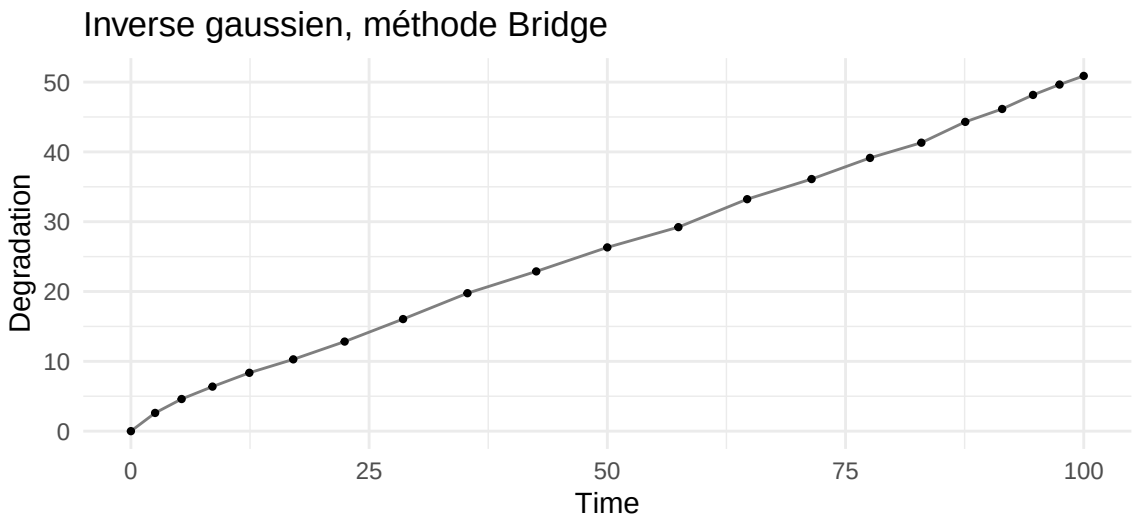
```
set.seed(443)
x <- sim_ig(t4, mu = 0.5, eta = 2, method = "Bridge")
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  2.540921  2.608959
## [3,]  5.322554  4.600284
## [4,]  8.562047  6.377572
## [5,] 12.431733  8.357797
## [6,] 17.042253 10.282031
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.609 1.991 1.777 1.980 1.924 2.546 3.224 3.707 3.121 3.439 2.900 4.005
## [13] 2.883 3.029 2.180 2.982 1.849 2.011 1.484 1.240
```

```
plot_traj(x, "Inverse gaussien, méthode Bridge")
```



Modèle 4 : variance-gamma

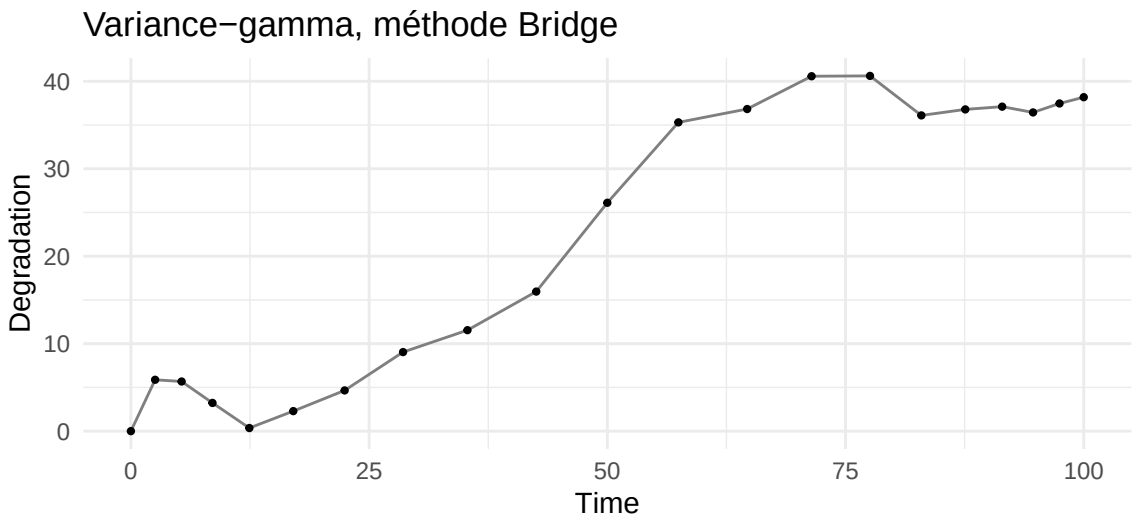
```
set.seed(444)
x <- sim_vg(t4, theta = 0.5, sigma = 1.5, nu = 2, method = "Bridge")
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  2.540921 5.8729629
## [3,]  5.322554 5.6758847
## [4,]  8.562047 3.2247562
## [5,] 12.431733 0.3556456
## [6,] 17.042253 2.2899966
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.873 -0.197 -2.451 -2.869  1.934  2.365  4.384  2.505  4.413 10.154
## [11]  9.194  1.525  3.749  0.038 -4.516  0.684  0.315 -0.661  1.024  0.719
```

```
plot_traj(x, "Variance-gamma, méthode Bridge")
```



Modèle 5 : Meixner

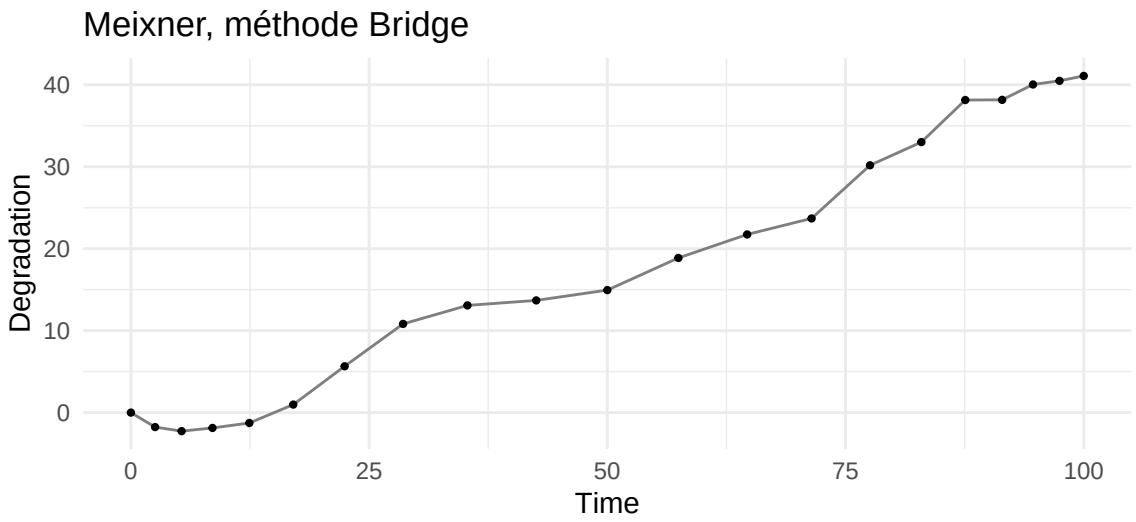
```
set.seed(445)
x <- sim_meixner(t4, a = 1, b = 0.3, d = 1.5, m = 0.2, method = "Bridge")
head(x)
```

```
##              t         X1
## [1,]  0.000000  0.0000000
## [2,]  2.540921 -1.7670458
## [3,]  5.322554 -2.2627026
## [4,]  8.562047 -1.8730068
## [5,] 12.431733 -1.2700594
## [6,] 17.042253  0.9794322
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -1.767 -0.496  0.390  0.603  2.249  4.674  5.167  2.259  0.611  1.258
## [11]  3.924  2.866  1.955  6.486  2.829  5.133  0.024  1.873  0.445  0.602
```

```
plot_traj(x, "Meixner, méthode Bridge")
```


Meixner, méthode Bridge
Degradation
40
30
20
10
0
0
25
50
75
100
Time

Cas 5 : une trajectoire, mixtes, beaucoup au milieu

Les inspections sont celles du cas 5 de la partie I.

Modèle 1 : gamma

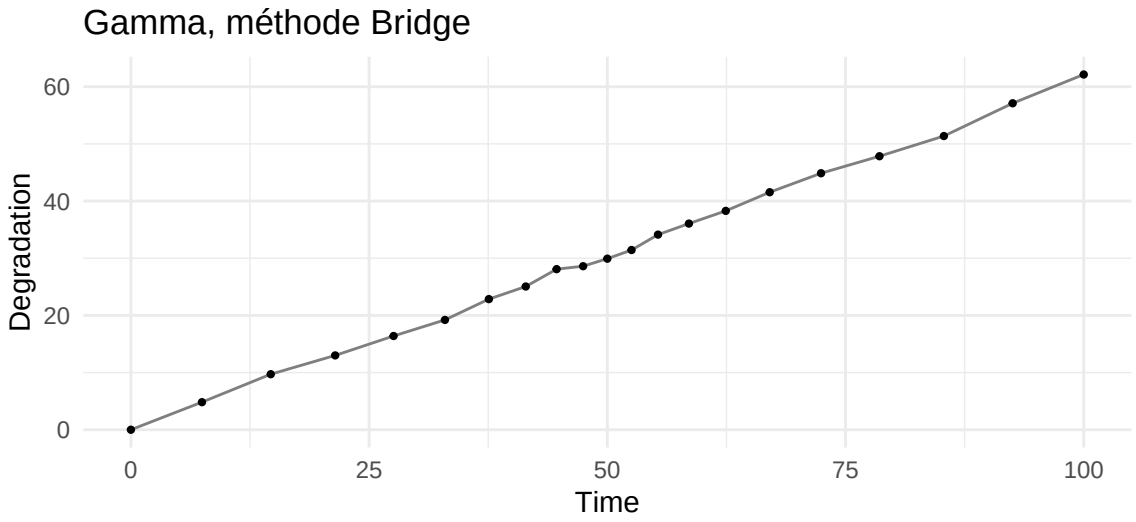
```
set.seed(451)
x <- sim_gamma(t5, a = 2, b = 3, method = "Bridge")
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  4.824765
## [3,] 14.677446  9.715350
## [4,] 21.437953 13.000505
## [5,] 27.568267 16.396633
## [6,] 32.957747 19.209455
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 4.825 4.891 3.285 3.396 2.813 3.639 2.204 3.035 0.517 1.318 1.512 2.696
## [13] 1.930 2.222 3.263 3.321 2.976 3.530 5.733 5.030
```

```
plot_traj(x, "Gamma, méthode Bridge")
```



Modèle 2 : Wiener

```
set.seed(452)
x <- sim_wiener(t5, mu = 1.5, sigma = 2, method = "Bridge")
head(x)
```

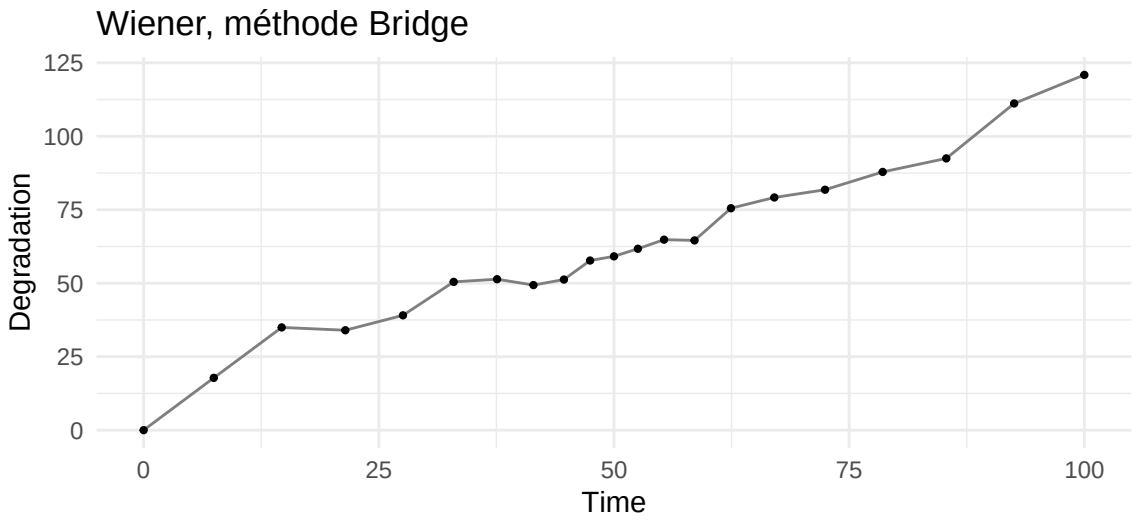
```
##              t       X1
## [1,]  0.000000  0.00000
## [2,]  7.459079 17.79564
## [3,] 14.677446 34.95383
## [4,] 21.437953 33.99292
```

```
## [5,] 27.568267 39.07028
## [6,] 32.957747 50.45291
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 17.796 17.158 -0.961  5.077 11.383  0.906 -2.000  1.887  6.476  1.420
## [11]  2.583  3.090 -0.253 10.957  3.659  2.633  6.058  4.601 18.698  9.721
```

```
plot_traj(x, "Wiener, méthode Bridge")
```



Modèle 3 : inverse gaussien

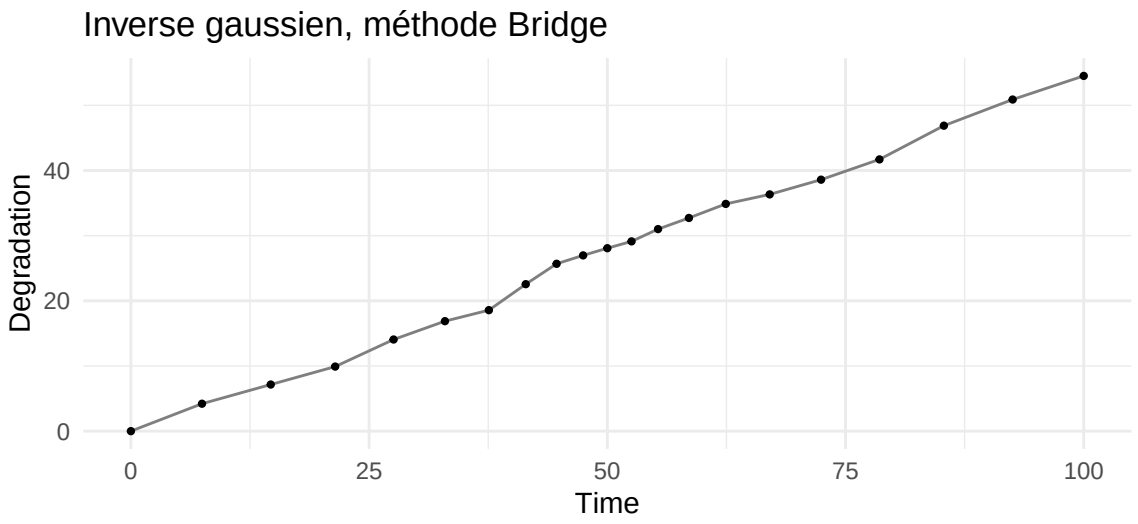
```
set.seed(453)
x <- sim_ig(t5, mu = 0.5, eta = 2, method = "Bridge")
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  4.208133
## [3,] 14.677446  7.153823
## [4,] 21.437953  9.908252
## [5,] 27.568267 14.063009
## [6,] 32.957747 16.876762
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 4.208 2.946 2.754 4.155 2.814 1.689 3.989 3.128 1.290 1.103 1.048 1.882
## [13] 1.711 2.157 1.460 2.260 3.109 5.176 4.014 3.626
```

```
plot_traj(x, "Inverse gaussien, méthode Bridge")
```



Modèle 4 : variance-gamma

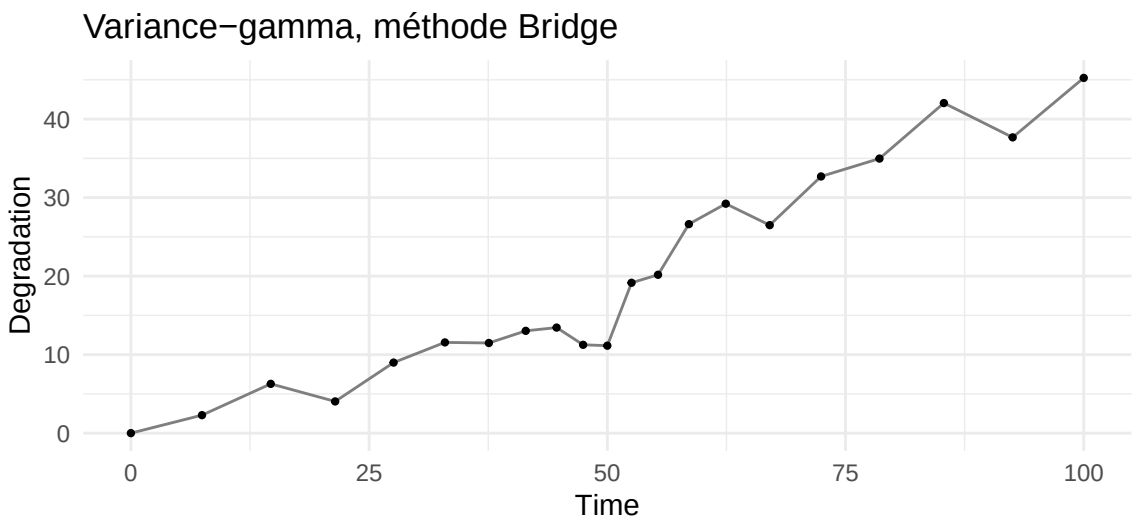
```
set.seed(454)
x <- sim_vg(t5, theta = 0.5, sigma = 1.5, nu = 2, method = "Bridge")
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  2.289563
## [3,] 14.677446  6.275705
## [4,] 21.437953  4.037208
## [5,] 27.568267  8.986054
## [6,] 32.957747 11.556520
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.290  3.986 -2.238  4.949  2.570 -0.074  1.547  0.415 -2.195 -0.110
## [11]  8.013  1.019  6.450  2.599 -2.728  6.200  2.276  7.073 -4.372  7.575
```

```
plot_traj(x, "Variance-gamma, méthode Bridge")
```



Modèle 5 : Meixner

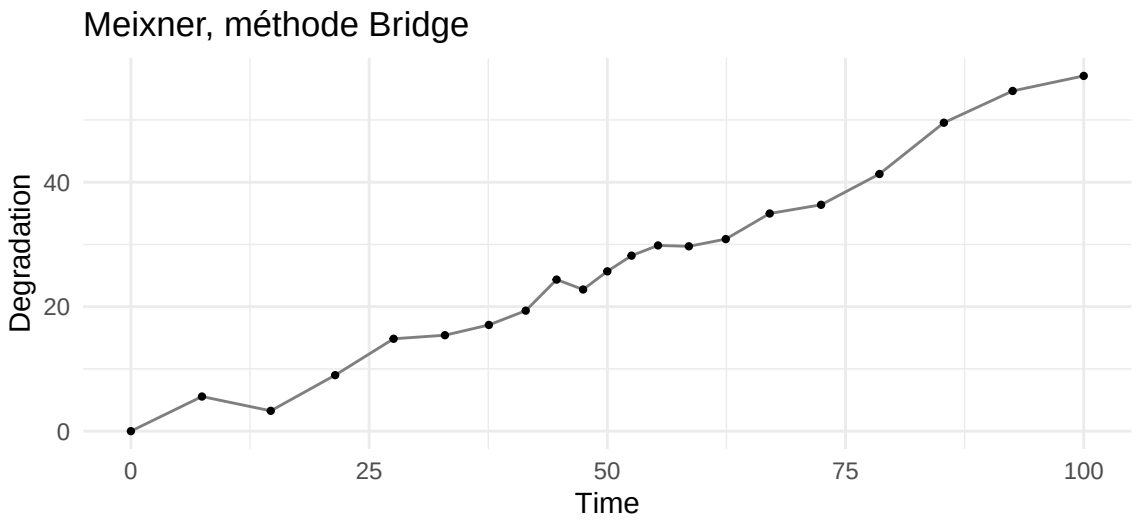
```
set.seed(455)
x <- sim_meixner(t5, a = 1, b = 0.3, d = 1.5, m = 0.2, method = "Bridge")
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  5.559707
## [3,] 14.677446  3.262350
## [4,] 21.437953  8.995337
## [5,] 27.568267 14.841616
## [6,] 32.957747 15.413291
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.560 -2.297  5.733  5.846  0.572  1.647  2.297  4.991 -1.591  2.919
## [11]  2.520  1.640 -0.138  1.159  4.113  1.397  4.951  8.237  5.105  2.418
```

```
plot_traj(x, "Meixner, méthode Bridge")
```


Meixner, méthode Bridge
Degradation
0
20
40
0
25
50
75
100
Time

Cas 6 : 5 trajectoires, inspections régulières

Les inspections sont celles du cas 1 de la partie II.

Modèle 1 : gamma

```
set.seed(511)
x <- sim_gamma(p1, a = 2, b = 3, method = "Bridge")
head(x[, 1:4])
```

```
##       t        X1        X2        X3
## [1,]  0  0.000000  0.000000  0.000000
## [2,]  5  1.923484  5.227193  3.792094
## [3,] 10  6.389738  7.759845  5.281737
## [4,] 15  8.754391 10.101173  7.681539
## [5,] 20 11.607538 14.234191 10.211011
## [6,] 25 14.364046 17.631993 12.909220
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.923 4.466 2.365 2.853 2.757 2.809 5.280 2.123 2.597 2.849 3.533 4.995
## [13] 3.462 2.688 3.082 3.837 2.911 5.168 2.168 4.743
```

```
plot_traj(x, "Gamma, méthode Bridge : 5 trajectoires")
```



Modèle 2 : Wiener

```
set.seed(512)
x <- sim_wiener(p1, mu = 1.5, sigma = 2, method = "Bridge")
head(x[, 1:4])
```

```
##       t        X1        X2        X3
## [1,]  0  0.000000  0.000000  0.000000
## [2,]  5  8.630464  5.300963  3.777183
## [3,] 10 24.741772 11.436122 15.789041
## [4,] 15 29.498452 10.079707 27.931407
```

```
## [5,] 20 32.167964 14.431923 34.579607
## [6,] 25 38.105141 17.544041 38.545649
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  8.630 16.111  4.757  2.670  5.937  1.899  9.004 11.869  7.120  3.002
## [11]  8.635  8.837  4.363  1.139  5.647  8.658 -0.617 15.038 11.095 10.377
```

```
plot_traj(x, "Wiener, méthode Bridge : 5 trajectoires")
```



Modèle 3 : inverse gaussien

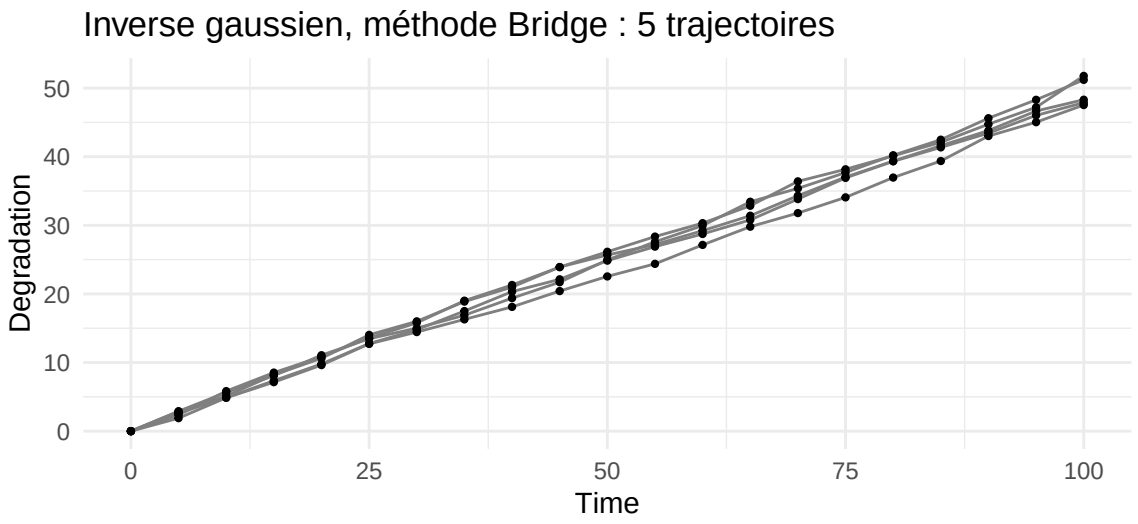
```
set.seed(513)
x <- sim_ig(p1, mu = 0.5, eta = 2, method = "Bridge")
head(x[, 1:4])
```

```
##       t        X1        X2        X3
## [1,]  0  0.000000  0.000000  0.000000
## [2,]  5  2.621690  2.900899  2.352945
## [3,] 10  4.879504  5.640333  5.831773
## [4,] 15  7.351249  8.094375  8.533054
## [5,] 20  9.794688 11.067981 10.833384
## [6,] 25 12.762827 13.472206 13.587171
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.622 2.258 2.472 2.443 2.968 2.024 2.716 2.833 1.797 2.726 2.033 1.848
## [13] 2.055 3.032 3.089 2.415 2.030 2.083 2.583 1.850
```

```
plot_traj(x, "Inverse gaussien, méthode Bridge : 5 trajectoires")
```



Modèle 4 : variance-gamma

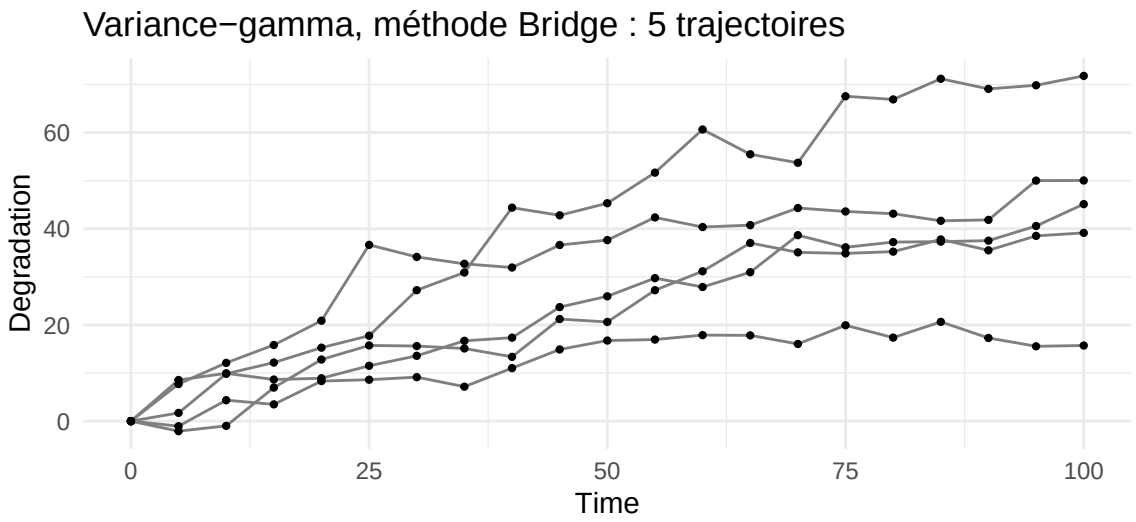
```
set.seed(514)
x <- sim_vg(p1, theta = 0.5, sigma = 1.5, nu = 2, method = "Bridge")
head(x[, 1:4])
```

```
##       t        X1        X2        X3
## [1,]  0  0.000000  0.000000  0.000000
## [2,]  5  8.554677  7.716347  1.710706
## [3,] 10  9.940078 12.114261  9.851185
## [4,] 15  8.652869 15.835531 12.177207
## [5,] 20  8.914483 20.887874 15.276886
## [6,] 25 11.512760 36.652946 17.759715
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  8.555  1.385 -1.287  0.262  2.598  2.070  3.120  0.653  6.358  2.245
## [11]  3.777 -1.844  3.077  7.700 -2.527  1.091  0.092  0.207  3.045  4.548
```

```
plot_traj(x, "Variance-gamma, méthode Bridge : 5 trajectoires")
```



Modèle 5 : Meixner

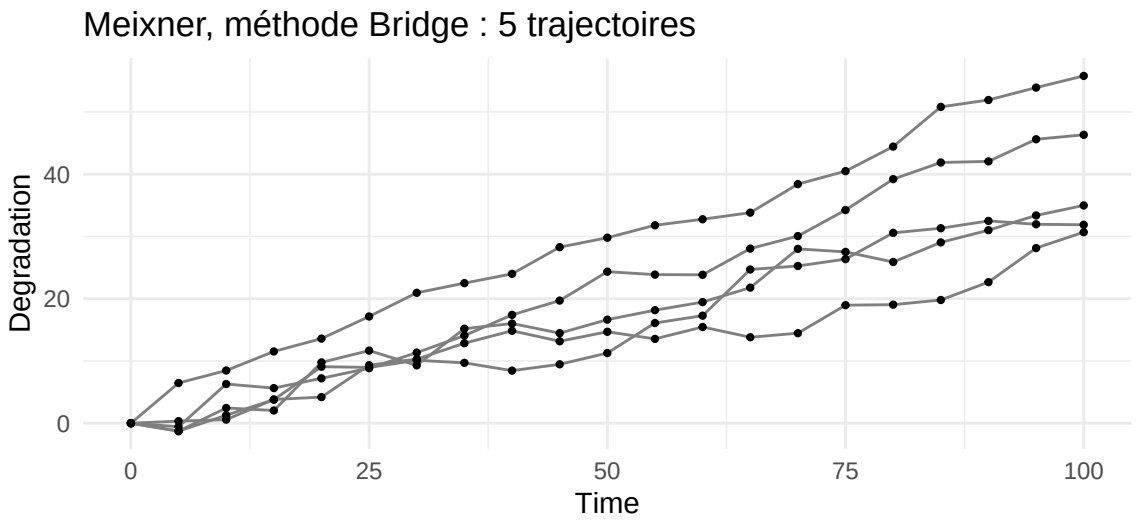
```
set.seed(515)
x <- sim_meixner(p1, a = 1, b = 0.3, d = 1.5, m = 0.2, method = "Bridge")
head(x[, 1:4])
```

```
##       t        X1        X2        X3
## [1,]  0 0.0000000  0.000000  0.000000
## [2,]  5 0.3348920 -1.263270 -1.270379
## [3,] 10 0.5799891  1.288124  2.449904
## [4,] 15 3.8120561  3.789686  2.047058
## [5,] 20 4.1974477  9.078235  9.775364
## [6,] 25 9.3261268  8.969583 11.665629
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.335  0.245  3.232  0.385  5.129  0.969  2.559  1.996 -1.681  1.521
## [11] -1.146  1.915 -1.646  0.653  4.499  0.081  0.749  2.873  5.467  2.558
```

```
plot_traj(x, "Meixner, méthode Bridge : 5 trajectoires")
```


Meixner, méthode Bridge : 5 trajectoires
Degradation
40
20
0
0
25
50
75
100
Time

Cas 7 : 5 trajectoires, pas décroissants

Les inspections sont celles du cas 2 de la partie II.

Modèle 1 : gamma

```
set.seed(521)
x <- sim_gamma(p2, a = 2, b = 3, method = "Bridge")
head(x[, 1:4])
```

```
##             t        X1       X2       X3
## [1,]  0.00000  0.000000  0.00000  0.00000
## [2,] 16.90921  8.850235 12.55949 11.79935
## [3,] 28.90648 17.460765 19.11553 20.06106
## [4,] 38.21229 23.732171 26.01877 24.23769
## [5,] 45.81569 26.730894 33.93502 30.29443
## [6,] 52.24427 31.969405 37.24288 34.04359
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 8.850 8.611 6.271 2.999 5.239 4.100 3.826 3.823 2.337 4.278 3.671 3.494
## [13] 1.016 5.203 1.225 2.644 1.500 2.180 1.095 1.226
```

```
plot_traj(x, "Gamma, méthode Bridge : 5 trajectoires")
```



Modèle 2 : Wiener

```
set.seed(522)
x <- sim_wiener(p2, mu = 1.5, sigma = 2, method = "Bridge")
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.00000
## [2,] 16.90921 25.20640 10.85472 22.61953
## [3,] 28.90648 42.32451 43.86137 41.50662
## [4,] 38.21229 66.65384 63.28238 48.39241
```

```
## [5,] 45.81569 79.61924 74.37585 67.07978
## [6,] 52.24427 81.96543 83.99152 73.30906
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 25.206 17.118 24.329 12.965  2.346 13.989  5.262  6.966  9.036  9.190
## [11]  4.683  4.422  9.869  8.539  4.617  3.385 -0.583  2.100 -1.713 -1.323
```

```
plot_traj(x, "Wiener, méthode Bridge : 5 trajectoires")
```



Modèle 3 : inverse gaussien

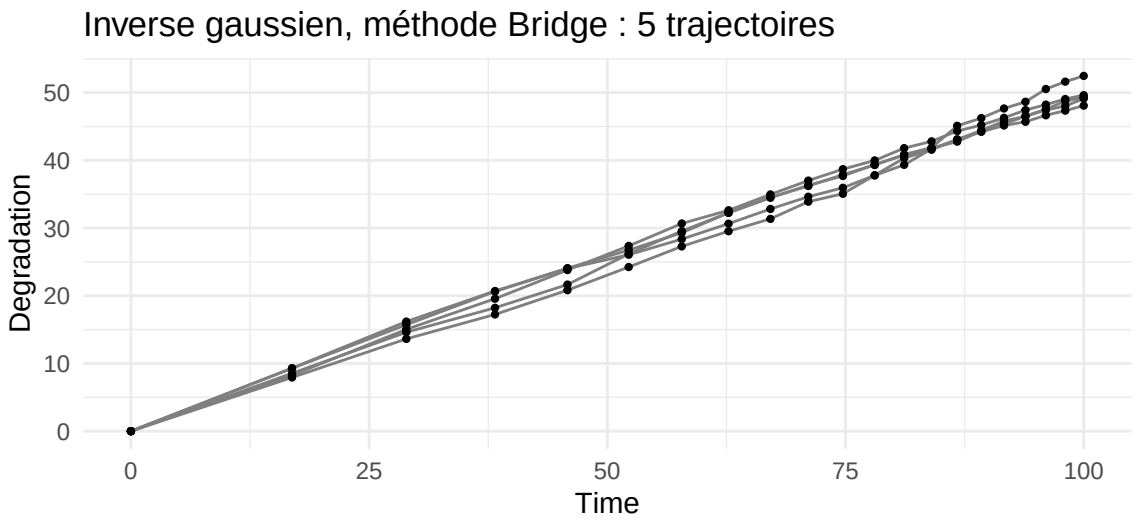
```
set.seed(523)
x <- sim_ig(p2, mu = 0.5, eta = 2, method = "Bridge")
head(x[, 1:4])
```

```
##             t        X1        X2        X3
## [1,]  0.00000  0.000000  0.000000  0.000000
## [2,] 16.90921  9.302803  8.541905  9.281964
## [3,] 28.90648 16.185305 14.609532 15.719741
## [4,] 38.21229 20.699407 18.219412 20.626694
## [5,] 45.81569 24.072384 21.650821 24.027608
## [6,] 52.24427 26.742643 26.188765 26.063094
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 9.303 6.883 4.514 3.373 2.670 2.562 2.944 2.283 1.682 1.677 1.418 1.562
## [13] 1.008 3.225 1.139 1.422 0.987 1.885 1.087 0.855
```

```
plot_traj(x, "Inverse gaussien, méthode Bridge : 5 trajectoires")
```



Modèle 4 : variance-gamma

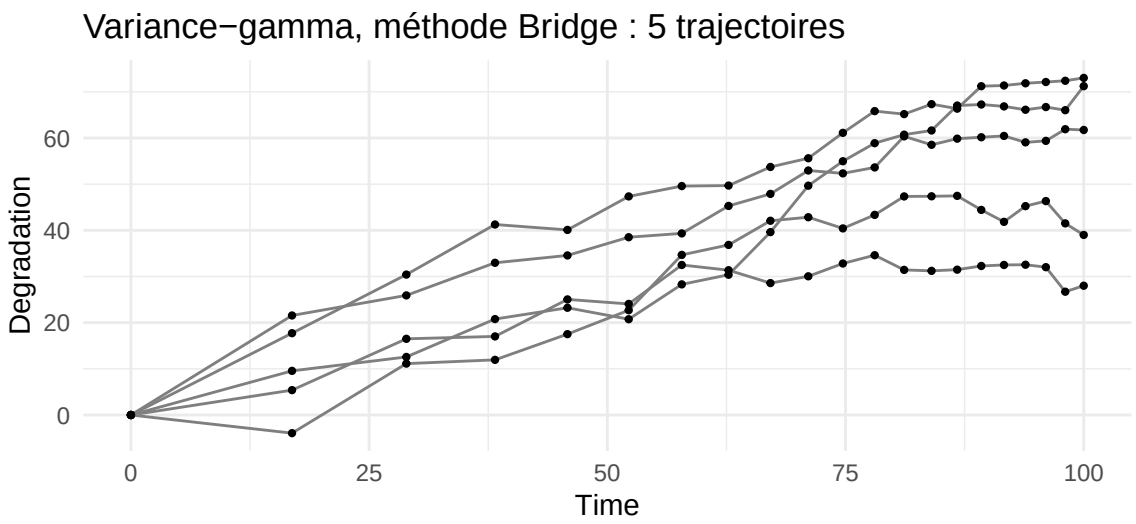
```
set.seed(524)
x <- sim_vg(p2, theta = 0.5, sigma = 1.5, nu = 2, method = "Bridge")
head(x[, 1:4])
```

```
##             t        X1        X2        X3
## [1,]  0.00000  0.000000  0.000000  0.000000
## [2,] 16.90921  9.559005  5.385369 -3.929168
## [3,] 28.90648 12.571264 16.498486 11.129751
## [4,] 38.21229 20.776954 17.018698 11.941802
## [5,] 45.81569 23.224605 25.035872 17.513795
## [6,] 52.24427 20.746512 24.054909 22.693467
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  9.559  3.012  8.206  2.448 -2.478  7.548  2.082  9.236 10.058  5.297
## [11]  3.907  1.843  0.905  5.404  0.224 -0.410 -0.727  0.591 -0.682  5.201
```

```
plot_traj(x, "Variance-gamma, méthode Bridge : 5 trajectoires")
```



Modèle 5 : Meixner

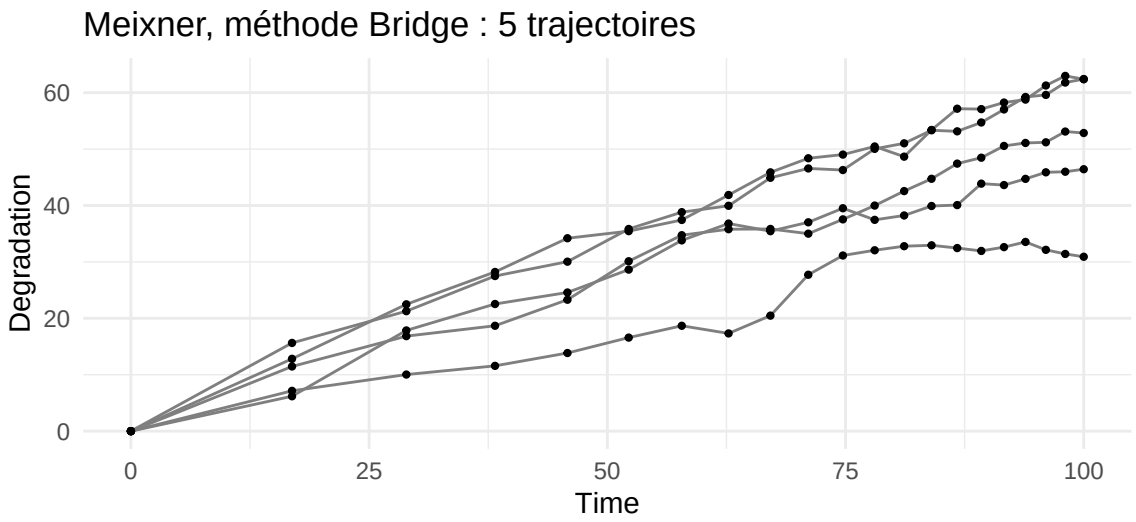
```
set.seed(525)
x <- sim_meixner(p2, a = 1, b = 0.3, d = 1.5, m = 0.2, method = "Bridge")
head(x[, 1:4])
```

```
##             t        X1       X2       X3
## [1,]  0.00000  0.000000  0.00000  0.00000
## [2,] 16.90921  6.181939 11.46190  7.14407
## [3,] 28.90648 17.851435 16.83948 10.03210
## [4,] 38.21229 22.536277 18.67842 11.57128
## [5,] 45.81569 24.577790 23.29189 13.84769
## [6,] 52.24427 28.634821 30.11715 16.58128
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  6.182 11.669  4.685  2.042  4.057  5.182  2.976 -1.347  1.574  2.491
## [11] -2.058  0.778  1.679  0.160  3.801 -0.248  1.096  1.160  0.106  0.443
```

```
plot_traj(x, "Meixner, méthode Bridge : 5 trajectoires")
```


Meixner, méthode Bridge : 5 trajectoires
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 8 : 5 trajectoires, pas croissants

Les inspections sont celles du cas 3 de la partie II.

Modèle 1 : gamma

```
set.seed(531)
x <- sim_gamma(p3, a = 2, b = 3, method = "Bridge")
head(x[, 1:4])
```

```
##              t       X1       X2       X3
## [1,]  0.000000 0.000000 0.000000 0.000000
## [2,]  1.940035 1.593415 0.855840 1.708145
## [3,]  3.974743 2.208597 1.454085 3.688863
## [4,]  6.113840 3.361383 2.242976 4.901654
## [5,]  8.368618 4.146268 3.752639 6.677950
## [6,] 10.752309 4.653457 6.030828 8.869401
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.593 0.615 1.153 0.785 0.507 1.306 0.355 1.841 2.064 2.100 1.827 2.630
## [13] 2.157 1.781 2.121 1.662 2.214 6.405 7.555 9.558
```

```
plot_traj(x, "Gamma, méthode Bridge : 5 trajectoires")
```



Modèle 2 : Wiener

```
set.seed(532)
x <- sim_wiener(p3, mu = 1.5, sigma = 2, method = "Bridge")
head(x[, 1:4])
```

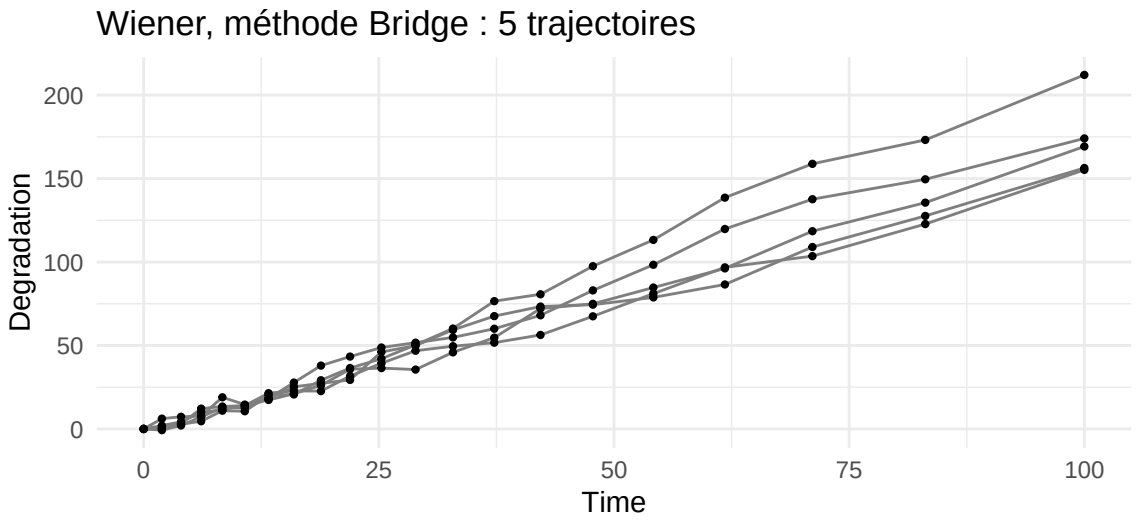
```
##              t          X1        X2         X3
## [1,]  0.000000  0.00000000  0.000000  0.0000000
## [2,]  1.940035  0.08412371  2.003715 -0.7128594
## [3,]  3.974743  2.82598841  4.275934  2.0759133
## [4,]  6.113840  4.61818629  9.945255  7.2620058
```

```
## [5,]  8.368618 10.94124975 11.447301 12.7475401
## [6,] 10.752309 10.60170861 14.212192 12.6402596
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.084  2.742  1.792  6.323 -0.340  9.579  0.495  8.506  7.293  5.584
## [11]  8.312  9.785 16.408  4.099 16.857 15.740 25.307 20.261 14.320 38.953
```

```
plot_traj(x, "Wiener, méthode Bridge : 5 trajectoires")
```



Modèle 3 : inverse gaussien

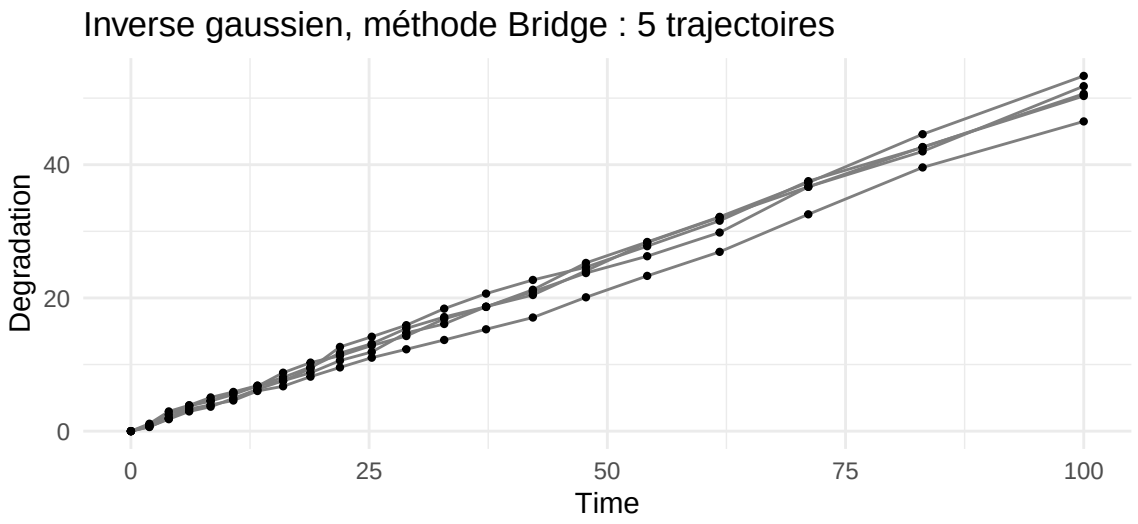
```
set.seed(533)
x <- sim_ig(p3, mu = 0.5, eta = 2, method = "Bridge")
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000 0.0000000 0.000000 0.0000000
## [2,]  1.940035 0.7442836 0.672474 0.6200534
## [3,]  3.974743 1.9259635 1.797640 1.9481482
## [4,]  6.113840 3.3161702 2.959427 3.7684788
## [5,]  8.368618 3.9460150 3.664520 4.5221940
## [6,] 10.752309 4.6008666 4.919258 5.4998812
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.744 1.182 1.390 0.630 0.655 1.429 0.703 1.445 1.392 1.458 1.264 1.396
## [13] 1.606 1.766 3.036 3.216 3.612 5.632 7.039 6.909
```

```
plot_traj(x, "Inverse gaussien, méthode Bridge : 5 trajectoires")
```



Modèle 4 : variance-gamma

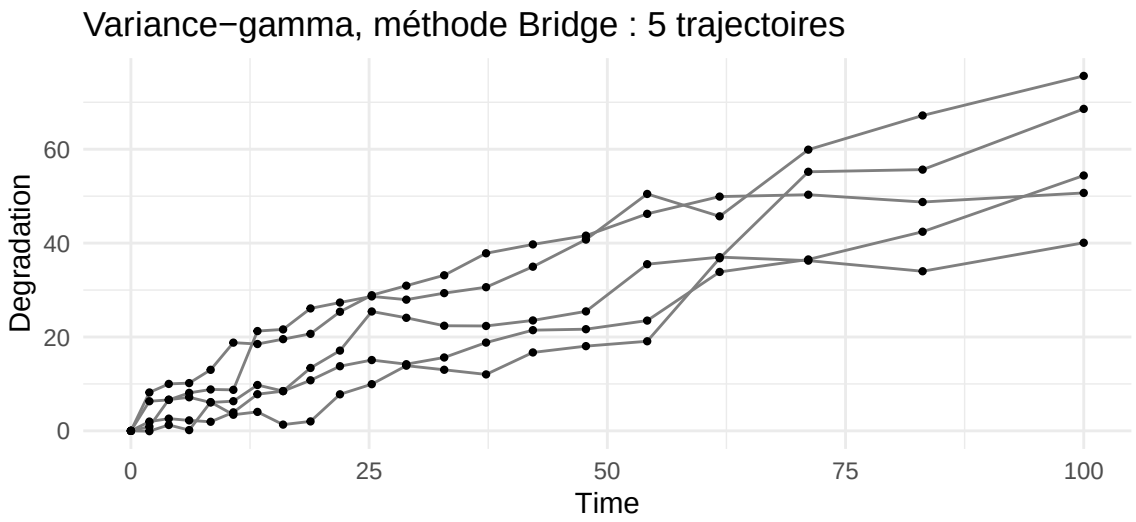
```
set.seed(534)
x <- sim_vg(p3, theta = 0.5, sigma = 1.5, nu = 2, method = "Bridge")
head(x[, 1:4])
```

```
##              t          X1        X2       X3
## [1,]  0.000000  0.00000000  0.000000 0.000000
## [2,]  1.940035 -0.05963254  8.180795 1.981404
## [3,]  3.974743  1.25218780 10.002406 2.603741
## [4,]  6.113840  0.16767655 10.173271 2.230321
## [5,]  8.368618  6.12547541 13.005860 1.936382
## [6,] 10.752309  3.43504229 18.796743 3.947320
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -0.060  1.312 -1.085  5.958 -2.690  0.604 -2.708  0.683  5.770  2.161
## [11]  3.940 -0.872 -0.989  4.680  1.346  1.036 17.682 18.426  0.457 12.939
```

```
plot_traj(x, "Variance-gamma, méthode Bridge : 5 trajectoires")
```



Modèle 5 : Meixner

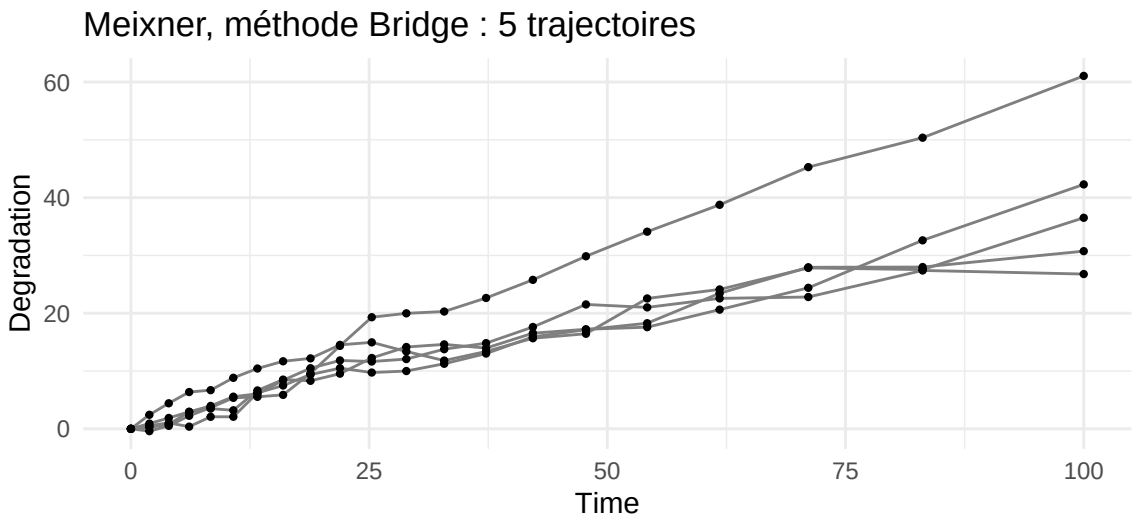
```
set.seed(535)
x <- sim_meixner(p3, a = 1, b = 0.3, d = 1.5, m = 0.2, method = "Bridge")
head(x[, 1:4])
```

```
##              t        X1       X2       X3
## [1,]  0.000000 0.0000000 0.000000 0.000000
## [2,]  1.940035 0.5535200 0.912742 2.428692
## [3,]  3.974743 1.0126826 1.876752 4.417595
## [4,]  6.113840 0.3761356 2.939767 6.363055
## [5,]  8.368618 2.0801818 3.959353 6.689141
## [6,] 10.752309 2.0868143 5.558063 8.825688
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.554  0.459 -0.637  1.704  0.007  4.146  1.262  1.889  1.112 -0.759
## [11]  0.260  1.258  1.769  2.884  1.239  1.122  5.176  4.505  0.037  2.751
```

```
plot_traj(x, "Meixner, méthode Bridge : 5 trajectoires")
```


Meixner, méthode Bridge : 5 trajectoires
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 9 : 5 trajectoires, mixtes, beaucoup au début et à la fin

Les inspections sont celles du cas 4 de la partie II.

Modèle 1 : gamma

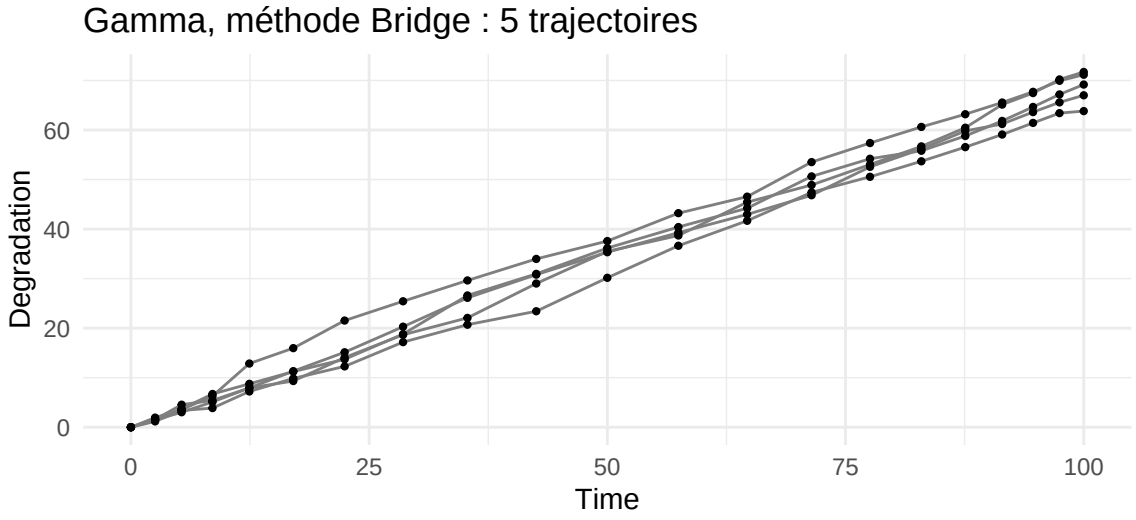
```
set.seed(541)
x <- sim_gamma(p4, a = 2, b = 3, method = "Bridge")
head(x[, 1:4])
```

```
##              t        X1        X2       X3
## [1,]  0.000000  0.000000  0.000000 0.000000
## [2,]  2.540921  1.940761  1.576777 1.177950
## [3,]  5.322554  3.555547  3.692252 3.385423
## [4,]  8.562047  6.300150  6.707820 3.861566
## [5,] 12.431733 12.856605  8.757016 7.221717
## [6,] 17.042253 15.965184 11.234889 9.844893
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.941 1.615 2.745 6.556 3.109 5.570 3.889 4.207 4.343 3.607 5.633 3.330
## [13] 6.980 3.846 3.255 2.561 2.365 2.158 2.233 1.218
```

```
plot_traj(x, "Gamma, méthode Bridge : 5 trajectoires")
```



Modèle 2 : Wiener

```
set.seed(542)
x <- sim_wiener(p4, mu = 1.5, sigma = 2, method = "Bridge")
head(x[, 1:4])
```

```
##              t        X1        X2         X3
## [1,]  0.000000  0.000000  0.000000  0.0000000
## [2,]  2.540921  2.990428  2.365571  0.9028729
## [3,]  5.322554 10.406900  4.764327  4.4478702
## [4,]  8.562047 16.821301  5.121398  3.8730080
```

```
## [5,] 12.431733 27.801905 17.392999 15.0643771
## [6,] 17.042253 35.636649 23.837251 21.4944076
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.990  7.416  6.414 10.981  7.835 13.162 18.763 11.216 11.769 16.653
## [11]  7.598 12.481 10.831 -0.017  7.019  7.546  6.407  9.576  7.533  4.876
```

```
plot_traj(x, "Wiener, méthode Bridge : 5 trajectoires")
```



Modèle 3 : inverse gaussien

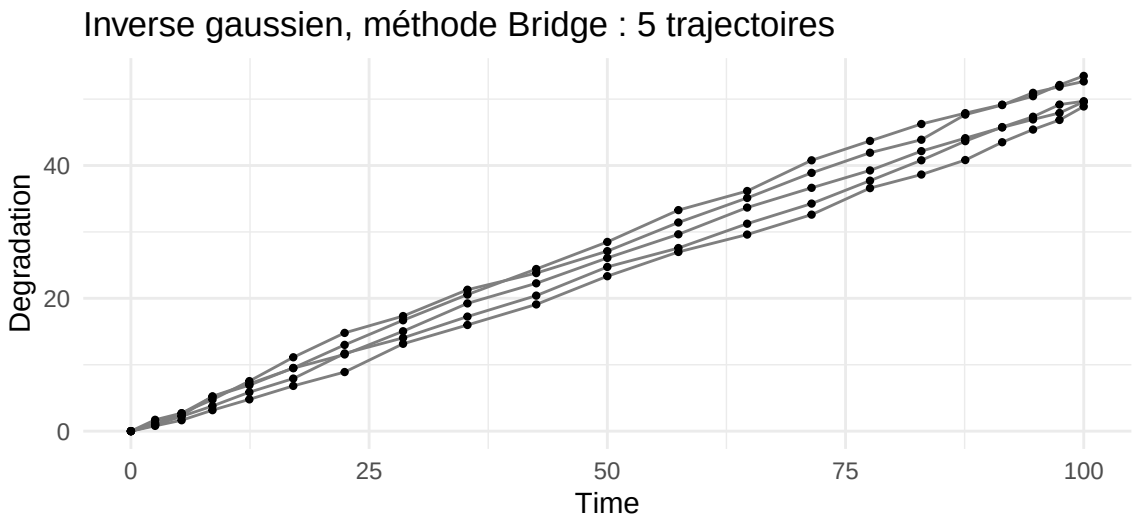
```
set.seed(543)
x <- sim_ig(p4, mu = 0.5, eta = 2, method = "Bridge")
head(x[, 1:4])
```

```
##              t       X1        X2        X3
## [1,]  0.000000 0.000000 0.0000000  0.000000
## [2,]  2.540921 1.227575 0.7931439  1.707584
## [3,]  5.322554 2.680013 1.6409722  2.705054
## [4,]  8.562047 5.190817 3.1602735  4.751396
## [5,] 12.431733 6.923978 4.7869655  7.532117
## [6,] 17.042253 9.517601 6.8125454 11.123545
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.228 1.452 2.511 1.733 2.594 3.472 3.708 3.868 3.847 4.079 4.792 2.876
## [13] 4.625 2.918 2.551 1.631 1.276 1.282 1.687 1.366
```

```
plot_traj(x, "Inverse gaussien, méthode Bridge : 5 trajectoires")
```



Modèle 4 : variance-gamma

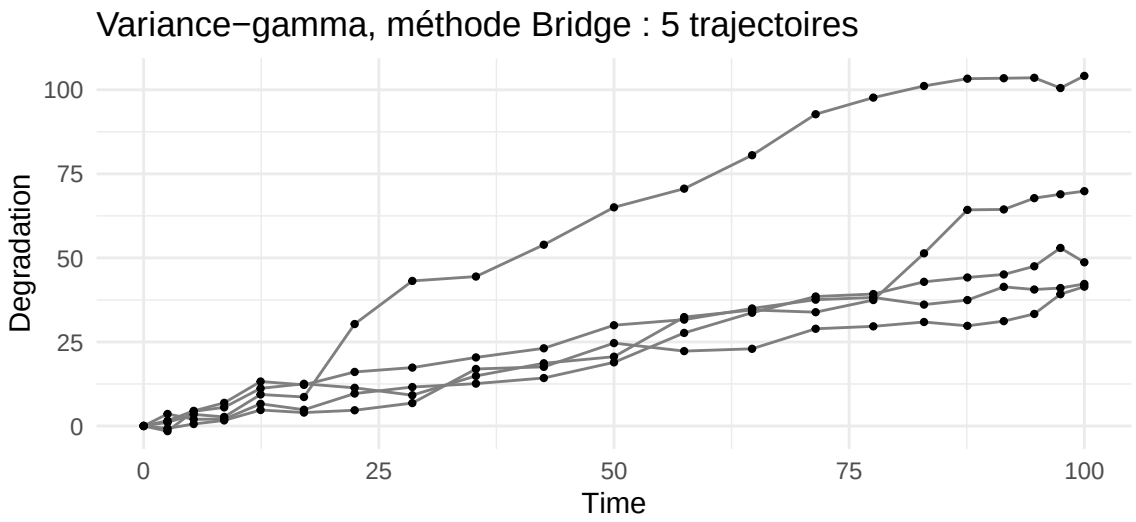
```
set.seed(544)
x <- sim_vg(p4, theta = 0.5, sigma = 1.5, nu = 2, method = "Bridge")
head(x[, 1:4])
```

```
##              t        X1        X2       X3
## [1,]  0.000000  0.000000  0.000000 0.000000
## [2,]  2.540921  1.562908 -1.535738 1.086260
## [3,]  5.322554  4.487151  4.373286 3.443775
## [4,]  8.562047  6.874411  5.532056 2.694894
## [5,] 12.431733 13.226729 11.193985 9.371615
## [6,] 17.042253 12.224249 12.547531 8.618425
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.563  2.924  2.387  6.352 -1.002  3.850  1.292  3.020  2.719  6.862
## [11]  1.669  3.321  2.613  0.662 -2.124  1.344  3.937 -0.800  0.442  1.216
```

```
plot_traj(x, "Variance-gamma, méthode Bridge : 5 trajectoires")
```



Modèle 5 : Meixner

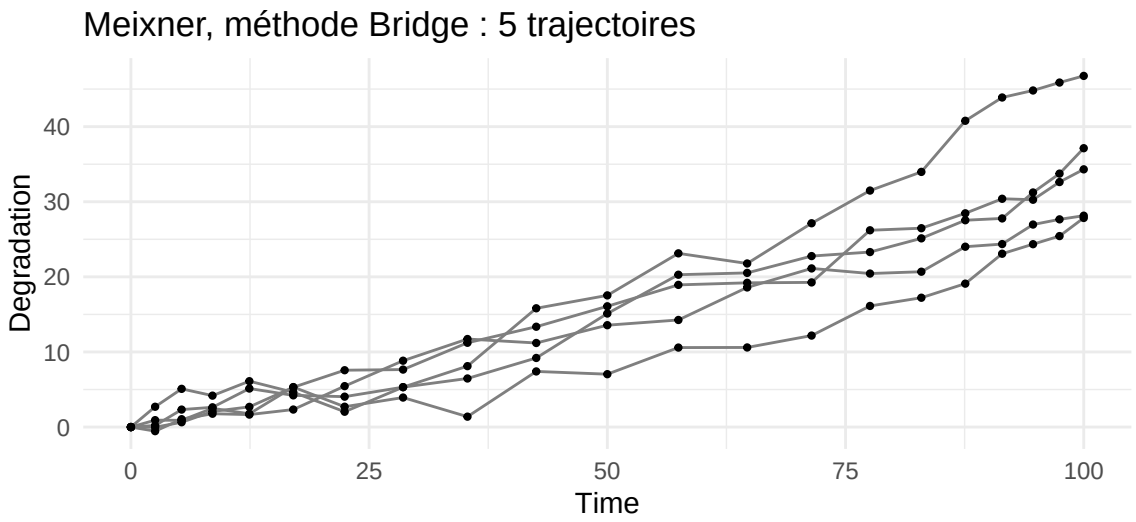
```
set.seed(545)
x <- sim_meixner(p4, a = 1, b = 0.3, d = 1.5, m = 0.2, method = "Bridge")
head(x[, 1:4])
```

```
##              t        X1          X2       X3
## [1,]  0.000000 0.0000000  0.00000000 0.000000
## [2,]  2.540921 0.2089684 -0.02018633 2.704348
## [3,]  5.322554 2.3280484  0.64136411 5.089662
## [4,]  8.562047 2.6319619  2.07979071 4.185797
## [5,] 12.431733 5.1320121  2.68809287 6.110201
## [6,] 17.042253 4.2237134  5.29722462 4.635469
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.209  2.119  0.304  2.500 -0.908 -0.175  1.264  2.800  7.703  1.718
## [11]  5.598 -1.349  5.357  4.351  2.486  6.805  3.103  0.933  1.056  0.884
```

```
plot_traj(x, "Meixner, méthode Bridge : 5 trajectoires")
```


Meixner, méthode Bridge : 5 trajectoires
Degradation
0
10
20
30
40
0
25
50
75
100
Time

Cas 10 : 5 trajectoires, mixtes, beaucoup au milieu

Les inspections sont celles du cas 5 de la partie II.

Modèle 1 : gamma

```
set.seed(551)
x <- sim_gamma(p5, a = 2, b = 3, method = "Bridge")
head(x[, 1:4])
```

```
##              t        X1        X2       X3
## [1,]  0.000000  0.000000  0.000000  0.00000
## [2,]  7.459079  5.063671  6.508734  5.85196
## [3,] 14.677446 11.259271  9.483939 10.06951
## [4,] 21.437953 17.328381 13.477721 13.93084
## [5,] 27.568267 22.692038 17.198616 18.21807
## [6,] 32.957747 25.675958 21.885855 21.63904
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 5.064 6.196 6.069 5.364 2.984 2.778 2.476 3.639 1.642 2.066 1.565 1.582
## [13] 1.746 4.453 2.220 2.772 5.493 3.379 4.646 4.385
```

```
plot_traj(x, "Gamma, méthode Bridge : 5 trajectoires")
```



Modèle 2 : Wiener

```
set.seed(552)
x <- sim_wiener(p5, mu = 1.5, sigma = 2, method = "Bridge")
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000  0.000000  0.00000  0.000000
## [2,]  7.459079  6.198118 14.09769  7.528178
## [3,] 14.677446  9.041830 27.68663 32.409574
## [4,] 21.437953 25.564524 34.08292 39.423504
```

```
## [5,] 27.568267 32.848039 46.88841 49.089781
## [6,] 32.957747 44.421071 54.60362 61.482914
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  6.198  2.844 16.523  7.284 11.573  5.569  8.352  2.004  9.175  2.665
## [11] -1.342  7.003  3.102  9.345 12.912  9.686 11.024  0.680 13.300  4.896
```

```
plot_traj(x, "Wiener, méthode Bridge : 5 trajectoires")
```



Modèle 3 : inverse gaussien

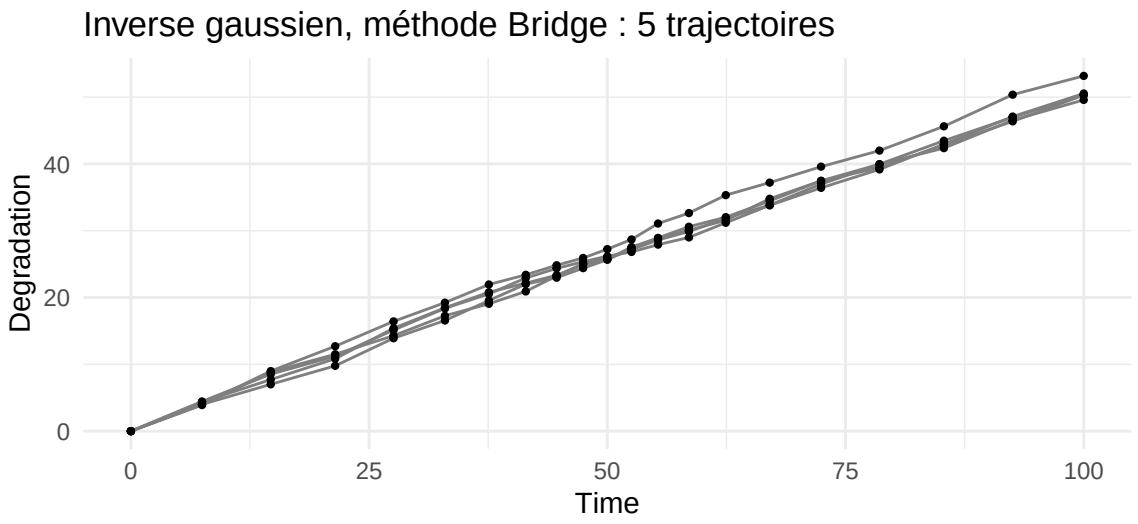
```
set.seed(553)
x <- sim_ig(p5, mu = 0.5, eta = 2, method = "Bridge")
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  7.459079  3.999548  4.363562  4.440128
## [3,] 14.677446  7.009505  7.714372  8.537071
## [4,] 21.437953  9.779341 10.845463 11.210881
## [5,] 27.568267 13.908552 15.398984 15.140182
## [6,] 32.957747 16.561040 18.462779 18.376759
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 4.000 3.010 2.770 4.129 2.652 2.988 2.463 0.959 1.430 1.253 1.690 1.538
## [13] 1.191 1.498 3.212 2.683 1.971 3.386 3.548 3.928
```

```
plot_traj(x, "Inverse gaussien, méthode Bridge : 5 trajectoires")
```



Modèle 4 : variance-gamma

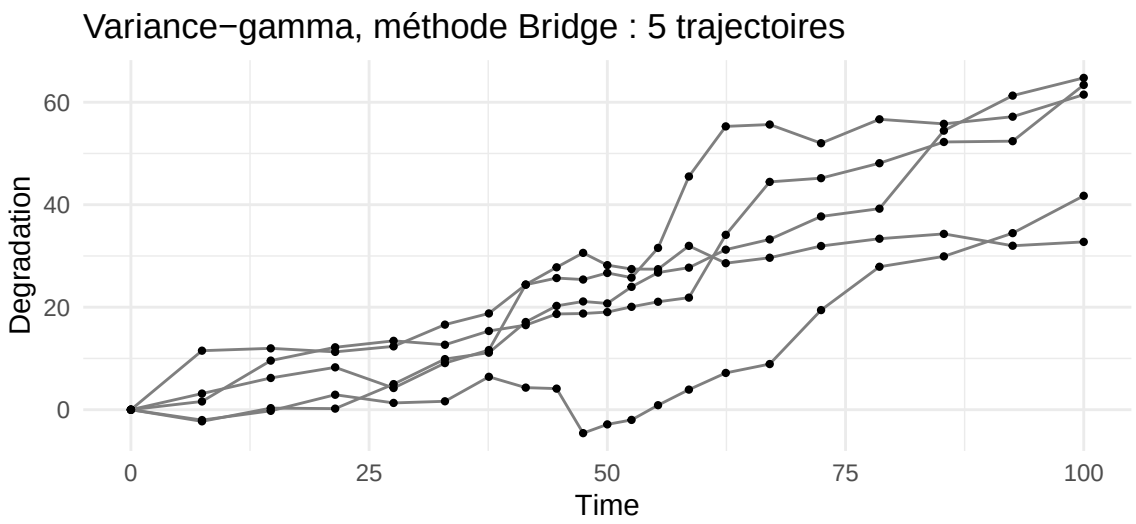
```
set.seed(554)
x <- sim_vg(p5, theta = 0.5, sigma = 1.5, nu = 2, method = "Bridge")
head(x[, 1:4])
```

```
##              t         X1       X2         X3
## [1,]  0.000000  0.0000000  0.00000  0.0000000
## [2,]  7.459079 -2.0245616 11.51484 -2.2726905
## [3,] 14.677446 -0.2188223 11.97109  0.3039157
## [4,] 21.437953  2.9184628 11.28425  0.2107721
## [5,] 27.568267  1.3208599 12.35878  4.9920393
## [6,] 32.957747  1.6406806 16.59837  9.8738825
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -2.025  1.806  3.137 -1.598  0.320  4.778 -2.112 -0.194 -8.687  1.708
## [11]  0.874  2.869  3.043  3.254  1.734 10.530  8.456  2.028  4.536  7.284
```

```
plot_traj(x, "Variance-gamma, méthode Bridge : 5 trajectoires")
```



Modèle 5 : Meixner

```
set.seed(555)
x <- sim_meixner(p5, a = 1, b = 0.3, d = 1.5, m = 0.2, method = "Bridge")
head(x[, 1:4])
```

```
##              t        X1         X2        X3
## [1,]  0.000000  0.000000  0.0000000  0.000000
## [2,]  7.459079  3.930971  0.4709974  2.564458
## [3,] 14.677446  8.605102  3.6718541  6.151996
## [4,] 21.437953 11.144342  5.0267061  9.612446
## [5,] 27.568267 13.449414  9.0364378 15.625212
## [6,] 32.957747 16.170417 14.0032023 15.369474
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.931  4.674  2.539  2.305  2.721  5.106  0.961  1.953  0.485  1.140
## [11] -1.355  0.753  1.875 -0.705  3.143  2.403  0.996  3.917 -0.458  3.379
```

```
plot_traj(x, "Meixner, méthode Bridge : 5 trajectoires")
```


Meixner, méthode Bridge : 5 trajectoires
Degradation
0
10
20
30
40
0
25
50
75
100
Time

Cas 11 : processus à échelle de temps

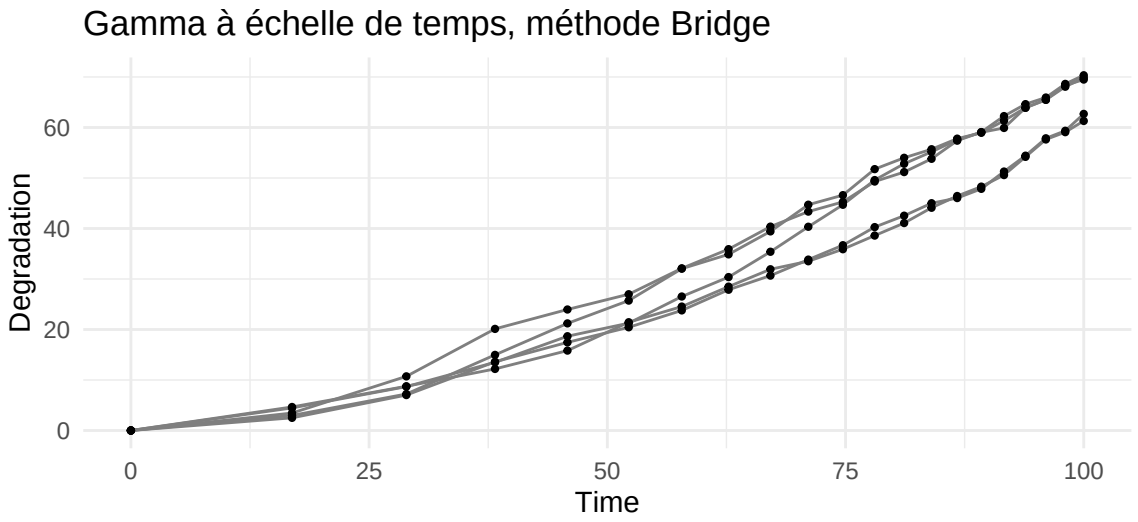
Les processus et la fonction `A` sont ceux de la partie III. Le pont est alors tiré sur l'échelle de temps `A(t)`.

Modèle 1 : gamma à échelle de temps

```
set.seed(421)
x <- sim_gamma(p2, b = 3, A = A, method = "Bridge")
head(x[, 1:4])
```

```
##             t        X1       X2        X3
## [1,]  0.00000  0.000000  0.00000  0.000000
## [2,] 16.90921  3.452498  4.46831  4.654112
## [3,] 28.90648 10.719155  8.68481  8.756487
## [4,] 38.21229 20.116906 13.50705 12.190685
## [5,] 45.81569 23.959211 18.66655 15.821413
## [6,] 52.24427 26.981648 21.25313 21.438965
```

```
plot_traj(x, "Gamma à échelle de temps, méthode Bridge")
```

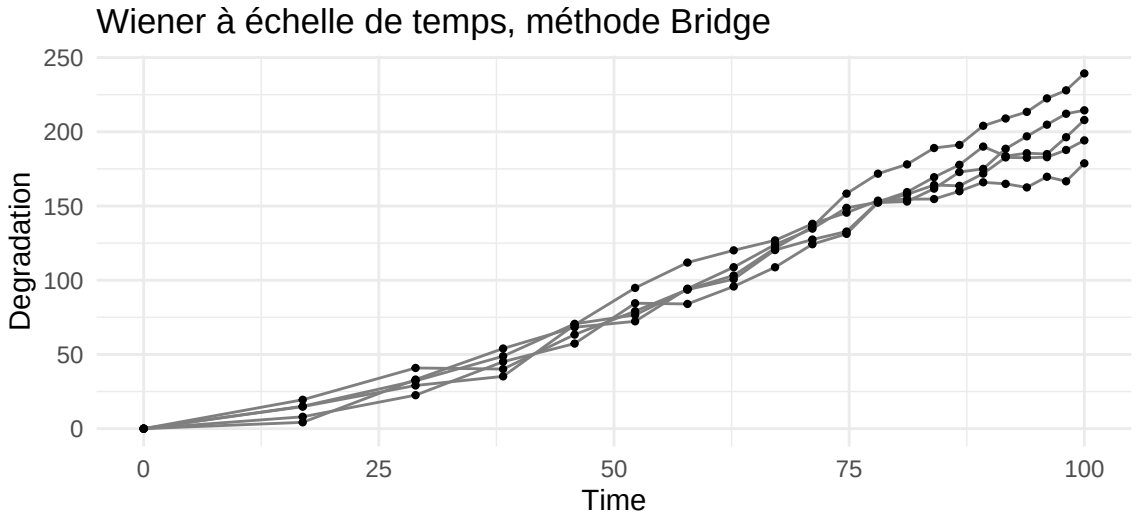


Modèle 2 : Wiener à échelle de temps

```
set.seed(422)
x <- sim_wiener(p2, sigma = 2, A = A, method = "Bridge")
head(x[, 1:4])
```

```
##             t        X1       X2        X3
## [1,]  0.00000  0.000000  0.00000  0.000000
## [2,] 16.90921  4.261326 15.09845  7.998852
## [3,] 28.90648 33.070877 32.18499 22.566928
## [4,] 38.21229 53.982407 48.81064 45.097343
## [5,] 45.81569 68.327226 70.62775 57.326368
## [6,] 52.24427 72.306392 76.64606 84.500329
```

```
plot_traj(x, "Wiener à échelle de temps, méthode Bridge")
```

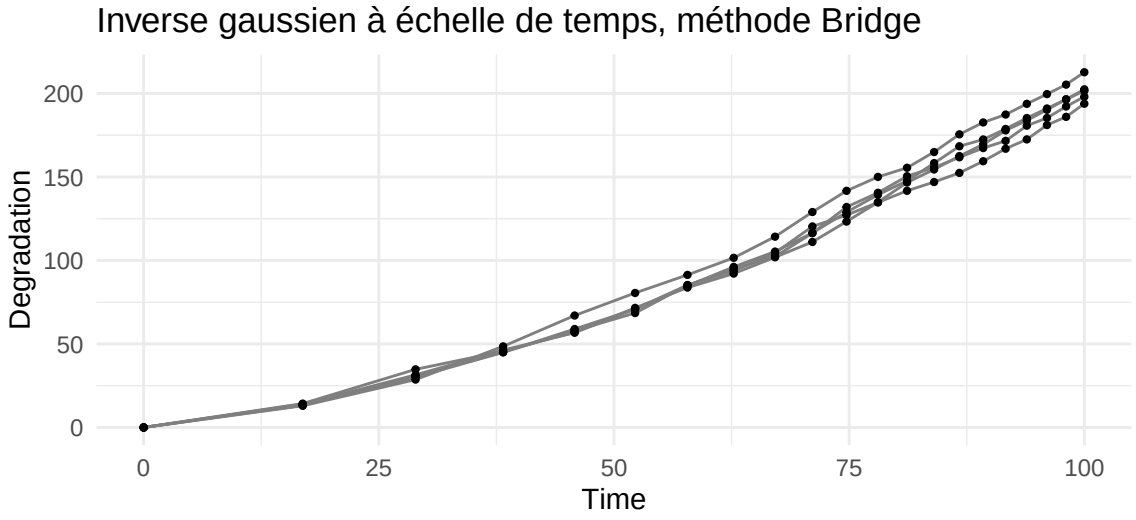


Modèle 3 : inverse gaussien à échelle de temps

```
set.seed(423)
x <- sim_ig(p2, eta = 2, A = A, method = "Bridge")
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.00000
## [2,] 16.90921 12.98125 13.89458 13.87690
## [3,] 28.90648 28.66407 31.49476 30.81953
## [4,] 38.21229 48.55013 46.31773 45.01009
## [5,] 45.81569 67.02454 56.72777 58.92792
## [6,] 52.24427 80.53375 70.91723 70.79056
```

```
plot_traj(x, "Inverse gaussien à échelle de temps, méthode Bridge")
```



Vérification : la densité des accroissements

Pour chaque processus, on simule 100 trajectoires inspectées tous les 5. L'histogramme montre les accroissements simulés avec `method = "Bridge"`, et la courbe rouge la densité exacte de l'accroissement sur cette durée, calculée sans simulation. L'histogramme doit suivre la courbe.

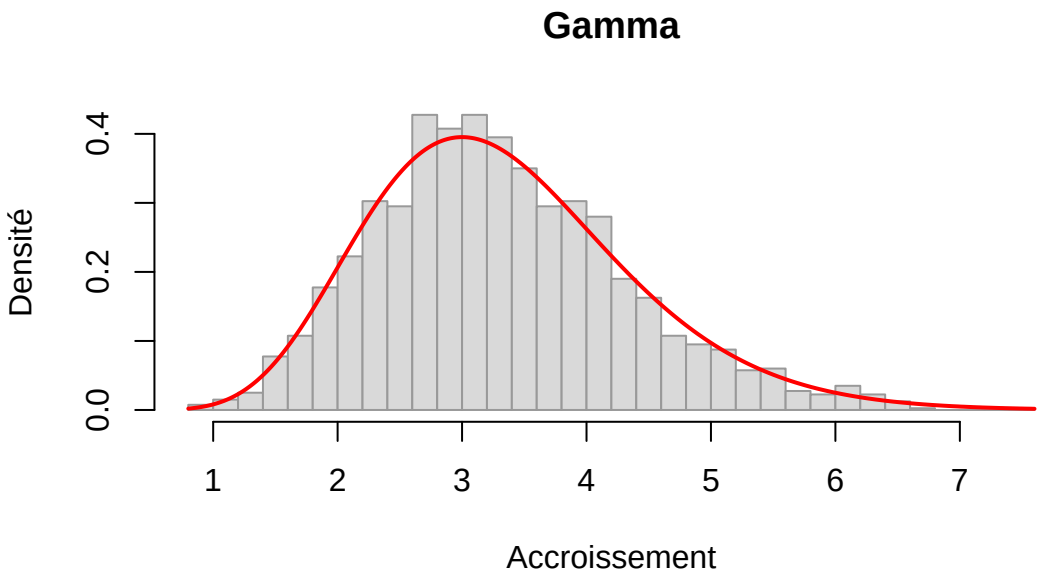
La forme de la densité dépend de la durée du pas : un accroissement gamma, par exemple, suit une loi gamma de forme `a * pas`, décroissante sur un pas court et en cloche sur un pas long.

```
pas <- 5
t6 <- seq(0, 100, by = pas)

trace_densite <- function(x, densite, titre) {
  dx <- traj_df(x, increments = TRUE)$dX
  dx <- dx[!is.na(dx)]
  hist(dx, breaks = 40, freq = FALSE, col = "grey85", border = "grey60",
       main = titre, xlab = "Accroissement", ylab = "Densité")
  curve(densite(x), add = TRUE, col = "red", lwd = 2, n = 500)
}
```

Modèle 1 : gamma

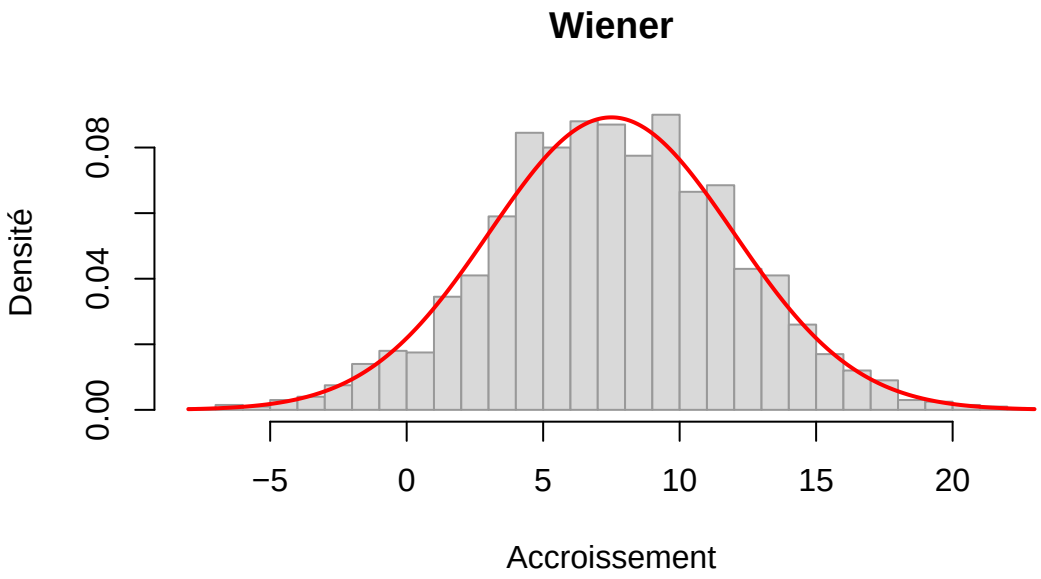
```
set.seed(431)
x <- sim_gamma(t6, a = 2, b = 3, n_traj = 100, method = "Bridge")
trace_densite(x, function(y) dgamma(y, shape = 2 * pas, rate = 3), "Gamma")
```



Modèle 2 : Wiener

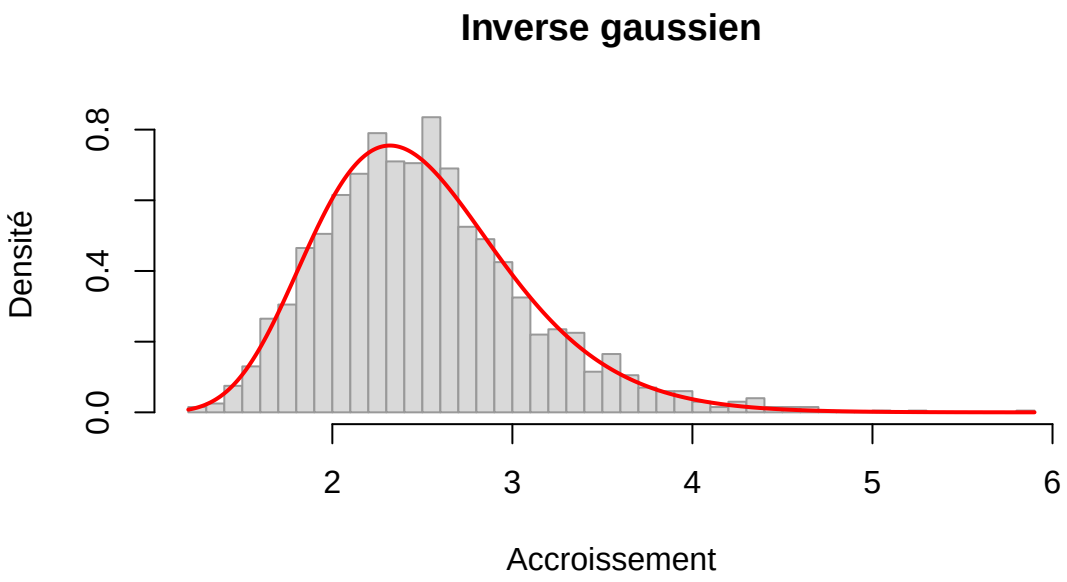
```
set.seed(432)
x <- sim_wiener(t6, mu = 1.5, sigma = 2, n_traj = 100, method = "Bridge")
```

```r
trace_densite(x, function(y) dnorm(y, mean = 1.5 * pas, sd = 2 * sqrt(pas)), "Wiener")
```



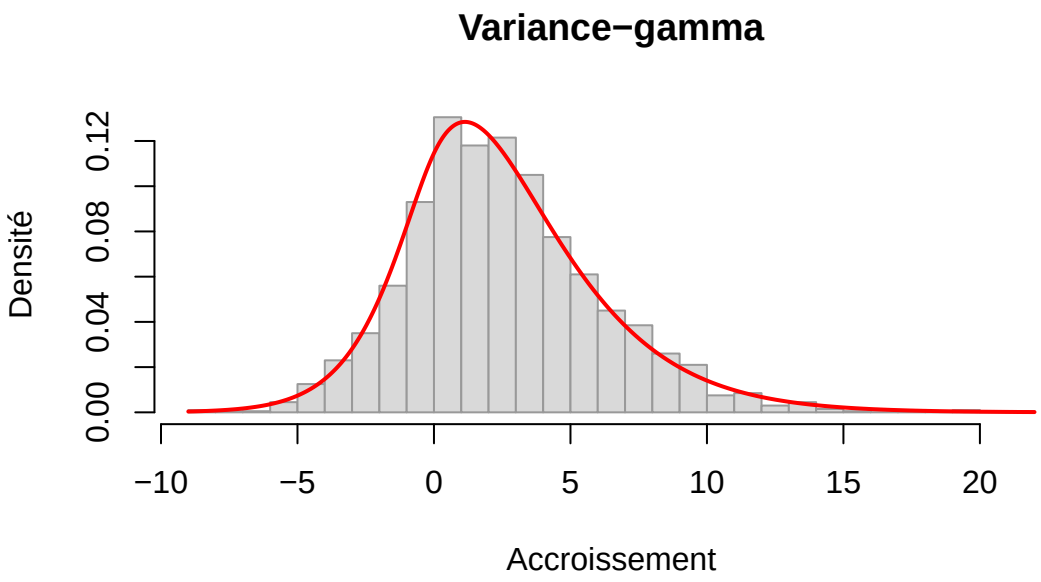
Modèle 3 : inverse gaussien

```r
set.seed(433)
x <- sim_ig(t6, mu = 0.5, eta = 2, n_traj = 100, method = "Bridge")
dig <- function(y) {
  m <- 0.5 * pas
  l <- 2 * pas^2
  sqrt(l / (2 * pi * y^3)) * exp(-l * (y - m)^2 / (2 * m^2 * y))
}
trace_densite(x, dig, "Inverse gaussien")
```



Modèle 4 : variance-gamma

```
set.seed(434)
x <- sim_vg(t6, theta = 0.5, sigma = 1.5, nu = 2, n_traj = 100, method = "Bridge")
dvg <- function(y, theta = 0.5, sigma = 1.5, nu = 2) {
  r <- sqrt(theta^2 + 2 * sigma^2 / nu)
  z <- abs(y) * r / sigma^2
  k <- pas / nu
  2 / (nu^k * sqrt(2 * pi) * sigma * gamma(k)) * (abs(y) / r)^(k - 0.5) *
    besselK(z, k - 0.5, expon.scaled = TRUE) * exp(theta * y / sigma^2 - z)
}
trace_densite(x, dvg, "Variance-gamma")
```



Modèle 5 : Meixner

```
set.seed(435)
x <- sim_meixner(t6, a = 1, b = 0.3, d = 1.5, m = 0.2, n_traj = 100, method = "Bridge")
loi <- Runuran::udmeixner(alpha = 1, beta = 0.3, delta = 1.5 * pas, mu = 0.2 * pas)
trace_densite(x, function(y) Runuran::ud(loi, y), "Meixner")
```

